VLSI Design

Contributed By:
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VLSI Design

Topic:

Introduction

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Module - I

Introduction

Historical perspective :-

As more and more complex functions are required in various data processing & Telecommunication devices, the need for higher integration density increases.

The level of integration (number of logic gates in a monolithic chip) has been steadily rising mainly due to the rapid progress in processing technology and interconnect technology.

Table: Evolution of logic complexity in IC

Era	Date	Complexity (logic blocks/chip)
Single transistor	1958	< 1
Unit logic (one gate)	1960	1
multi-function	1962	2 - 4
complex function	1964	5 - 20
medium scale Integration (MSI)	1967	20 - 200
Large scale Integration (LSI)	1972	200 - 2000
very large scale Integration (VLSI)	1978	2,000 - 20,000
Ultra large scale Integration (ULSI)	1989	> 20,000

The monolithic integration of a large number of functions on a single chip provides:

1> less area/volume i.e compactness

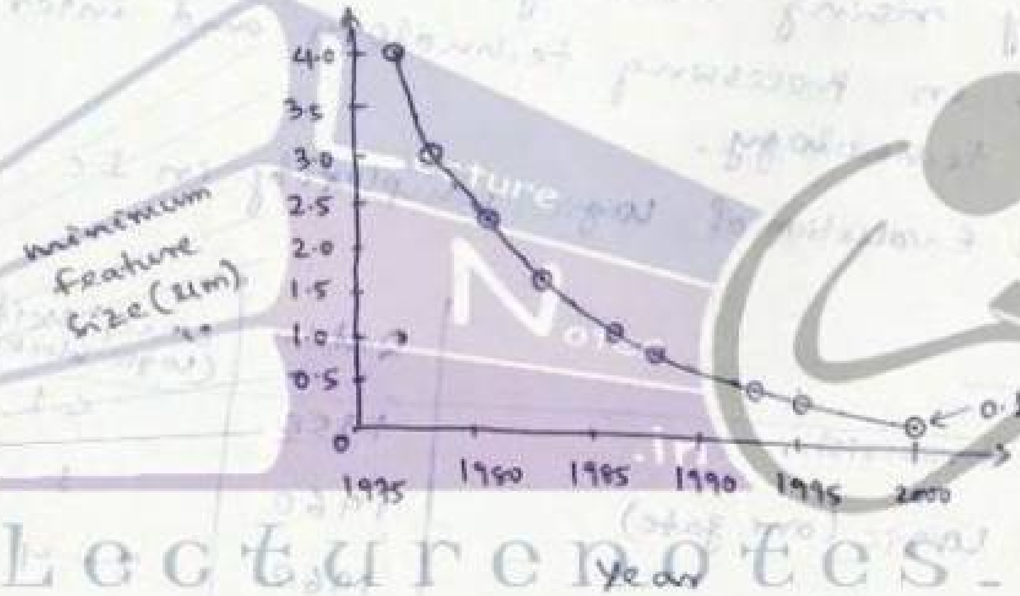
2> Less power consumption

3> less Testing requirements at system level

4> Higher reliability due to better on-chip interconnects

5> Higher speed due to short interconnect length

6> Significant cost reduction.



- ~~According~~ [Evolution of minimum feature size in integrated circuits]

- According to the International Technology Roadmap for semiconductors (ITRS), mos transistors with a feature size of 70nm by 2008 and 32nm by ~~2015~~ 2012 was expected to be available. allowing device densities of (about) 3 billion transistors/chip and 8-10 billion transistors/chip respectively.

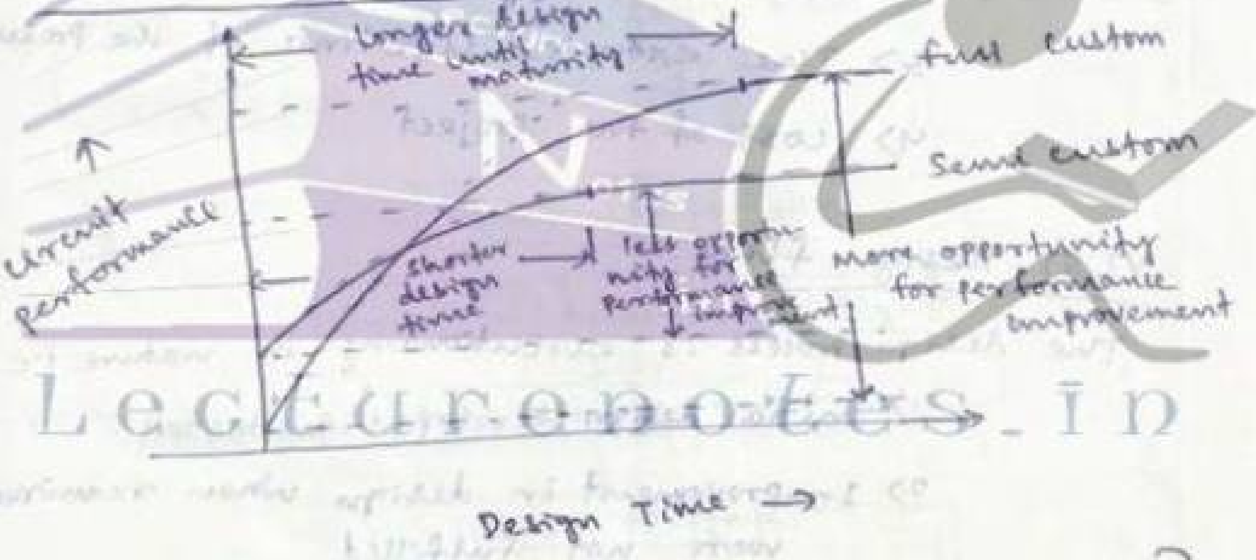
- According to Gordon moore's prediction, the transistor count / chip will be nearly doubled every two years (moore's law, 1960s).

figure 1.3 : level of integration vs time for memory chip and logic chips (microprocessor)

for memory chip :
rate of increase = 1.5 times every year

for logic chip :
rate of increase = 1.25 times every year.

VLSI Design methodologies :-



[Fig: Impact of different VLSI design styles upon the design cycle time and the achievable circuit performance]

- A full-custom design style is the one where the geometry and placement of every transistor can be optimized individually and it requires a longer time to reach design maturity.

- So the final product has a high level of circuit performance (high processing speed, low power dissipation) and better area utilisation. But it results in higher cost due to larger design time.

In contrast, "semi-custom" design style has a shorter design time but provides less circuit performance than full-custom design.

- The choice of a particular design style for a VLSI product depends on

1> The performance requirements
2> The technology used
3> The expected lifetime of the product
4> cost of the project

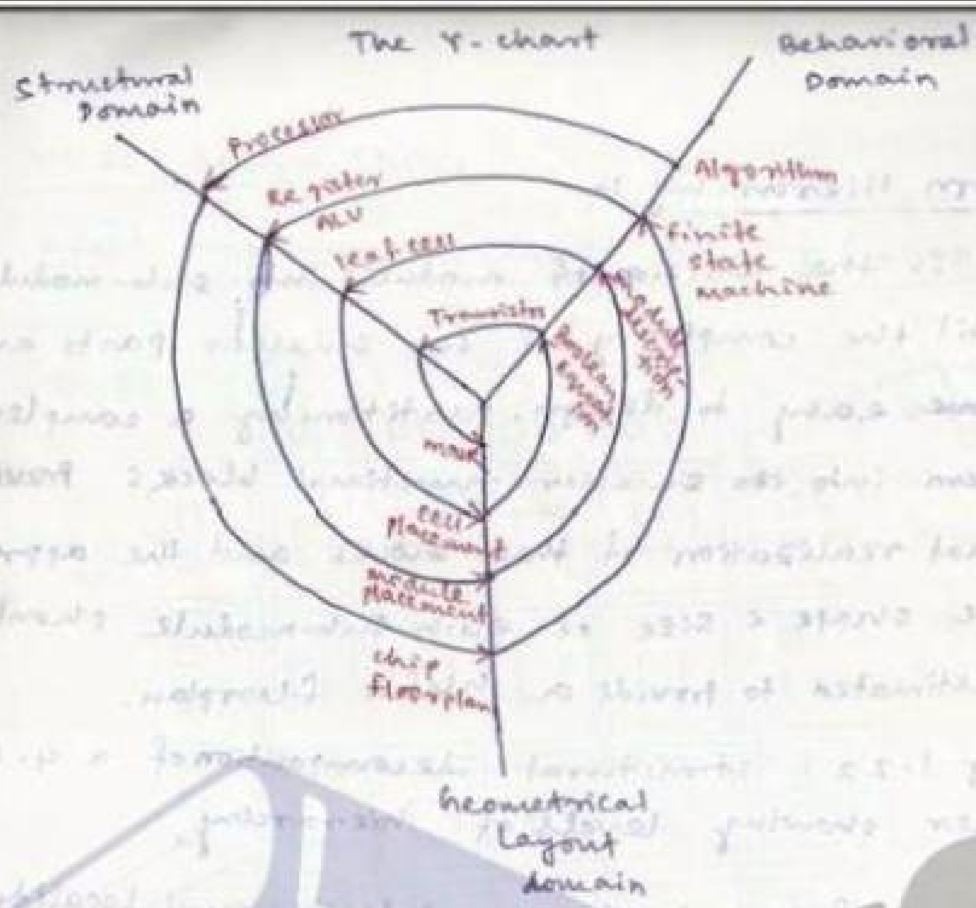
VLSI Design flow :-

The design process is evolutionary in nature i.e.

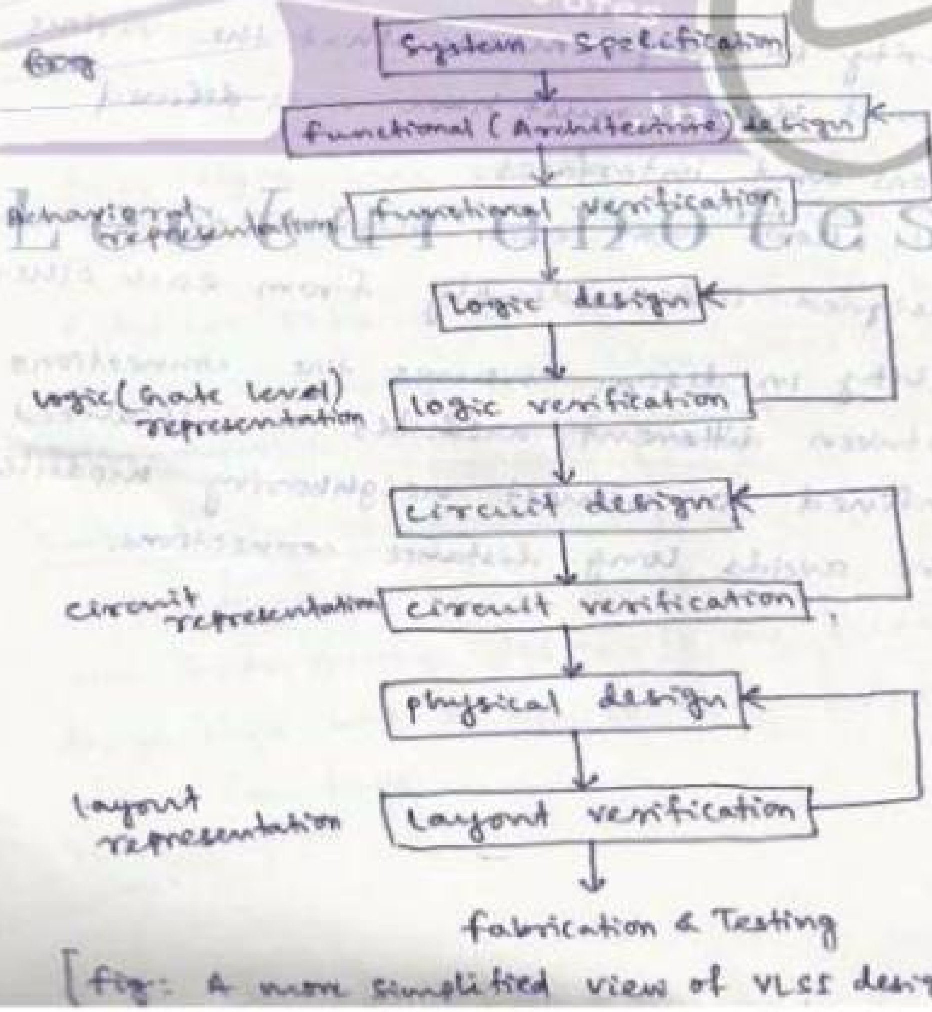
1> Initial design developed & tested
2> Improvement in design when requirements were not fulfilled
3> when improvements are too costly or not possible then requirements are revised & impact analysis was made.

- A Y-chart illustrates a simplified design flow in 3 different axes.

1> Behavioral domain
2> Structural domain
3> Geometrical layout domain



[fig: Simplified VLSI design flow in 3 domains (Y-chart representation)]



[fig: A more simplified view of VLSI design flow]

Design Hierarchy :-

It is the division of modules into sub-modules until the complexity of the smaller parts are ~~become~~ easy to design. partitioning a complex system into ~~the~~ smaller functional blocks provides actual realization of these blocks and the approximate shape & size of each sub-module should be estimated to provide a useful floorplan.

fig 1.22 : structural decomposition of a 4-bit adder showing levels of hierarchy.

concepts of Regularity, modularity and locality :-

⇒ Regularity is the concept of ~~repea~~ repeting a functional block throughout the design.

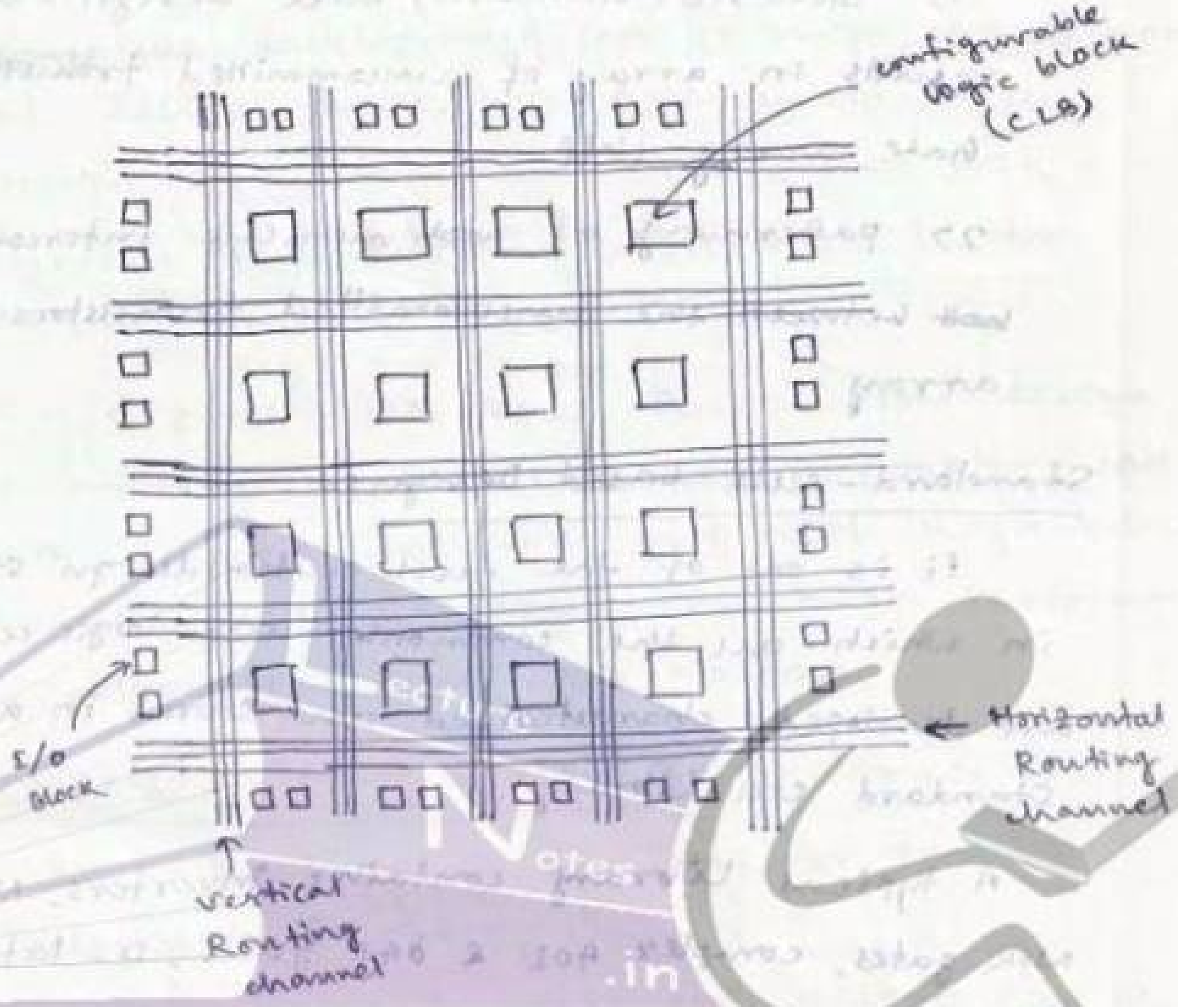
⇒ modularity in design means that the various functional blocks must have well defined functions and interfaces.

It allows that each block or module can be designed independently from each other,

⇒ locality in design ensures the connections ~~are~~ between different modules are localized i.e confined to atmost neighboring modules therefore avoids long distance connections.

VLSI Design Styles:-

Field Programmable Gate Array (FPGA):-



This design style ~~[illegible]~~ is a fast prototyping and cost-effective chip design process.

A typical FPGA chip consists of I/O buffers, array of configurable logic blocks (CLBs) and Programmable interconnect structures.

Gate Array Design:-

In terms of fast prototyping capability, the Gate Array design is the second best design style with a typical turn-around time of a few days.

Gate Array implementation requires 2 manufacturing steps:-

1> Generic (standard) mask design :- which results in array of uncommitted transistor on Gate array chip.

2> patterning of ~~mett~~ metallic interconnects ~~bett~~ between the uncommitted transistors of the array.

Standard - cells based Design:-

It is one of the full custom design styles in which all the commonly used logic cells are developed, characterized and stored in a standard cell library.

A typical library contains inverters, NAND, NOR gates, complex AOI & OAI gates, D - latches and flip-flops.

Each cell is characterized according to characteristics like :

1> Delay time vs Load capacitance

2> Circuit Simulation model

3> Timing Simulation model

4> fault Simulation model

5> Cell data for place-and-route

6> mask data

Full custom design :-

In this design style the entire mask design is done new without use of any library.

Since the development cost is high, a concept of design reuse is popular to reduce design cycle time and development cost.

Typical full custom design style is the design of memory cell

- In digital CMOS VLSI, full custom design is rarely used due to high laboratory cost with exception including design of high-volume products like memory chips, high performance microprocessors and FPGA masters.

Computer-Aided Design Technology :-

CAD Technology is essential for timely development of ICs.

CAD Technology for VLSI chip design can be categorized into :

1> High level synthesis
2> Logic synthesis
3> Circuit optimization
} synthesis tools

4> Layout —— Layout tools

5> Simulation
6> Design rules checking
7> Formal verification
} simulation and verification tools.

Synthesis tools :-

The synthesis tools like HDL (verilog & VHDL) are helpful in automation of design phase in top level of design hierarchy as well as estimates the low level design features such as chip area, signal delay etc.

Tools for circuit optimization does transistor ~~Layout tools~~ :- sizing for minimization of delay process variations, noise and reliability issues.

Layout tools :-

It includes floorplanning, place-and-route, module generation etc.

Simulation and verification Tools :-

The simulation technology, which is the most mature area of VLSI CAD, includes many tools like:

1> circuit - level simulation (SPICE or its derivatives eg HSPICE)

2> Timing level simulation

3> Logic level simulation

4> Behavioral Simulation

5> Device - level Simulation

6> Process Simulation

The purpose of all these simulation tools are to investigate whether the designed circuit meets the required specifications at all the stages of the designed process or not.

circuit-level simulation is used to determine nominal and worst-case gate delays to identity delay-critical signal paths or elements and to predict the influence of parasitic effects upon circuit behaviour.

Logic simulation is performed mainly to verify the functionality of the circuit i.e to determine if the designed circuit actually has the desired logic behaviour.

—o—

VLSI Design

Topic:

Fabrication Of MOSFET

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fabrication of MOSFETs

Introduction :-

Create n-well regions and channel-stop regions

↓

Grow field oxide and gate oxide (thin oxide)

↓

Deposit and pattern polysilicon layer

↓

Implant source and drain regions, substrate contacts

↓

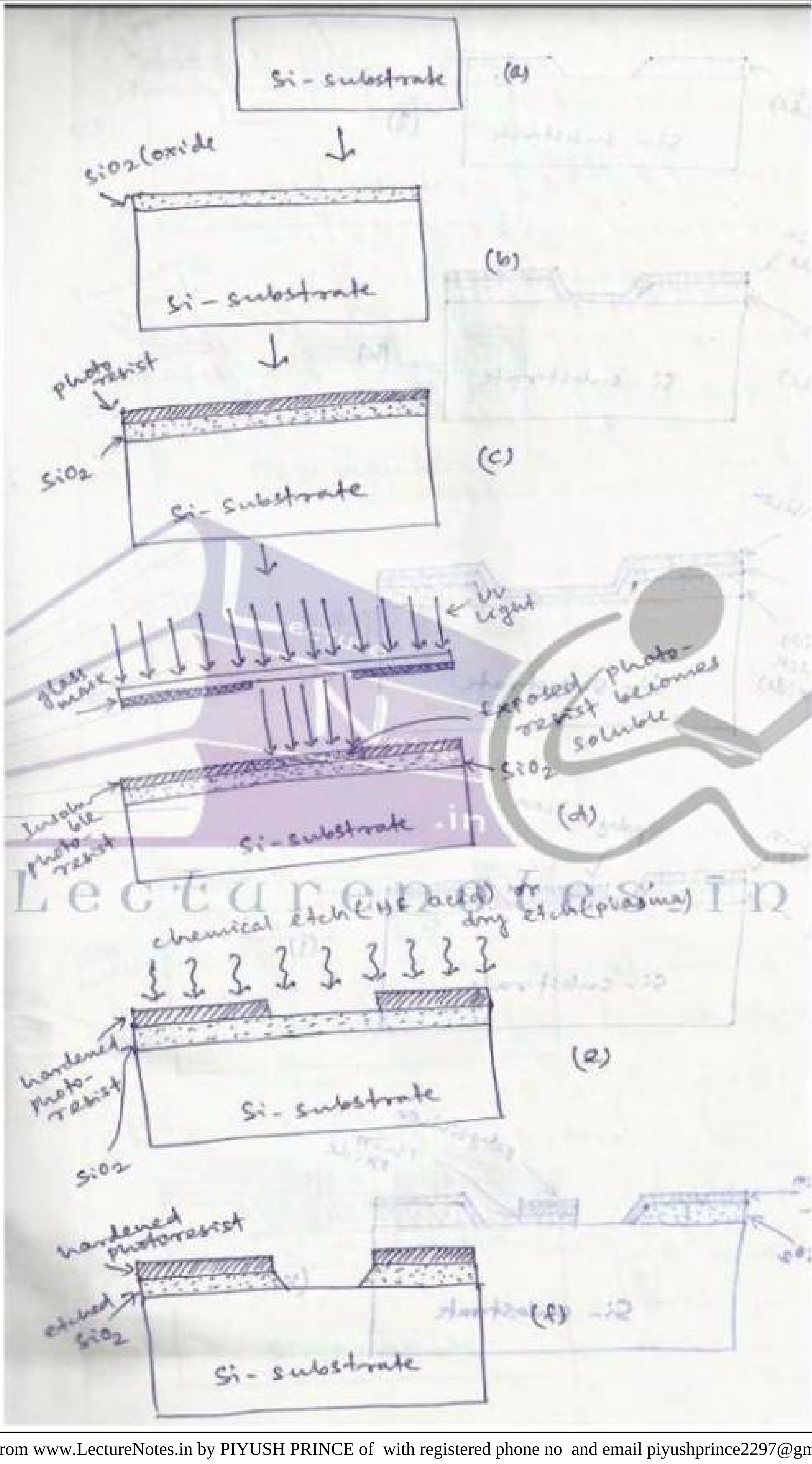
create contact windows, deposit and pattern metal layer

(process semence for the fabrication of the n-well cmos integrated circuit)

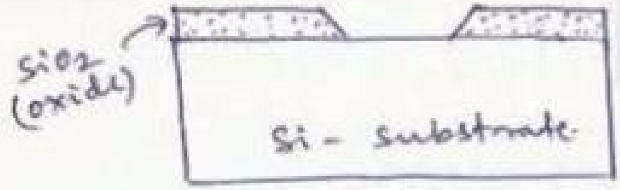
fabrication process flow: Basic Steps

An integrated circuit is a set of patterned layers of doped silicon, polysilicon metal and insulating silicon dioxide.

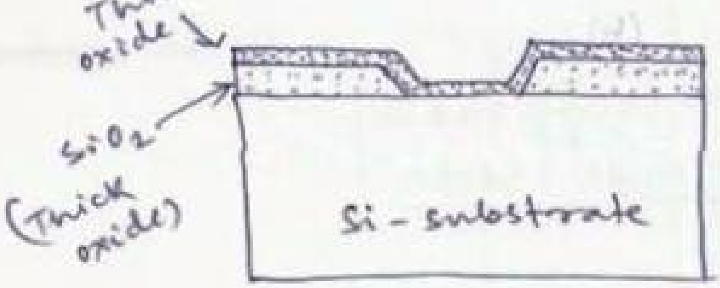
- The process of transferring a pattern to a layer of the IC chip is called lithography (Typically photolithography vs EBL)



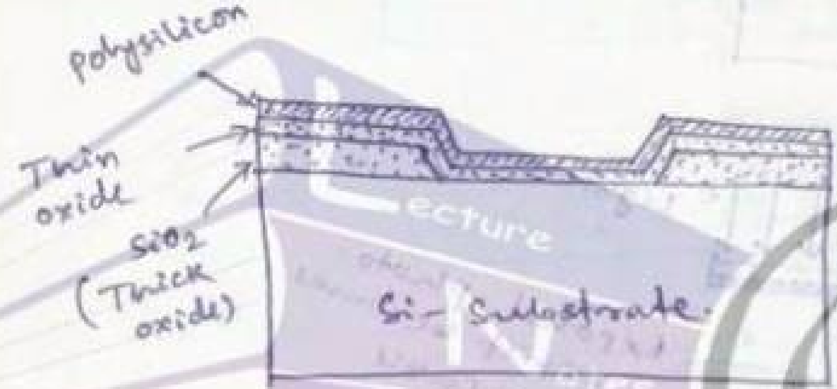
Si - substrate
(a)
SiO2 (oxide
Si - substrate
(b)
photo resist
SiO2
Si - substrate
(c)
UV light
glass mask
Exposed photo-resist becomes soluble
SiO2
Insoluble photo-resist
Si - substrate
(d)
chemical etch (HF acid) or dry etch (plasma)
hardened photo-resist
SiO2
Si - substrate
(e)
hardened photoresist
etched SiO2
Si - substrate
(f)


SiO2 (oxide)
Si - substrate

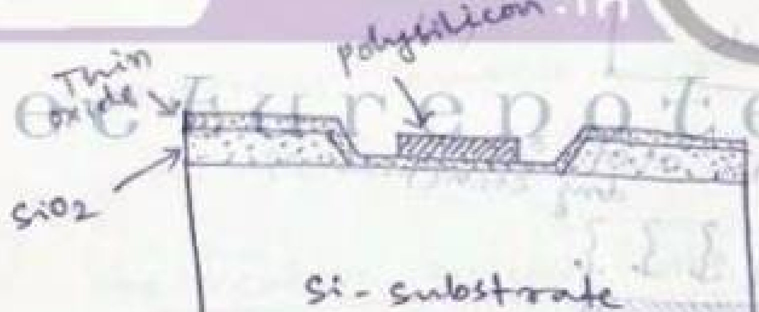
(g)


Thin oxide
SiO2 (Thick oxide)
Si - substrate

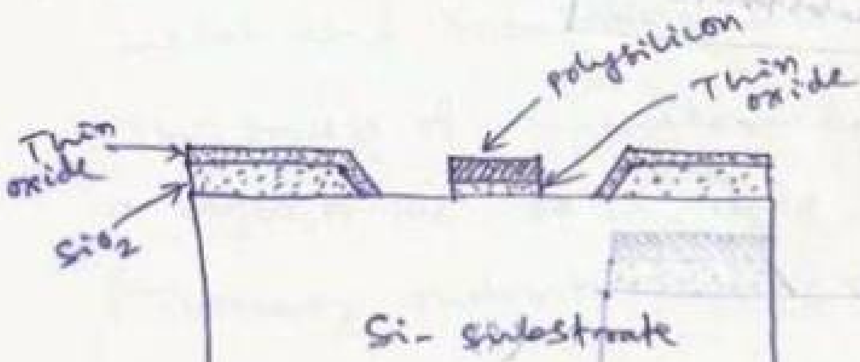
(h)


polysilicon
Thin oxide
SiO2 (Thick oxide)
Si - substrate

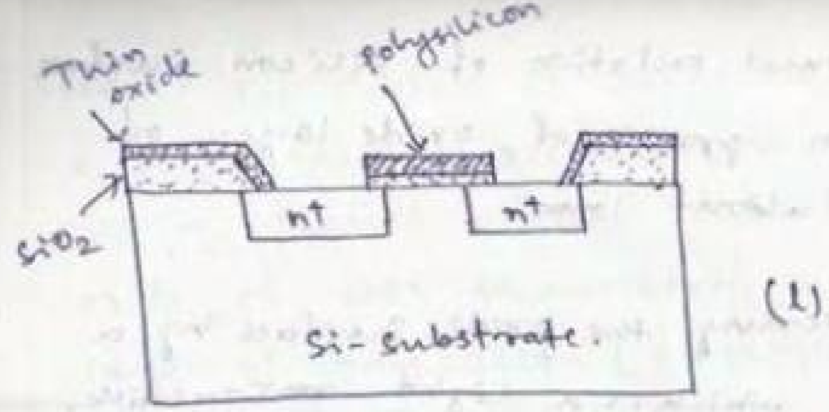
(i)

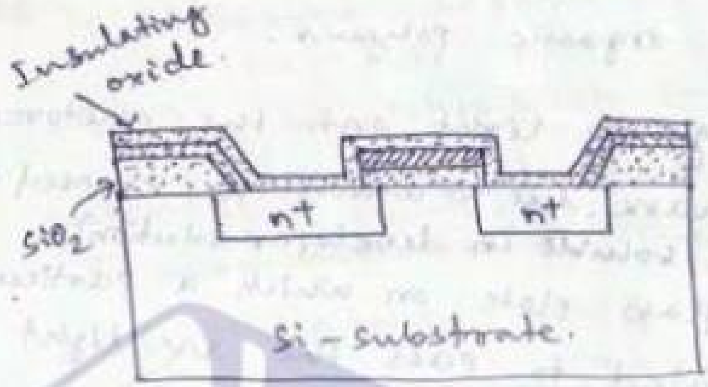

Thin oxide
polysilicon
SiO2
Si - substrate

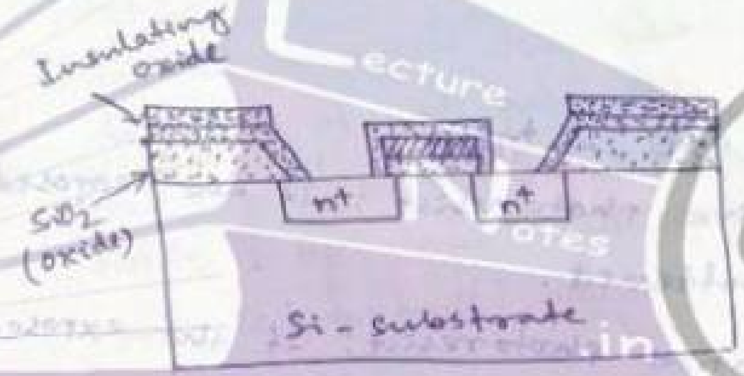
(j)

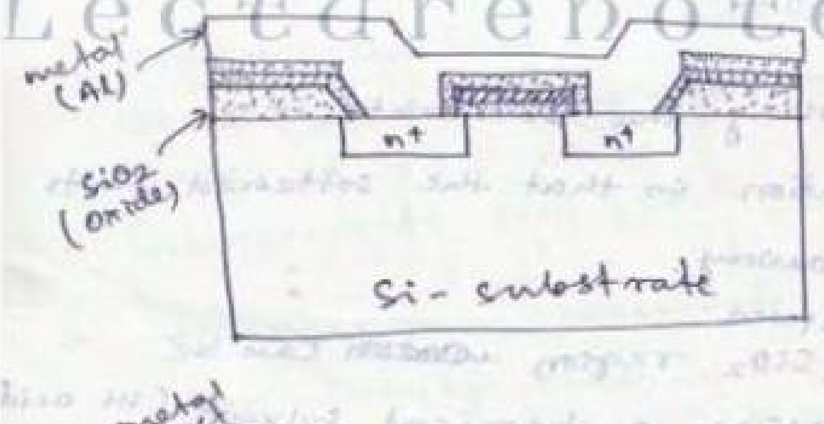

Thin oxide
polysilicon
Thin oxide
SiO2
Si - substrate

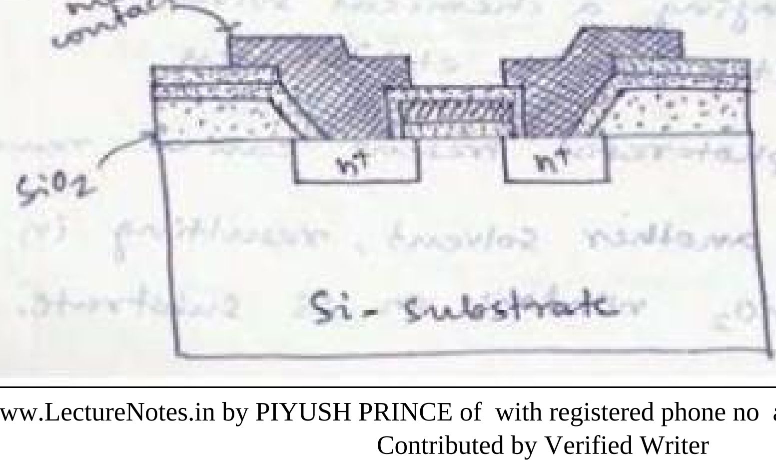
(k)


Thin oxide
polysilicon
SiO_2
n^+
n^+
(l)
Si-substrate.


Insulating oxide
SiO_2
n^+
n^+
(m)
Si-substrate.


Insulating oxide
SiO_2 (oxide)
n^+
n^+
(n)
Si-substrate


metal (Al)
SiO_2 (oxide)
n^+
n^+
(o)
Si-substrate


metal contact
SiO_2
n^+
n^+
(p)
Si-substrate

step-1: Thermal oxidation of Silicon surface results in growth of oxide layer of thickness about 1mm.

step-2: covering the oxide surface by a photoresist which is a light- ~~se~~sensitive, acid-resistant organic polymer.

step-3: passing uv light onto the photoresist through a mask, due to which the exposed part is softened (i.e soluble in developer solution).

* mask is a glass plate on which a particular pattern is coated to pass the uv light selectively.

* photoresists are 2 types:
 1> positive photoresist: if the exposed parts get softened.
 2> Negative photoresist: if the exposed parts get hardened.

step 4: passing the substrate into developer solution so that the softened parts are etched away.

step 5: The exposed SiO_2 region ~~[illegible]~~ can be removed applying a chemical solvent (HF acid) or by dry etch (plasma etch) process.

Then the photoresist present can be removed by applying another solvent, resulting in creation of SiO_2 window on Si substrate.

Step 6: Top surface is deposited with a thin & high quality oxide layer which will work as a ~~perface~~ perfect insulator ~~for~~ as gate oxide of mos transistor. on top of that a layer of polysilicon (polycrystalline silicon) is ~~deposited~~ deposited.

step 7: The polysilicon layer is patterned & etched to form interconnects and gate of mos transistor. The exposed thin gate oxide is etched away and then source and drain regions (2 n^+ regions) are formed through the bare Si substrate by diffusion of n^+ carriers. or ion implantation.

step 8: deposition of another layer of insulating ~~oxide~~ SiO_2 and patterning & etching to provide contact window for source & drain (n^+ regions).

step 9: surface is covered with evaporated aluminum. finally the metal layer is patterned and etched to form metal interconnects over source & drain.

* for ~~accuracy~~ accurate generation of high-density patterns in sub-micron devices, Electron beam lithography (EBL) is used instead of optical lithography.

Device Isolation Techniques:-

The mos transistors in an IC must be electrically isolated from each other during fabrication. This can be achieved by creating dedicated ~~regi~~ fabrication regions called "active areas" separated by thick oxide layer barriers called "field oxide".

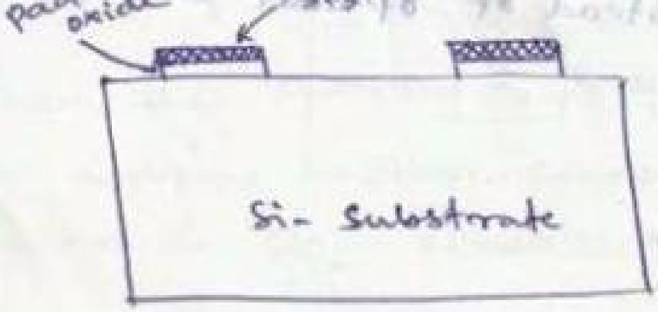
Need for device isolation:

1> Prevent unwanted conduction paths bet^n devices
2> Prevent creation of inversion layers outside the channel region
3> reduce leakage current.

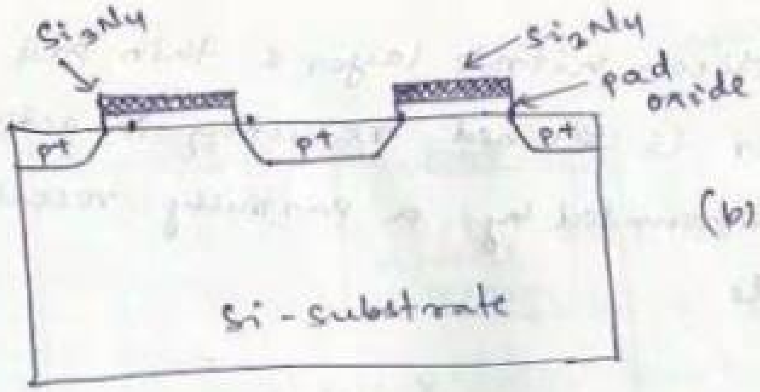
Local oxidation of silicon (LOCOS):-

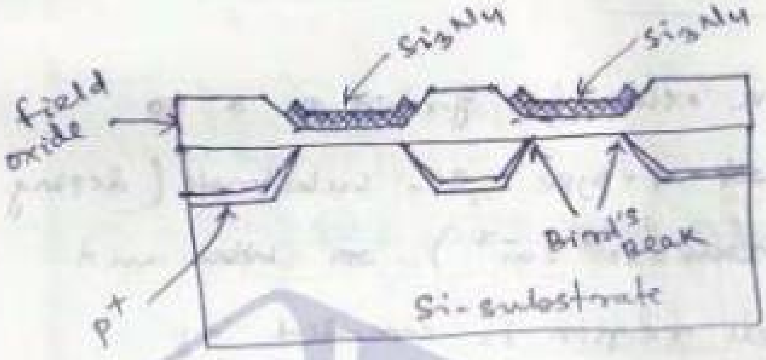
It is a technique of ~~set~~ selectively growing the field oxide in specific regions instead of selectively etching away the active areas after oxide growth on the entire surface.

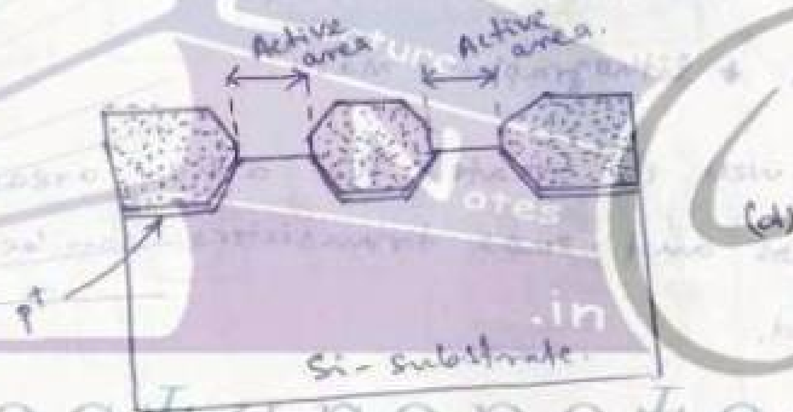
Steps for LOCOS Process:-



(a)







Step-1: deposition of a thin pad oxide on Silicon ~~st~~ substrate followed by deposition and patterning of a silicon nitride layer to form the active areas.

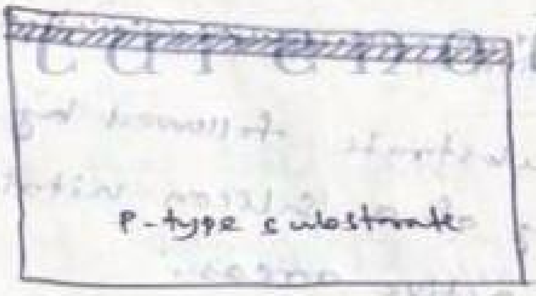
Step-2:- Thick field oxide is grown in the area not covered with silicon nitride. This field oxide forms a lateral extension under the nitride layer called bird's beak region. This region is responsible for reduction of active area.

Step 3:- Silicon nitride layer & thin pad oxide layer is etched resulting in active areas surrounded by a partially recessed field oxide.

The CMOS n-well process:-

→ An initial thick oxide is grown on a ~~po~~ moderately doped P-type Si- substrate (doping conc. of less than $10^{15}\ cm^{-3}$) ~~on wa~~ and then a n-well region is created by diffusion or ion implantation of phosphorus atoms through a lithographic mask.

- once n-well is created the active areas of the nmos and pmos transistors can be fabricated.



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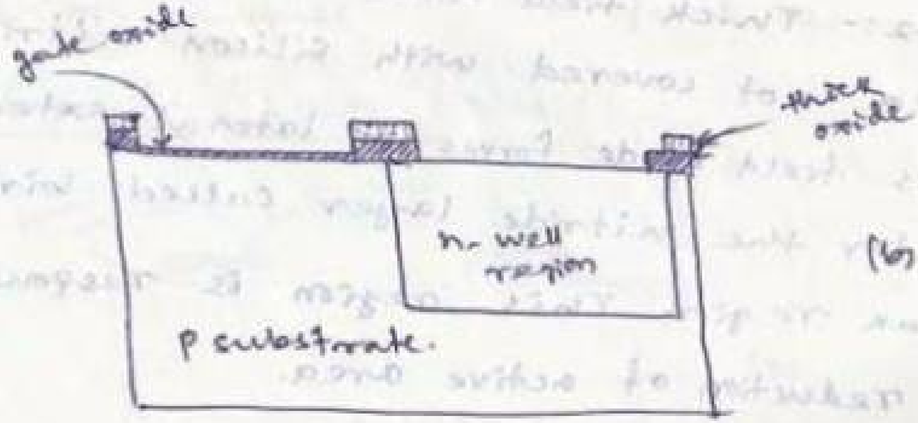


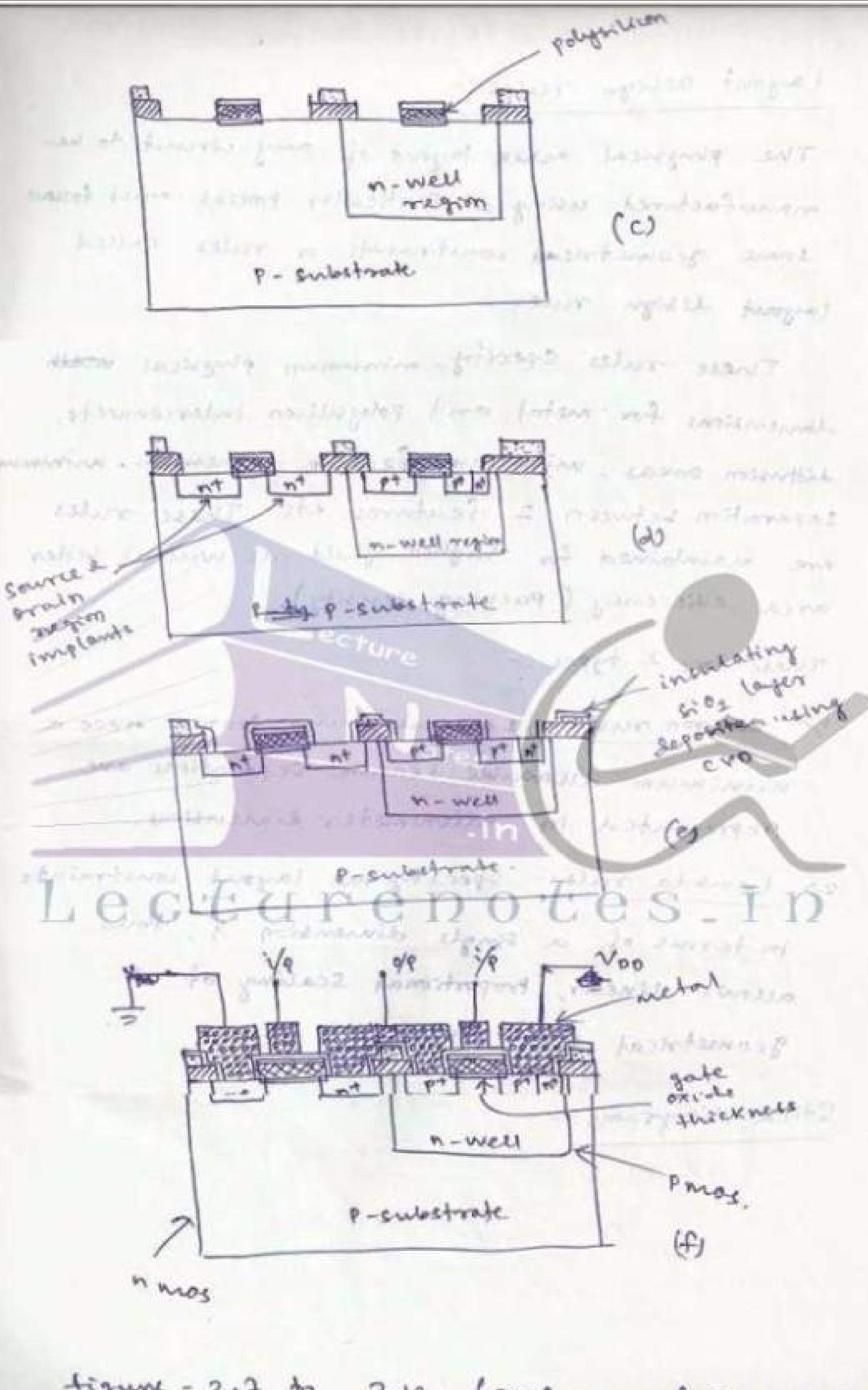


figure - 2.7 to 2.12 (reference of 3D view)

Layout Design rules :-

The physical mask layout of any circuit to be manufactured using a particular process must follow some geometrical constraints or rules called layout design rule.

These rules specify minimum physical ~~width~~ dimensions for metal and polysilicon interconnects, diffusion areas, minimum feature dimensions, minimum separation between 2 features. etc. These rules are maintained for higher yield as well as better area efficiency (packing density).

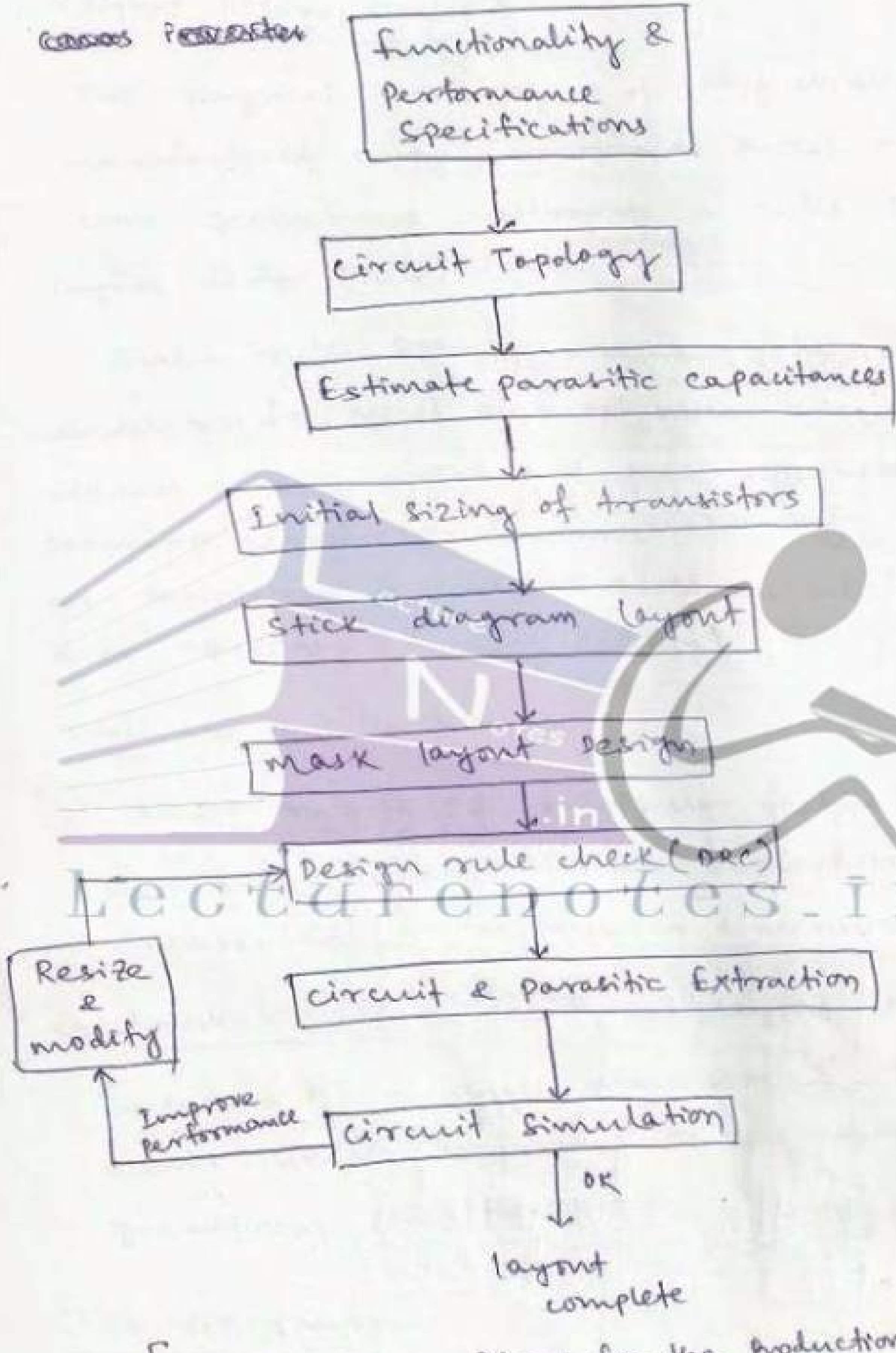
These are 2 types :-

1) micron rule :- If minimum feature sizes & minimum allowable feature separations are represented in micrometer dimensions.

2) lambda rule :- specify the layout constraints in terms of a single dimension 'λ', thus allows linear, proportional scaling of all geometrical constraints.

Stick diagrams :-

full-customs mask layout design:-



[Typical design flow for the Production of a mask layout] mosis layout design rules:-

figure 2.13, page 68 & 69

figure 2.16 & 2.17, page 72 & 73

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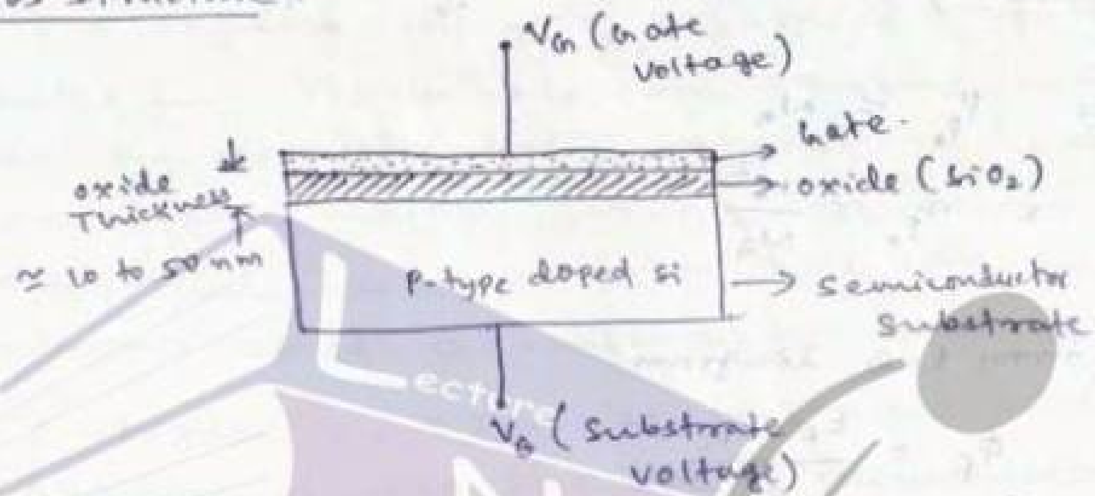
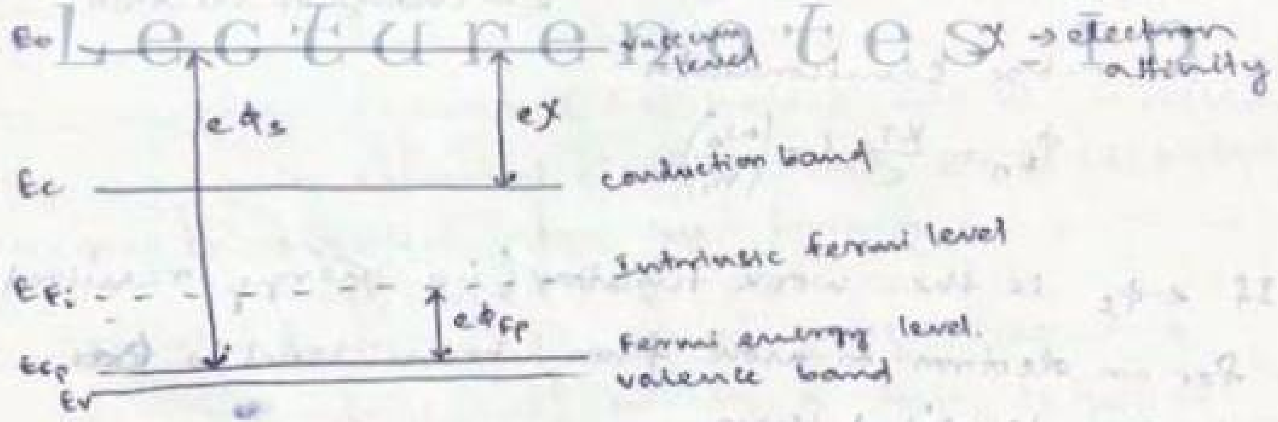
MOS Transistor

MOS structure:-



MOS structure acts as a ||el plate capacitor with gate and substrate as 2 plates & oxide as dielectric.



[E-B diagram of a p-type si substrate]

According to law of mass action, the equilibrium mobile carrier concentrations are related as $n \times p = n_i^2$

VLSI Design

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MOS Transistors

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mosis layout design rules:-

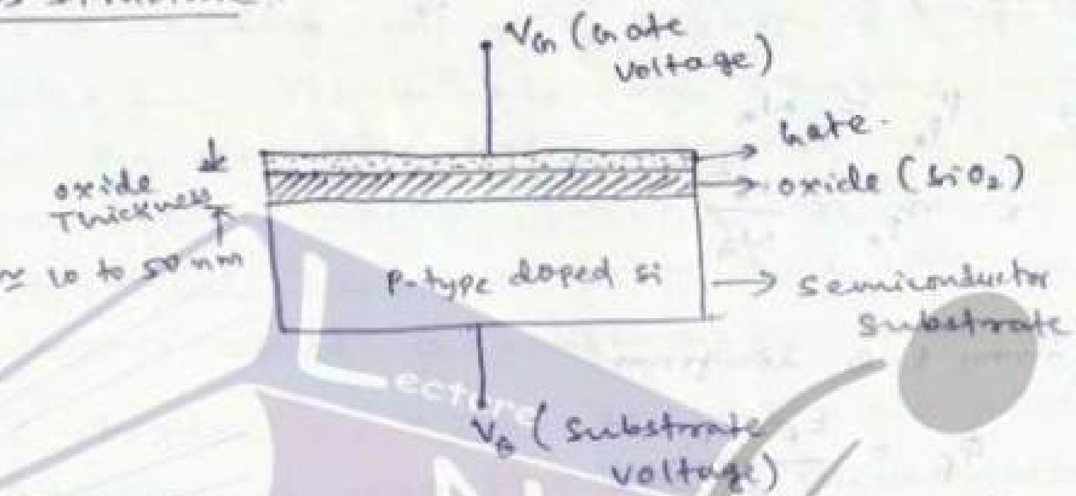
figure 2.13, page 68 & 69

figure 2.16 & 2.17, page 72 & 73

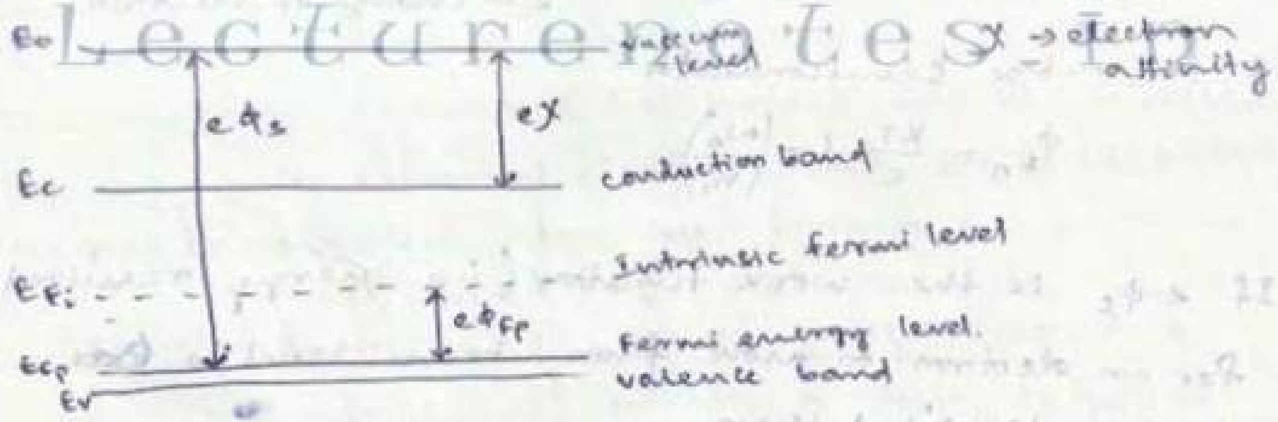
——o——

MOS Transistor

MOS structure:-



MOS structure acts as a $\parallel^{el}$ plate capacitor with gate and substrate as 2 plates & oxide as dielectric.



[E-B diagram of a p-type Si substrate]

According to law of mass action, the equilibrium mobile carrier concentrations are related as $n \times p = n_i^2$

where n & p are electron and hole concentration in any ~~material~~ semiconductor and n_i denotes the intrinsic carrier concentration of that material.

so $n = \frac{n_i^2}{p}$ or $p = \frac{n_i^2}{n}$

for a p-type Si substrate, majority carrier (hole concentration) is denoted as $p_{p_0} \simeq N_A$

so $p_{p_0} \simeq N_A$

& $n_{p_0} \simeq \frac{n_i^2}{N_A}$

Again from E-B diagram

$$\phi_F = \frac{E_F - E_{Fi}}{e}$$

for p-type semiconductor,

$$\phi_{Fp} = \frac{KT}{e} \ln\left(\frac{n_i}{N_A}\right)$$, K → Boltzmann constant.

e → charge of electron

& for n-type semiconductor

$$\phi_{Fn} = \frac{KT}{e} \ln\left(\frac{N_D}{n_i}\right)$$

If $e\phi_s$ is the work function (i.e energy required for an electron to move from fermi level to ~~free~~ vaccum level) & then

$$e\phi_s = e\chi + (E_c - E_F)$$

(fig 3.3 & 3.4, page 86)

mos system under External bias :-

Depending on the polarity and magnitude of V_G, 3 different operating regions are there for mos. They are :

1> accumulation
2> depletion
3> inversion

→ If a negative voltage V_G ($V_G < V_{FB}$) is applied to gate, the holes from p-substrate are attracted near to the semiconductor- oxide interface. The hole concentration (majority carrier conc.) at the interface exceeds the equilibrium hole conc. This condition is called "accumulation".

(figure 3.5 : Cross-sectional view and E-B diagram of mos structure in accumulation region.)

→ If a small/negligible +ve gate bias V_G is applied to the gate electrode with substrate bias $V_B = 0V$, the +ve surface potential causes the energy band to bend downward near the surfaces. The majority carriers (i.e holes) will be repelled by the gate ~~potential~~ electric field. so a depletion region is created near the interface.

(fig 3.6 : page 88.)

let there are dQ no. of holes present below the interface within a thin layer of thickness dx.

$$dQ = -eN_A\,dx.$$

Then according to poisson eqn.

$$d\phi_s = -x\,\frac{dQ}{\epsilon_{si}} = \frac{eN_A x}{\epsilon_{si}}\cdot dx$$

$$\int_{\phi_F}^{\phi_s} d\phi_s = \int_0^{x_d} \frac{eN_A x}{\epsilon_{si}}\,dx$$

$$\Rightarrow \phi_s - \phi_F = \frac{eN_A x_d^2}{2\epsilon_{si}}$$

$\Rightarrow$ width/depth of depletion region near the interface is

$$\boxed{x_d = \sqrt{\frac{2\epsilon_{si}|\phi_s - \phi_F|}{eN_A}}}$$

W_{dep}

$$V_{ox} = \frac{-Q_{dep}}{C_{ox}} = \frac{eN_a W_{dep}}{C_{ox}} = \frac{\sqrt{2eN_a\epsilon_s\phi_s}}{C_{ox}}$$

& depletion region charge density

$$Q_{dep} = -eN_A x_d = -\sqrt{\frac{e^2 N_A^2 \times 2\epsilon_{si}|\phi_s - \phi_F|}{eN_A}}$$

$$\boxed{Q_{dep} = -\sqrt{2eN_A\epsilon_{si}|\phi_s - \phi_F|}}$$

again $eN_a W_{dep} = \sqrt{2eN_a\epsilon_s\phi_s}$

$\Rightarrow \phi_s = \frac{eN_a W_{dep}^2}{2\epsilon_s}$

& $V_{g} = V_{fb} + V_{ox} + \phi_s$

$= V_{fb} + \frac{eN_a x_d}{C_{ox}} + \frac{eN_a x_d^2}{2\epsilon_s}$

$\rightarrow$ If the gate bias is a larger +ve voltage, the downward bending of the energy bands near interface will further increase for which the midgap energy E_{Fi} (or E_i) falls below E_{Fp}. due to this the surface semiconductor becomes n-type within the thin layer. The electron density at interface is more than hole density. This n-type region created near the interface due to +ve gate bias is called inversion layer and the condition is called surface inversion.

(fig 3.7, page 90)

The maximum depletion width is achieved when the surface potential 'ϕ_s' is equal and opposite to the bulk/substrate potential 'ϕ_F':

$$\boxed{x_{d_m} = \sqrt{\frac{2\epsilon_{si}|2\phi_F|}{eN_A}}}$$

Structure and operation of MOS transistor (MOSFET):

(fig 3.8, page 91, structure of n-channel E-MOSFET)

(fig 3.9 - page 92, circuit symbol for n-channel & p-channel E-MOSFET)

(fig 3.10 - formation of depletion region in an n-channel E-MOSFET in surface depletion mode of operation i.e $V_{GS} < V_{T0}$)

(fig 3.11 & 3.12 - E-B diagram & formation of inversion layer (channel) in n-channel E-MOSFET in surface inversion mode i.e $V_{GS} > V_{T0}$)

Threshold Voltage:-

The work function difference ϕ_{GC} between the gate and channel is given by

$$\phi_{GC} = \phi_{F(substrate)} - \phi_M \quad \text{for metal gate}$$

$$\text{or } \phi_{GC} = \phi_{F(substrate)} - \phi_{F(gate)} \quad \text{for polysilicon gate}$$

The depletion region charge density at surface inversion condition ($\phi_s = -\phi_F$) is given by

$$Q_{B0} = -\sqrt{2eN_A\epsilon_{si}|-2\phi_F|} \quad \text{with } V_{SB} = 0 \text{ i.e zero substrate bias.}$$

If the substrate (Body) is biased at a diff. voltage V_{SB} (source-to-substrate voltage), then

$$Q_B = -\sqrt{2eN_A\epsilon_{si}|-2\phi_F + V_{SB}|}$$

Then the threshold voltage V_{T0} for zero substrate bias is given by

$$\boxed{V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}}$$

where $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

& for non-zero substrate bias voltage,

$$V_T = \phi_{GC} - 2\phi_F - \frac{Q_B}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

$$\boxed{V_T = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} - \frac{Q_B - Q_{B0}}{C_{ox}}}$$

$$\Rightarrow \boxed{V_T = V_{T0} - \frac{Q_B - Q_{B0}}{C_{ox}}}$$

Again $$\frac{Q_B - Q_{B0}}{C_{ox}} = -\frac{\sqrt{2eN_A\epsilon_{si}}}{C_{ox}} \times \left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)$$

$$\Rightarrow V_T = V_{T0} + \frac{\sqrt{2eN_A\epsilon_{si}}}{C_{ox}} \times \left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)$$

$$\boxed{V_T = V_{T0} + \gamma\left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)}$$

where $\gamma = \frac{\sqrt{2eN_A\epsilon_{si}}}{C_{ox}}$ → substrate-bias coefficient.

MOSFET operation :

(figure 3.14 : (a), (b) & (c) : operation of n-channel MOSFET in linear region, edge of saturation & beyond saturation region.)

page - 100

→ A MOSFET consists of a MOS capacitor with two p-n junctions placed immediately adjacent to the channel region which is controlled by MOS gate.

→ when $0 < V_{GS} < V_{T0}$, the region between source and drain is depleted, so no charge carrier flows through the region betn source & drain.

→ when $V_{GS} > V_{T0}$, an n-type conducting channel is established betn source & drain due to surface inversion.

→ with $V_{GS} > V_{T0}$ & $V_{DS} = 0$, thermal equilibrium exists in the channel for which $I_D = 0$

→ If V_{DS} is slightly increased, a current I_D proportional to V_{DS} will flow from drain to source through the conducting channel. This mode of operation is called linear mode.

→ with increase in V_{DS}, the magnitude of I_D will increase as well as the width of conducting channel near drain region decreases.

→ for $V_{DS} = V_{Dsat}$, the width of channel at drain becomes zero and a maximum/ Saturated drain current flows through the depletion region. This point of operation is called pinch-off point.

→ for $V_{DS} > V_{D_{sat}}$, the effective channel length decreases and a depleted region is formed adjacent to the drain & this depletion region grows towards source with increase in V_{DS}. This mode of operation is called saturation mode or saturation region.

MOSFET Current - Voltage characteristics :-

gradual channel Approximation (GCA) :-

for establishing the MOSFET i-v relationship.

fig 3.15 (page 101): cross-sectional view of an n-channel transistor

Here $V_S = V_B = 0V$.

$V_{GS} > V_{T_0}$ to create a conducting inversion layer between the source & drain

⇒ The ~~channel~~ channel ~~length~~ coordinate & channel voltage (V_c) varies from $y=0$ to $y=L$ i.e

$$\left.\begin{aligned} V_c(y=0) &= V_c = 0 \\ V_c(y=L) &= V_{DS} \end{aligned}\right\} \text{boundary cond}^n.$$

→ If $Q_I(y)$ is the total mobile electron charge in the surface inversion layer

$$Q_I(y) = -C_{ox}\left[V_{GS} - V_c(y) - V_{T_0}\right]$$

→ Assuming a channel width of 'W' and a constant surface mobility μ_n, the differential resistance of a differential channel 'dy' is given by

$$dR = - \frac{dy}{W \cdot \mu_n Q_1(y)}$$

& differential channel voltage with channel or drain current I_D is

$$dV_c = I_D \cdot dR = - \frac{I_D \, dy}{W \cdot \mu_n Q_1(y)}$$

$$\Rightarrow I_D \, dy = - W \mu_n \cdot Q_1(y) \cdot dV_c$$

integrating over the boundary conditions.

$$\int_0^L I_D \, dy = - W \cdot \mu_n C_{ox} \int_0^{V_{DS}} (V_{GS} - V_c - V_{T0}) \, dV_c$$

$$\Rightarrow I_D \cdot L = W \mu_n C_{ox} \left[\int_0^{V_{DS}} (V_{GS} - V_{T0}) dV_c - \int_0^{V_{DS}} V_c \, dV_c \right]$$

$$\Rightarrow I_D = \mu_n \cdot C_{ox} \times \frac{W}{L} \left[(V_{GS} - V_{T0}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

Drain current in linear region :-

$$\boxed{I_D = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L} \left[2 (V_{GS} - V_{T0}) V_{DS} - V_{DS}^2 \right]} \quad \text{—— (1.1)}$$

$$I_D = \frac{k_n'}{2} \cdot \frac{W}{L} \left[2 (V_{GS} - V_{T0}) V_{DS} - V_{DS}^2 \right] \quad \text{—— (1.2)}$$

$$I_D = \frac{k_n}{2} \left[2 (V_{GS} - V_{T0}) V_{DS} - V_{DS}^2 \right] \quad \text{—— (1.3)}$$

Where $K_n' = \mu_n \cdot C_{ox}$

$$\& \; K_n = \mu_n \cdot C_{ox} \cdot \left(\frac{W}{L}\right) = K_n'\left(\frac{W}{L}\right)$$

Ex 3.4

Drain current in Saturation region :-

In Saturation region

$$V_{DS} \geq V_{D_{sat}} \text{ or } (V_{GS} - V_{T0})$$

So putting $V_{D_{sat}} = (V_{GS} - V_{T0})$ in eqn (1.1)

$$I_{D_{sat}} = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L}\left[2(V_{GS}-V_{T0})(V_{GS}-V_{T0}) - (V_{GS}-V_{T0})^2\right]$$

$$\boxed{I_{D_{sat}} = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L}\left[(V_{GS} - V_{T0})^2\right]} \quad \text{--- } 2.1$$

$\&$ $I_D = 0$ for $V_{GS} < V_{T0}$

figure 3.17 & 3.18, page 106

current-voltage eqn for n-channel & p-channel MOSFET

page 111.

MOSFET Scaling and Small-geometry Effects :-

Scaling means reduction of dimensions of MOSFET, which leads to change in operational characteristics

2 types of Scaling techniques :-

1> full scaling or constant-field scaling.

2> constant voltage Scaling

1> full scaling :-

This scaling keeps the internal electric fields constant while all the dimensions are scaled down by a factor of S $(S > 1)$

~~Quantities~~	Before Scaling	After Scaling
~~channel length~~		
channel length	L	$L' = L/S$
channel width	W	$W' = W/S$
Gate oxide thickness	t_{ox}	$t'_{ox} = t_{ox}/S$
junction depth	x_j	$x'_j = x_j/S$
power supply voltage	V_{DD}	$V'_{DD} = V_{DD}/S$
Threshold voltage	V_{T0}	$V'_{T0} = V_{T0}/S$
Doping density	N_D N_A	$N'_D = S \cdot N_D$ $N'_A = S \cdot N_A$

$$C'_{ox} = \frac{\epsilon_{ox}}{t'_{ox}} = S \cdot \frac{\epsilon_{ox}}{t_{ox}} = S \cdot C_{ox} \quad \text{--- (1)}$$

$$k'_n = \mu_n \cdot C'_{ox}\left(\frac{W'}{L'}\right) = S \cdot \mu_n \cdot C_{ox}\left(\frac{W}{L}\right)$$

$$= S \cdot k_n$$

$$I'_{D(linear)} = \frac{K'_n}{2}\left[2\left(V'_{GS} - V'_{T0}\right)\cdot V'_{DS} - V'^2_{DS}\right]$$

$$= \frac{S \cdot K_n}{2}\left[2\left(\frac{V_{GS}}{S} - \frac{V_{T0}}{S}\right)\cdot \frac{V_{DS}}{S} - \left(\frac{V_{DS}}{S}\right)^2\right]$$

$$= \frac{I_{D(linear)}}{S} \qquad \text{—— (2.1)}$$

$$I'_{D(saturation)} = \frac{K'_n}{2}\left(V'_{GS} - V'_{T0}\right)^2 = \frac{I_{D(sat)}}{S} \qquad \text{—— (2.2)}$$

Power dissipation of the MOSFET before scaling

$$P = I_D \times V_{DS}$$

& after scaling $P' = I'_D \cdot V'_{DS} = \frac{I_D}{S} \times \frac{V_{DS}}{S} = \frac{P}{S^2}$ —— (3)

Power density after scaling

$$\frac{P'}{Area'} = \frac{P/S^2}{W' \cdot L'} = \frac{P/S^2}{W \cdot L/S^2} = \frac{P}{Area} \text{—— (4)}$$

i.e power density remains constant.

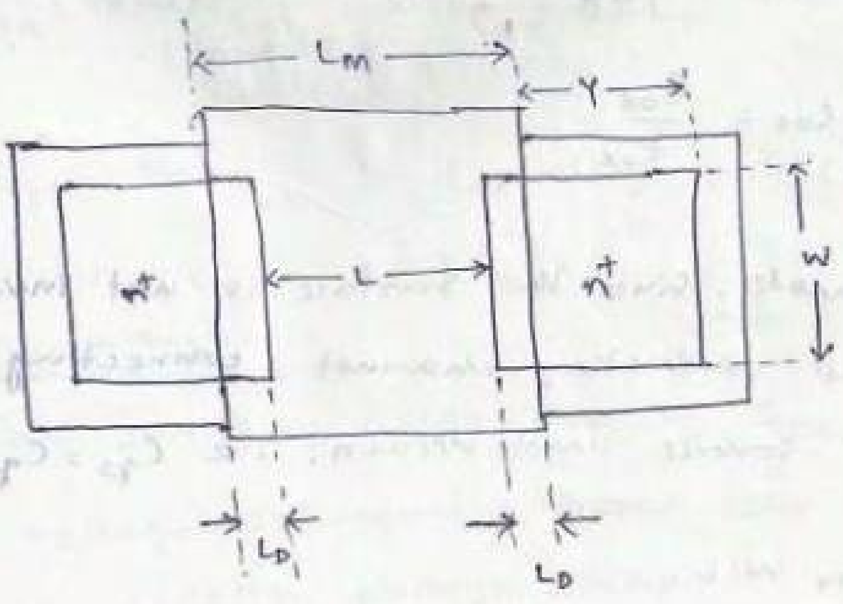
→ full scaling is more attractive because of the significant reduction in the power dissipation after scaling.

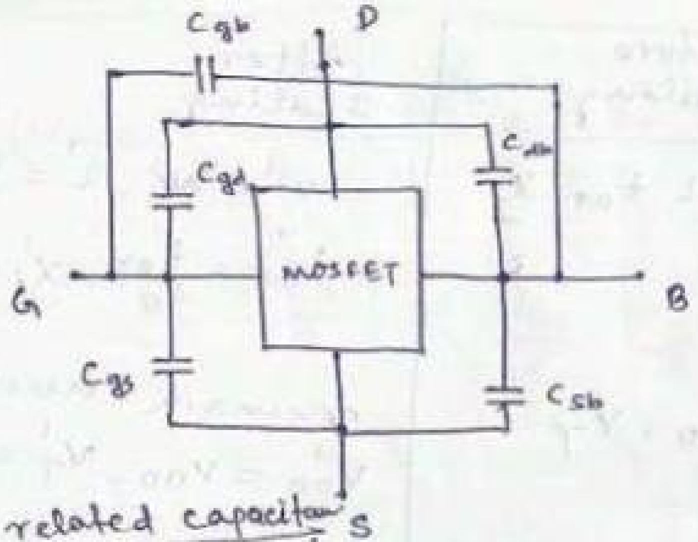
<u>constant - voltage scaling</u> :-

The limitation to full ~~costom~~ scaling is the scaling down of all the voltage levels which is not always possible.

Quantity	Before scaling	After scaling
Dimensions	W, L, t_{ox}, x_j	$W' = W/s$, $L' = L/s$. $t'_{ox} = \frac{t_{ox}}{s}$, $x'_j = x_j/s$
voltages	V_{DD}, V_T	remain unchanged $V'_{DD} = V_{DD}$, $V'_T = V_T$
Doping densities	N_A, N_D	$N'_A = s^2 \cdot N_A$, $N'_D = s^2 \cdot N_D$
oxide capacitance	C_{ox}	$C'_{ox} = s \cdot C_{ox}$
Drain current	I_D	$I'_D = s \cdot I_D$
power dissipation	P	$P' = s \cdot P$
power density	$P/Area$	$P'/Area' = s^3 \cdot (P/Area)$

MOSFET Capacitance :-





Oxide related capacitance

Capacitance	cutoff	linear	saturation
C_{gb} (total)	$(C_{ox} WL)$	0	0
C_{gd} (total)	$(C_{ox} W \cdot L_D)$	$\left(\frac{1}{2} C_{ox} WL + C_{ox} \cdot W \cdot L_D\right)$	$(C_{ox} \cdot W \cdot L_D)$
C_{gs} (total)	$(C_{ox} \cdot W \cdot L_D)$	$\left(\frac{1}{2} C_{ox} WL + C_{ox} \cdot W \cdot L_D\right)$	$\left(\frac{2}{3} C_{ox} WL + C_{ox} W \cdot L_D\right)$

$$C_{GS}(\text{overlap}) = C_{GD}(\text{overlap}) = C_{ox} \cdot W \cdot L_D$$

$$\text{where } C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

→ In cutoff mode, since the surface is not inverted, there is no conducting channel connecting the surface to source and drain. i.e $C_{gs} = C_{gd} = 0$ & $C_{gb} = C_{ox} \cdot W \cdot L$.

→ In linear-mode operation, the inverted channel extends across the MOSFET, betn the source & the drain. i.e $C_{gb} = 0$.

& gate-to-channel capacitance is equally shared by source & drain. i.e

$$C_{gs} \simeq C_{gd} \simeq \frac{1}{2} C_{ox} W \cdot L$$

$$C_{gd}(\text{total})_{\text{linear}} = \frac{1}{2} C_{ox} W \cdot L + C_{ox} W \cdot L_D$$

$$\& \quad C_{gs}(\text{total})_{\text{linear}} = \frac{1}{2} C_{ox} \cdot WL + C_{ox} W \cdot L_D .$$

→ In saturation mode, the inversion layer at the drain is not complete. so $C_{gd} = 0$

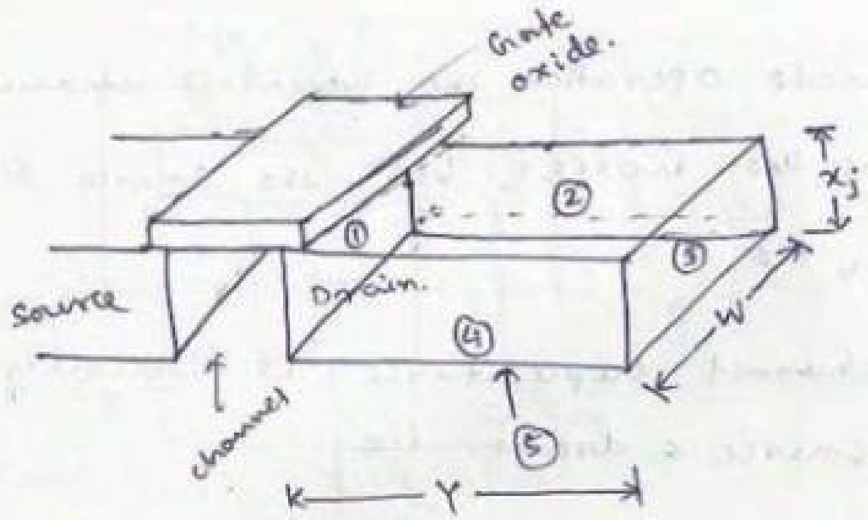
$$\& \quad C_{gs} \simeq \frac{2}{3} C_{ox} \cdot W \cdot L$$

so $C_{gd}(\text{total})_{\text{sat}} = C_{ox} \cdot W \cdot L_D$

$$C_{gs}(\text{total})_{\text{sat}} = C_{ox} \cdot WL_D + \frac{2}{3} C_{ox} WL$$

Junction capacitance :-

The junction capacitances C_{sb} and C_{db} are ~~voltage dependent~~ dependent on the depletion charge present in the junction due to the respective potential difference betn the substrate & respective source or drain.



Junction	Area	Type
1	$W \cdot x_j$	n^+/p
2	$Y \cdot x_j$	n^+/p^+
3	$W \cdot x_j$	n^+/p^+
4	$Y \cdot x_j$	n^+/p^+
5	$W \cdot Y$	n^+/p

Derivation of depletion capacitance

For a reverse-biased abrupt PN junction, the depletion layer thickness

$$x_d = \sqrt{\frac{2\epsilon_{si}}{e} \cdot \frac{N_A + N_D}{N_A \cdot N_D} \cdot \left(V_{bi} - V_R\right)} \quad \text{—— (1)}$$

$$= \left[\frac{2\epsilon_{si}}{eN}\left(V_{bi} - V_R\right)\right]^{1/2}, \quad \text{where } N = N_A \| N_D = \frac{N_A N_D}{N_A + N_D}$$

where $V_{bi} = \frac{kT}{e} \cdot \ln\left(\frac{N_A N_D}{n_i^2}\right)$ & $V_R \rightarrow$ reverse bias potential depletion region charge stored in an area 'A' is

$$Q_j = A \cdot e N x_d = A \cdot e\left(\frac{N_A N_D}{N_A+N_D}\right)\left[\frac{2\epsilon_{si}}{e} \cdot \frac{N_A N_D}{N_A+N_D}(V_{bi}-V_R)\right]^{1/2}$$

$$Q_j = A\left[2\epsilon_{si} e N (V_{bi}-V_R)\right]^{1/2} \quad \text{——— (2)}$$

Then junction capacitance associated with depletion region

$$C_j(V_R) = \left|\frac{dQ_j}{dV_R}\right| = \frac{d}{dV_R}\left\{A\left[2\epsilon_{si} e N (V_{bi}-V_R)\right]^{1/2}\right\}$$

$$= A \cdot \left[\frac{\epsilon_{si} e N}{2}\right]^{1/2} \cdot (V_{bi}-V_R)^{-1/2}$$

——— (3)

$$\boxed{C_j(V_R) = \frac{A \cdot C_{j0}}{\left(1-\frac{V_R}{V_{bi}}\right)^{1/2}}}, \text{ where } C_{j0} = \left[\frac{\epsilon_{si} \cdot e N}{2 V_{bi}}\right]^{1/2}$$

In general $\boxed{C_j(V_R) = \frac{A \cdot C_{j0}}{\left(1-\frac{V_R}{V_{bi}}\right)^m}}$ ——— (4)

where $m = 1/2$, for abrupt junction profile.

$m = 1/3$, for linearly graded junction profile

⇒ Equivalent large-signal capacitance

$$C_{eq} = \frac{\Delta Q}{\Delta V} = \frac{Q_j(V_2) - Q_j(V_1)}{(V_2 - V_1)} = \frac{1}{(V_2 - V_1)} \int_{V_1}^{V_2} C_j(V_R) \cdot dV_R$$

$$= \frac{A}{V_2 - V_1} \cdot \left(\frac{\epsilon_{si} e N}{2}\right)^{1/2} \cdot \int_{V_1}^{V_2} (V_{bi}-V_R)^{-1/2} \cdot dV_R$$

$$= -\frac{A}{V_2 - V_1} \cdot \left[\frac{\epsilon_{si} e N}{2}\right]^{1/2} \cdot 2\left|(V_{bi}-V_R)^{1/2}\right|_{V_1}^{V_2}$$

$$\boxed{C_{eq} = \frac{2A}{(V_1 - V_2)}\left[\frac{\epsilon_{si} e N}{2}\right]^{1/2} \cdot \left\{(V_{bi}-V_2)^{1/2} - (V_{bi}-V_1)^{1/2}\right\}} \quad \text{——— (5)}$$

If $C_{eq} = A \cdot C_{j_0} \cdot K_{eq}$ ——— (6)

Then $K_{eq} = \dfrac{2\sqrt{V_{bi}}}{(V_1 - V_2)} \cdot \left\{ \sqrt{V_{bi} - V_2} - \sqrt{V_{bi} - V_1} \right\}$

& $C_{j_0} = \left[\dfrac{\epsilon_{Si}\, e N}{2 V_{bi}} \right]^{1/2}$

Ex : 3.7, page 137.

Ex : 3.8, page 139

Exercise problem : 3.1 to 3.17.

— 0 —

VLSI Design

Topic:

MOS Inverter-Statistic Characteristics

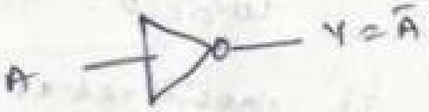
Contributed By:

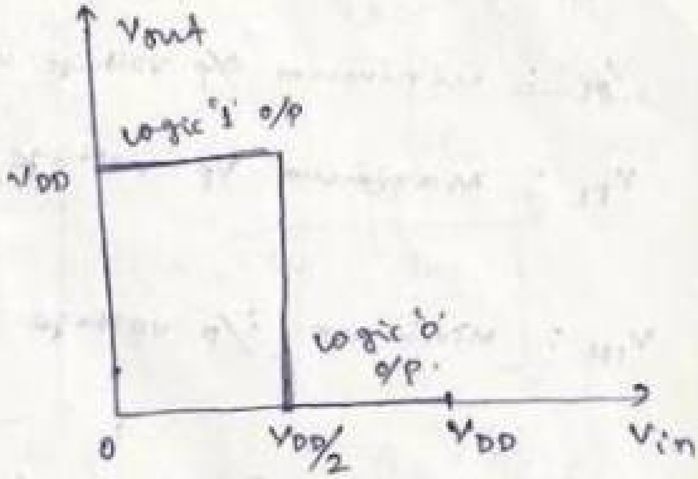
Verified Writer

module - II

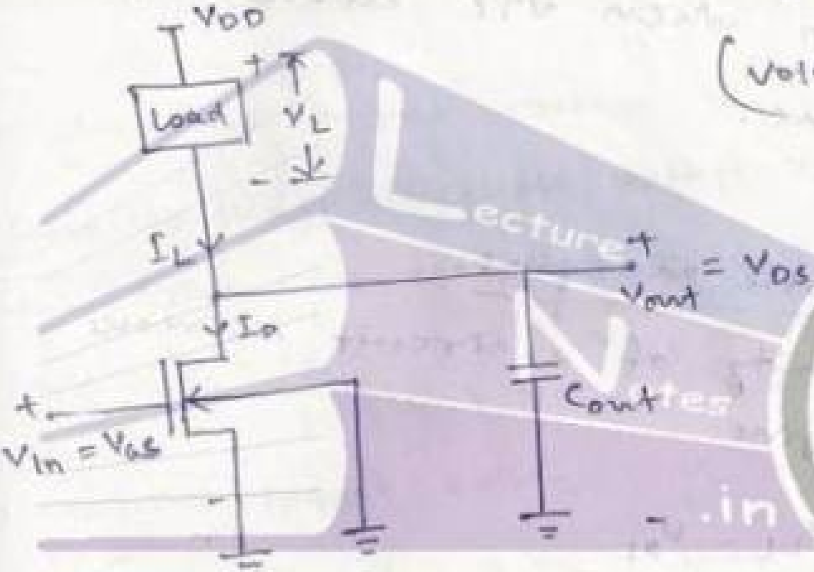
MOS inverters - static characteristics :-

Introduction :

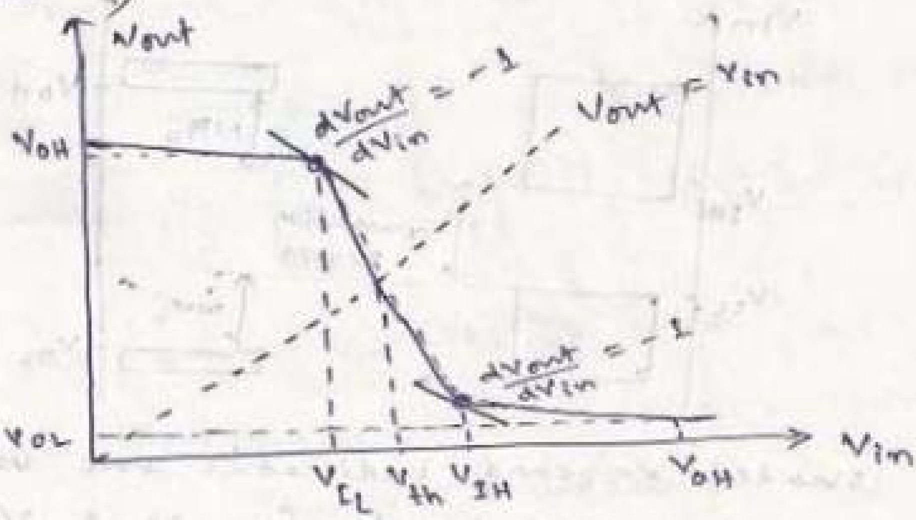




(voltage transfer characteristics of ideal inverter)



(General circuit structure of n-mos inverter with nmos as driver transistor)



V_{OH} : maximum o/p voltage when o/p level is logic '1'

V_{OL} : minimum o/p voltage when o/p level is logic '0'.

V_{IL} : Maximum i/p voltage which can be considered as logic '0'.

V_{IH} : minimum i/p voltage which is considered as logic '1'.

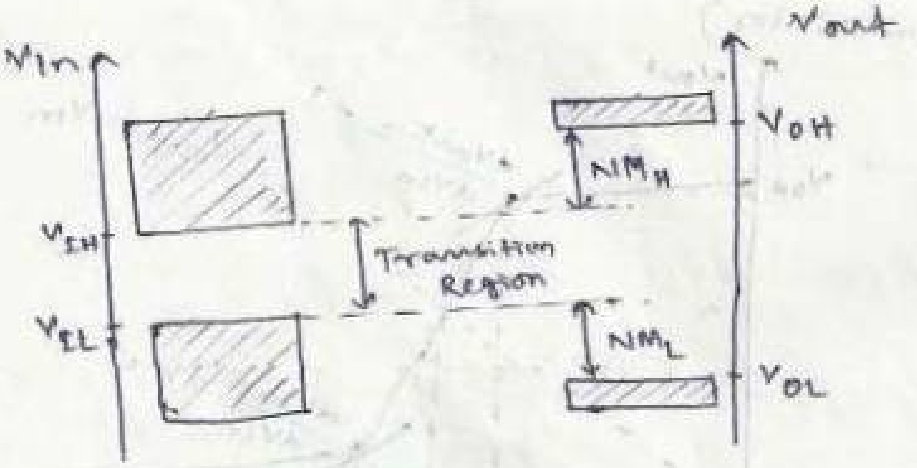
V_{th} : value of V_{in} when VTC curve cuts $V_{out} = V_{in}$ line.

Noise immunity & Noise margins:

The noise immunity of a circuit increases with noise margins.

$$NM_L = V_{IL} - V_{OL}$$

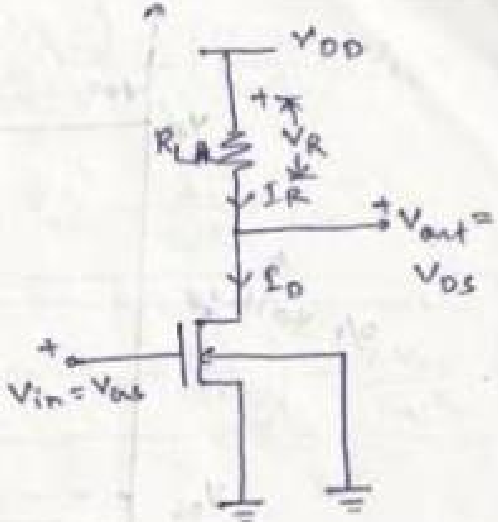
$$NM_H = V_{OH} - V_{IH}$$



Shaded regions indicates the allowable high & low levels for i/p & o/p signals.

Resistive-load Inverter :-

Input voltage Range	operating mode
$V_{in} < V_{T0}$	cutoff
$V_{T0} \le V_{in} < V_{out} + V_{T0}$	Saturation
$V_{in} \geq V_{out} + V_{T0}$	Linear.



$V_{SB} = 0V$ &
$V_T = V_{T0}$

→ when the input voltage 'V_{in}' is smaller than threshold voltage 'V_{th}', the driver transistor is in cutoff mode.

i.e $I_D = I_R = 0$ & $(V_{out} = V_{DS}) = V_{DD}$

→ when input voltage 'V_{in}' is increased beyond 'V_{T0}' till '$V_{out} + V_{T0}$', then it establishes a relation $V_{out} + V_{T0} > V_{in}$ & $V_{in} > V_{T0}$

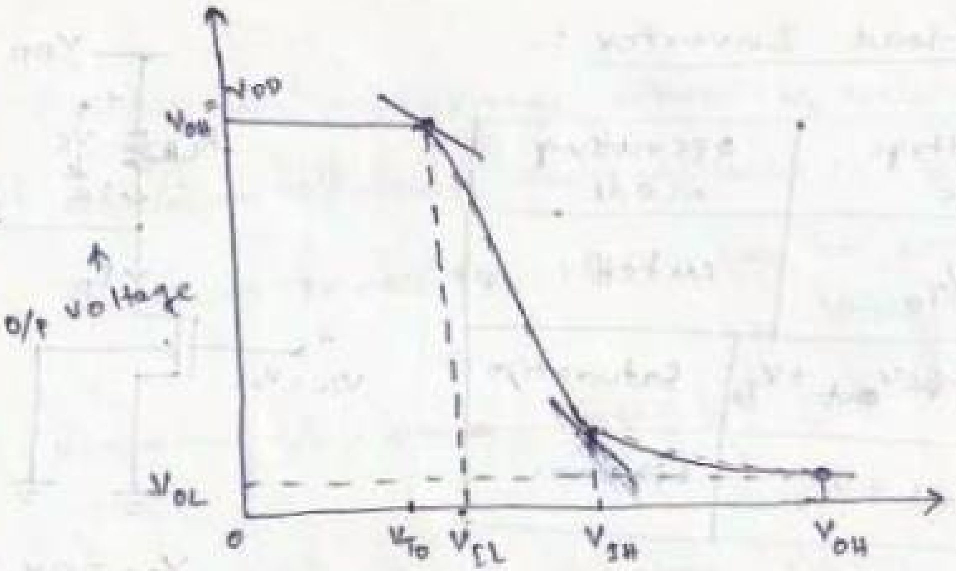
or $V_{out} > (V_{in} - V_{T0})$ which is positive

So the driver MOSFET is in saturation, with

$$I_D = I_R = \frac{K_n}{2}(V_{in} - V_{T0})^2 = \frac{\mu_n C_{ox}}{2} \times \frac{W}{L}(V_{in} - V_{T0})^2$$

→ when input voltage 'V_{in}' is beyond '$V_{out} + V_{T0}$', the driver transistor drives larger current and ~~the~~ V_{out} starts decreasing. This is the Linear mode of operation. ($V_{in} \geq V_{out} + V_{T0}$)

$$I_D = I_R = \frac{K_n}{2}\left[2(V_{in} - V_{T0})V_{out} - V_{out}^2\right]$$



Calculation of V_{OH} :-

we know $V_{out} = V_{DD} - I_D \cdot R_L$

when $V_{in} < V_{T0}$ (low), cutoff mode,

$I_R = I_D = 0$.

So $V_{out} = V_{OH} = V_{DD} - 0 \times R_L = V_{DD}$

$\Rightarrow \boxed{V_{OH} = V_{DD}}$

Calculation of V_{OL} :-

when $V_{in} = V_{OH} = V_{DD}$, $V_{out} = V_{OL}$ &

since $V_{in} - V_{T0} > V_{out}$, driver transistor operates in linear region.

So $I_R \times R_L = V_{DD} - V_{out} \Rightarrow \boxed{I_R = \frac{V_{DD} - V_{out}}{R_L}}$

$= \frac{V_{DD} - V_{OL}}{R_L}$

$$\frac{V_{DD} - V_{OL}}{R_L} = \frac{k_n}{2}\left[2\left(V_{DD} - V_{T0}\right)V_{OL} - V_{OL}^2\right]$$

$$\Rightarrow 2(V_{DD}-V_{T0})V_{OL} - V_{OL}^2 = \frac{2(V_{DD})}{k_n R_L} - \left(\frac{2}{k_n R_L}\right) V_{OL}$$

$$\Rightarrow V_{OL}^2 - 2\left(V_{DD} - V_{T0} + \frac{1}{k_n \cdot R_L}\right) V_{OL} + \frac{2}{k_n R_L} \cdot V_{DD} = 0$$

$$\Rightarrow \boxed{V_{OL} = V_{DD} - V_{T0} + \frac{1}{k_n \cdot R_L} - \sqrt{\left(V_{DD} - V_{T0} + \frac{1}{k_n R_L}\right)^2 - \frac{2V_{DD}}{k_n \cdot R_L}}}$$

Calculation of V_{IL}:-

when $V_{in} = V_{IL}$, $\frac{dV_{out}}{dV_{in}} = -1$ & $V_{out} > V_{in} - V_{T0}$.

So the driver transistor operates in saturation.

i.e $I_R = I_D = \frac{V_{DD} - V_{out}}{R_L} = \frac{k_n}{2} \cdot (V_{in} - V_{T0})^2$

differentiate wrt V_{in} we get

$$-\frac{1}{R_L} \cdot \frac{dV_{out}}{dV_{in}} = \frac{k_n}{2} \cdot 2(V_{in} - V_{T0}) \cdot (1)$$

$$= \frac{1}{R_L} = k_n\left[V_{IL} - V_{T0}\right]$$

$$\Rightarrow V_{IL} - V_{T0} = \frac{1}{k_n \cdot R_L}$$

$$\Rightarrow \boxed{V_{IL} = V_{T0} + \frac{1}{k_n \cdot R_L}}$$

$$\& \quad V_{out}\Big|_{(V_{in} = V_{IL})} = V_{DD} - I_D \cdot R_L$$

$$= V_{DD} - \frac{k_n R_L}{2}\left[V_{T0} + \frac{1}{k_n \cdot R_L} - V_{T0}\right]^2$$

$$\boxed{V_{out}\Big|_{(V_{in} = V_{IL})} = V_{DD} - \frac{1}{2k_n \cdot R_L}}$$

Calculation of V_{IH} :-

when $V_{in} = V_{IH}$, $\frac{dV_{out}}{dV_{in}} = -1$ & $V_{out} < V_{in} - V_{T0}$

So the driver transistor is in linear region.

$$\frac{V_{DD} - V_{out}}{R_L} = \frac{k_n}{2}\left[2(V_{in} - V_{T0})V_{out} - V_{out}^2\right] \quad \text{—— (1)}$$

differentiate w.r.t V_{in} we get

$$-\frac{1}{R_L} \cdot \frac{dV_{out}}{dV_{in}} = \frac{k_n}{2}\left[2(V_{in} - V_{T0}) \cdot \frac{dV_{out}}{dV_{in}} + 2V_{out} \cdot (1) - 2V_{out} \cdot \frac{dV_{out}}{dV_{in}}\right]$$

$$\Rightarrow -\frac{1}{R_L} \cdot (-1) = \frac{k_n}{2}\left[2(V_{IH} - V_{T0})(-1) + 2V_{out} + 2V_{out}\right]$$

$$\Rightarrow \frac{1}{R_L} = k_n\left[-(V_{IH} - V_{T0}) + 2V_{out}\right]$$

$$\Rightarrow \frac{1}{k_n \cdot R_L} = 2V_{out} - V_{IH} + V_{T0}$$

$$\Rightarrow \boxed{V_{IH} = V_{T0} + 2V_{out} - \frac{1}{k_n \cdot R_L}} \quad \text{—— (2)}$$

Putting ② in ① we get

$$\frac{V_{DD} - V_{out}}{R_L} = \frac{k_n}{2}\left[2\left(V_{T0} + 2V_{out} - \frac{1}{k_n R_L} - V_{T0}\right)V_{out} - V_{out}^2\right]$$

$$\Rightarrow V_{out} = \sqrt{\frac{2}{3} \times \frac{V_{DD}}{k_n R_L}}$$

Putting in eqn ② we get

$$\boxed{V_{IH} = V_{T0} + \sqrt{\frac{8}{3} \times \frac{V_{DD}}{k_n R_L}} - \frac{1}{k_n R_L}}$$

Calculation of V_{th}:-

When $V_{in} = V_{th}$, $V_{out} > (V_{in} - V_{T0})$, i.e driver transistor operates in saturation region.

i.e $\frac{V_{DD} - V_{out}}{R_L} = \frac{k_n}{2} \cdot (V_{in} - V_{T0})^2$

& $V_{in} = V_{out} = V_{th}$

$$\Rightarrow \frac{V_{DD} - V_{th}}{R_L} = \frac{k_n}{2}(V_{th} - V_{T0})^2$$

$$\Rightarrow (V_{DD} - V_{th}) = \frac{k_n R_L}{2}\left[V_{th}^2 + V_{T0}^2 - 2V_{T0} \cdot V_{th}\right]$$

$$\Rightarrow V_{th}^2 + V_{T0}^2 - 2V_{T0} \cdot V_{th} = \frac{2(V_{DD} - V_{th})}{k_n \cdot R_L}$$

$$\Rightarrow V_{th}^2 + \left(\frac{2}{k_n R_L} - 2V_{T0}\right)V_{th} + V_{T0}^2 - \frac{2V_{DD}}{k_n R_L} = 0$$

$$\Rightarrow \boxed{V_{th} = V_{T0} - \frac{1}{k_n R_L} - \sqrt{\frac{2}{k_n R_L}\left[V_{DD} - V_{T0} + \frac{1}{k_n R_L}\right]}}$$

$$\boxed{\begin{aligned} NM_L &= V_{IL} - V_{OL} \\ NM_H &= V_{OH} - V_{IH} \end{aligned}}$$

(fig: 5.9, page 182, VTC of resistive-load inverter for different values of $k_n \cdot R_L$)

⇒ Average DC power consumption of the inverter

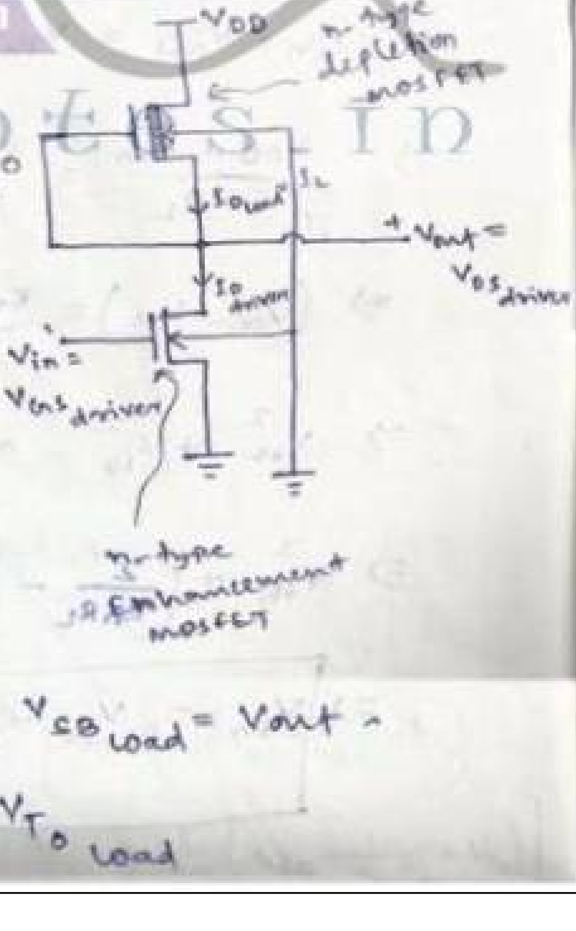
$$\boxed{P_{DC}(avg) = \frac{V_{DD}}{2} \times \frac{V_{DD} - V_{OL}}{R_L}}$$

Inverters with n-type MOSFET Load:-

Inverters with resistive load requires larger chip area compared to inverters with n-type MOSFET as active load.

Depletion type n-MOSFET-load nMOS inverter:-

$V_{T0_{load}} < 0$	$V_{GS_{driver}} = V_{in}$	$V_{GS_{load}} = 0$
$V_{T0_{driver}} > 0$	$V_{DS_{driver}} = V_{out}$	
$V_{GS_{load}} = 0$		



→ Since $V_{GS_{load}} > V_{T_{load}}$, the load transistor is always conducting.

→ Since $V_{S_{load}} \neq 0$ for which $V_{SB_{load}} = V_{out}$,

we can conclude $V_{T_{load}} \neq V_{T0_{load}}$

i.e $V_{T_{load}} = V_{T_{0_{load}}} + \gamma \left(\sqrt{|2\phi_F| + V_{out}} - \sqrt{|2\phi_F|} \right)$

where $\gamma = \dfrac{\sqrt{2 e N_A \epsilon_{si}}}{C_{ox}}$ $\longrightarrow$ Substrate-bias coefficient

⇒ operating mode of load transistor

a) when $V_{out} < V_{DD} + V_{T_{load}}$

the load transistor is in saturation.

with $V_{DS_{load}} > V_{GS_{load}} - V_{T_{load}}$. (or $(V_{DD} - V_{out}) > |V_{T_{load}}|$)

$$I_{D_{load}} = \frac{K_{n_{load}}}{2} \left[\left(V_{GS_{load}} - V_{T_{load}} \right)^2 \right]$$

$$= \frac{K_{n_{load}}}{2} \left[\left(0 - V_{T_{load}} \right)^2 \right]$$

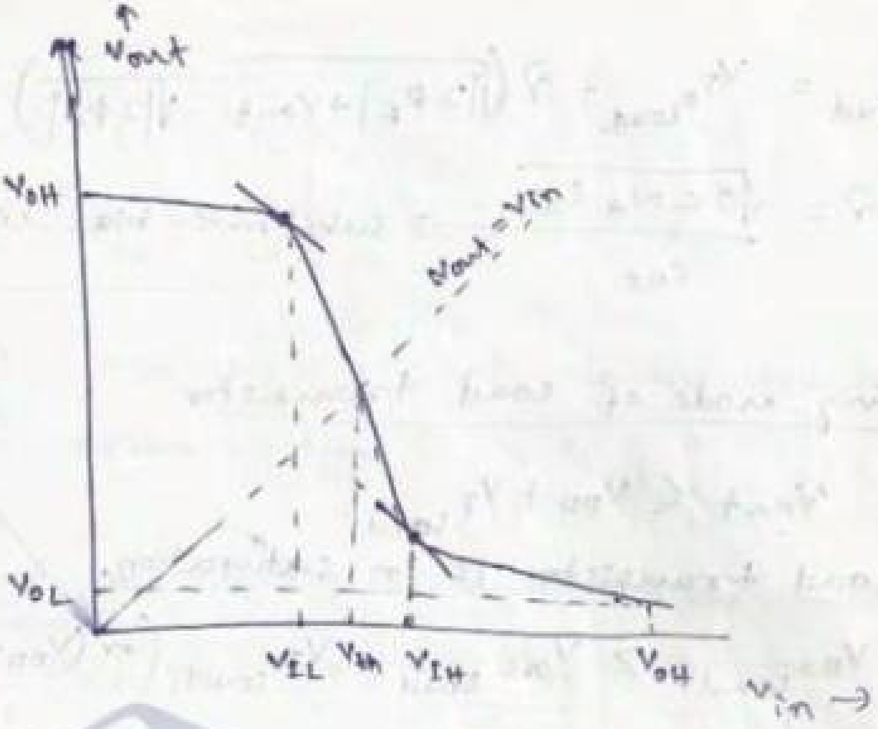
$$\Rightarrow \boxed{I_{D_{load}} = \frac{K_{n_{load}}}{2} \cdot V_{T_{load}}^2}$$

b) when $V_{out} > V_{DD} + V_{T_{load}}$, load transistor is in linear mode i.e

$$I_{D_{load}} = \frac{K_{n_{load}}}{2} \left[2 \left(V_{GS_{load}} - V_{T_{load}} \right) \cdot V_{DS_{load}} - V_{DS_{load}}^2 \right]$$

$$= \frac{K_{n_{load}}}{2} \left[2 \left(0 - V_{T_{load}} \right) \cdot (V_{DD} - V_{out}) - (V_{DD} - V_{out})^2 \right]$$

$$\Rightarrow \boxed{I_{D_{load}} = \frac{K_{n_{load}}}{2} \left[2 |V_{T_{load}}| (V_{DD} - V_{out}) - (V_{DD} - V_{out})^2 \right]}$$



(VTC for depletion-load inverter ckt)

V_{in}	V_{out}	Driver operating Region	Load operating Region
V_{OL}	V_{OH}	cutoff	linear
V_{IL}	$\simeq V_{OH}$	Saturation	linear
V_{IH}	Small	linear	Saturation
V_{OH}	V_{OL}	linear	Saturation

for all above conditions, $I_{D_{load}} = I_{D_{driver}}$.

Calculation of V_{OH} :-

when $V_{in} = V_{OL}$, $V_{out} = V_{OH}$,

$I_{D_{driver}} = 0$ (cutoff)

$$I_{D_{load}} = \frac{k_{n\,load}}{2}\left[2\,|V_{T\,load}|\,(V_{DD}-V_{OH}) - (V_{DD}-V_{OH})^2\right]$$

$$\Rightarrow \frac{k_{n\,load}}{2}\left[2|V_{T\,load}|(V_{DD}-V_{OH})-(V_{DD}-V_{OH})^2\right]=0$$

$$\Rightarrow (V_{DD}-V_{OH})^2 = 2|V_{T\,load}|(V_{DD}-V_{OH})$$

$$\Rightarrow V_{DD}-V_{OH} = 2|V_{T\,load}|$$

Since $|V_{T\,load}|$ is neglegible compared to magnitude of V_{DD} & V_{OH}.

we get $V_{DD}-V_{OH} \simeq 0$

$$\Rightarrow \boxed{V_{OH} \simeq V_{DD}}$$

Calculation of V_{OL} :-

when $V_{in} = V_{OH}$, $V_{out} = V_{OL}$

$$I_{D\,driver} = \frac{k_{n\,driver}}{2}\left[2(V_{OH}-V_{T0})\cdot V_{OL} - V_{OL}^2\right]$$

$$I_{D\,load} = \frac{k_{n\,load}}{2}\left[(V_{load}-V_{T\,load})\right]^2$$

$$I_{D\,load} = \frac{k_{n\,load}}{2}|V_{T\,load}|^2$$

putting $I_{D\,driver} = I_{D\,load}$ we get

$$k_{n\,driver}\left[2(V_{DD}-V_{T0})V_{OL}-V_{OL}^2\right] = k_{n\,load}\cdot|V_{T\,load}|^2$$

$$\Rightarrow V_{OL}^2 - 2(V_{DD}-V_{T0})V_{OL} + \left(\frac{k_{n\,load}}{k_{n\,driver}}\right)\cdot|V_{T\,load}|^2 = 0$$

$$\Rightarrow \boxed{V_{OL} = V_{DD}-V_{T0}-\sqrt{(V_{DD}-V_{T0})^2-\left(\frac{k_{n\,load}}{k_{n\,driver}}\right)\cdot|V_{T\,load}|^2}}$$

Calculation of V_{IL} :-

when $V_{in} = V_{IL}$, $V_{out} \simeq V_{OH}$ & $\frac{dV_{out}}{dV_{in}} = -1$.

$$I_{D_{driver}} = \frac{K_{n_{driver}}}{2}\left[V_{in} - V_{T_0}\right]^2$$

$$I_{D_{load}} = \frac{K_{n_{load}}}{2}\left[2|V_{T_{load}}|(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2\right]$$

$$\Rightarrow \frac{K_{n_{driver}}}{2}\left[V_{in} - V_{T_0}\right]^2 = \frac{K_{n_{load}}}{2}\left[2|V_{T_{load}}|(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2\right]$$

differentiating wrt V_{in} we get

$$K_{n_{driver}}(V_{in} - V_{T_0}) = \frac{K_{n_{load}}}{2}\left[2|V_{T_{load}}| \cdot \left(\frac{-dV_{out}}{dV_{in}}\right) + 2(V_{DD} - V_{out}) \times \frac{d|V_{T_{load}}|}{dV_{in}} - 2(V_{DD} - V_{out})\left(-\frac{dV_{out}}{dV_{in}}\right)\right]$$

Then assuming $\frac{d|V_{T_{load}}|}{dV_{in}} = 0$ & putting $V_{in} = V_{IL}$ & $\frac{dV_{out}}{dV_{in}} = -1$, we get

$$K_{n_{driver}} \times (V_{IL} - V_{T_0}) = \frac{K_{n_{load}}}{2}\left[2|V_{T_{load}}| - 2(V_{DD} - V_{out})\right]$$

$$\Rightarrow V_{IL} - V_{T_0} = \frac{K_{n_{load}}}{K_{n_{driver}}}\left[|V_{T_{load}}| - V_{DD} + V_{out}\right]$$

$$\Rightarrow \boxed{V_{IL} = V_{T0} + \frac{k_{n\,load}}{k_{n\,driver}}\left[V_{out} - V_{DD} + |V_{T\,load}|\right]}$$

assuming $V_{out} \simeq V_{OH} \simeq V_{DD}$ we get

$$V_{IL} = V_{T0} + \frac{k_{n\,load}}{k_{n\,driver}} \times |V_{T\,load}|$$

Calculation of V_{IH} :-

when $V_{in} = V_{IH}$, V_{out} is small, for which driver is in linear & load is in saturation mode.

again $\frac{dV_{out}}{dV_{in}} = -1$.

$$I_{D\,driver} = \frac{k_{n\,driver}}{2}\left[2(V_{in} - V_{T0})V_{out} - V_{out}^2\right]$$

$$I_{D\,load} = \frac{k_{n\,load}}{2}\left[-V_{T\,load}\right]^2$$

$$\Rightarrow \frac{k_{n\,driver}}{2}\left[2(V_{in} - V_{T0})V_{out} - V_{out}^2\right] = \frac{k_{n\,load}}{2} \times V_{T\,load}^2$$

differentiating wrt V_{in} we get,

$$k_{n\,driver}\left[2V_{out} + 2(V_{in} - V_{T0}) \times \left(\frac{dV_{out}}{dV_{in}}\right) - 2V_{out} \times \left(\frac{dV_{out}}{dV_{in}}\right)\right]$$

$$= k_{n\,load} \; 2 \cdot |V_{T\,load}| \cdot \frac{dV_{T\,load}}{dV_{in}}$$

$$\Rightarrow V_{out} + (V_{in} - V_{T0}) \cdot \left(\frac{dV_{out}}{dV_{in}}\right) - V_{out}\left(\frac{dV_{out}}{dV_{in}}\right)$$

$$= \frac{k_{n\,load}}{k_{n\,driver}} \cdot |V_{T\,load}| \cdot \frac{dV_{T\,load}}{dV_{out}} \times \left(\frac{dV_{out}}{dV_{in}}\right)$$

$$\Rightarrow V_{out} - V_{IH} + V_{T0} + V_{out} = -\frac{k_{n\,load}}{k_{n\,driver}}|V_{T\,load}|\left(\frac{dV_{T\,load}}{dV_{out}}\right)$$

$$\Rightarrow \boxed{V_{IH} = V_{T0} + 2V_{out} + \left(\frac{k_{n\,load}}{k_{n\,driver}}\right) \cdot \left|V_{T\,load}\right| \cdot \left(\frac{dV_{T\,load}}{dV_{out}}\right)}$$

~~again~~ assuming $V_{out} \simeq V_{OL}$

$$V_{IH} = V_{T0} + 2V_{OL} + \left(\frac{k_{n\,load}}{k_{n\,driver}}\right) \left|V_{T\,load}\right| \cdot \left(\frac{dV_{T\,load}}{dV_{out}}\right)$$

where $V_{T\,load} = V_{T0\,load} + \gamma\left(\sqrt{|2\phi_F| + V_{out}} - \sqrt{|2\phi_F|}\right)$

so $$\frac{dV_{T\,load}}{dV_{out}} = 0 + \frac{\gamma}{2}\left\{|2\phi_F| + V_{out}\right\}^{-1/2} \times 1$$

$$= \frac{\gamma}{2\sqrt{|2\phi_F| + V_{out}}}$$

Putting the values of $\gamma = \frac{\sqrt{2eN_A\epsilon_{si}}}{C_{ox}}$

$\phi_F \simeq -0.7\,V$ for silicon, $V_{out} \simeq V_{OL}$, we can solve for V_{IH}.

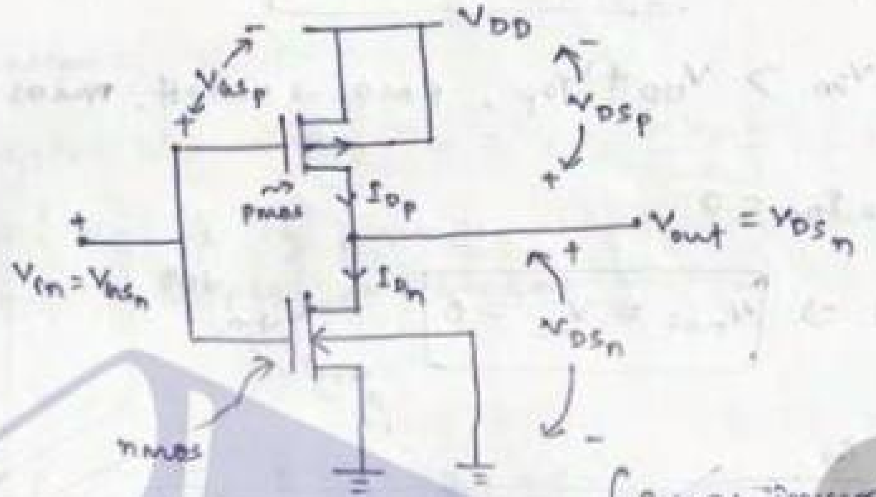
$\Rightarrow$ Average DC power consumption

$$\boxed{P_{DC} = \frac{V_{DD}}{2} \times \frac{k_{n\,load}}{2}\left|V_{T\,load}\right|^2}$$

(Ex 5.3, page 195)

CMOS inverter :-

It consists of a ~~[illegible]~~ enhancement type n-MOS and enhancement type P-MOS, operating in complementary mode, with input voltage applied to ~~[illegible]~~ the gate terminal of both the transistors.



(cmos inverter ckt)

→ For high input, the nmos transistor ~~[illegible]~~ acts as driver ~~[illegible]~~ which pulls down the output node while the pmos transistor acts as load.

→ For low input, the pmos transistor acts as driver which pulls up the output node while the nmos transistor acts as load.

Advantages of cmos inverter over other :-

1> steady-state power dissipation of the cmos inverter is negligible.

2> The VTC of cmos inverter exhibits full output voltage swing betn 0V and V_{DD}. with very sharp transition region. (VTC of cmos inverter looks like an ideal VTC).

⇒ from the ckt

$V_{GS_n} = V_{in}$

$V_{DS_n} = V_{out}$

$V_{GS_p} = -(V_{DD} - V_{in}) = (V_{in} - V_{DD})$

$V_{DS_p} = -(V_{DD} - V_{out}) = (V_{out} - V_{DD})$

Since $V_{SB} = 0$, $V_{T_n} = V_{T_{on}}$ & $V_{T_p} = V_{T_{op}}$

$\Rightarrow$ when $V_{in} < V_{T_{on}}$, nmos ~~is on~~ $\rightarrow$ cutoff, pmos $\rightarrow$ Linear

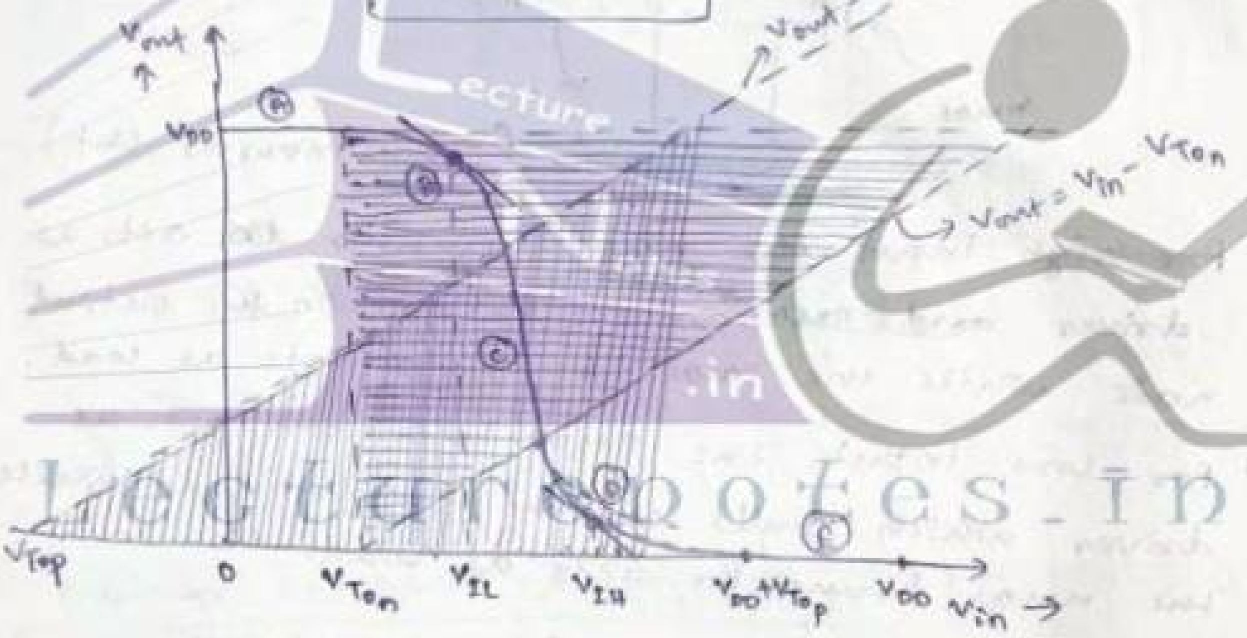
i.e $I_{D_n} = I_{D_p} = 0$

$V_{DS_p} = 0 \Rightarrow \boxed{V_{out} = V_{OH} = V_{DD}}$

$\Rightarrow$ when $V_{in} > V_{DD} + V_{T_{op}}$, pmos $\rightarrow$ cutoff, nmos $\rightarrow$ Linear.

$I_{D_n} = I_{D_p} = 0$

$V_{DS_n} = 0 \Rightarrow \boxed{V_{out} = V_{OL} = 0}$



Region	V_{in}	V_{out}	nmos	pmos
A	$< V_{Ton}$	V_{OH}	cutoff	linear
B	V_{IL}	high (i.e. $\simeq V_{OH}$)	Saturation	linear
C	V_{th}	V_{th}	Saturation	Saturation
D	V_{IH}	low (i.e. $\simeq V_{OL}$)	Linear	Saturation
E	$> (V_{DD} + V_{Top})$	V_{OL}	Linear	cutoff

⇒ nmos transistor operates in saturation when

$$V_{out} \geq V_{in} - V_{T0_n} \quad \text{or} \quad V_{DS_n} \geq V_{GS_n} - V_{T0_n}$$

⇒ pmos transistor operates in saturation when

$$V_{out} \leq V_{in} - V_{T0p} \quad \text{or} \quad V_{DS_p} \leq V_{GS_p} - V_{T0p}$$

Calculation of V_{IL}:-

In region 'B', when $V_{in} = V_{IL}$, $\frac{dV_{out}}{dV_{in}} = -1$, V_{out} is high ($V_{out} \simeq V_{OH}$), nmos operates in saturation mode & pmos operates in linear mode.

$$I_{D_n} = I_{D_p}$$

$$\Rightarrow \frac{K_n}{2}(V_{GS_n} - V_{T0_n})^2 = \frac{K_p}{2}\left[2(V_{GS_p} - V_{T0p})V_{DS_p} - V_{DS_p}^2\right]$$

$$\Rightarrow \frac{K_n}{2}(V_{in} - V_{T0_n})^2 = \frac{K_p}{2}\left[2(V_{in} - V_{DD} - V_{T0p})(V_{out} - V_{DD}) - (V_{out} - V_{DD})^2\right]$$

differentiating wrt V_{in} we get

$$K_n \times 2(V_{in} - V_{T0_n}) = K_p\left[2(V_{in} - V_{DD} - V_{T0p})\left(\frac{dV_{out}}{dV_{in}}\right) + 2(V_{out} - V_{DD}) - 2(V_{out} - V_{DD})\left(\frac{dV_{out}}{dV_{in}}\right)\right]$$

$$\Rightarrow K_n(V_{IL} - V_{T0_n}) = K_p\left[-V_{IL} + V_{DD} + V_{T0p} + V_{out} - V_{DD} + V_{out} - V_{DD}\right]$$

$$\Rightarrow K_n\left[V_{IL} - V_{T0n}\right] = K_p\left[2V_{out} - V_{IL} + V_{T0p} - V_{DD}\right]$$

$$\Rightarrow \left(\frac{K_n}{K_p}\right)V_{IL} - \left(\frac{K_n}{K_p}\right)V_{T0_n} = 2V_{out} - V_{IL} + V_{T0p} - V_{DD}$$

$$\Rightarrow V_{IL}\left[1 + \frac{K_n}{K_p}\right] = 2V_{out} + V_{T0p} - V_{DD} + \left(\frac{K_n}{K_p}\right)V_{T0n}$$

$$\Rightarrow \boxed{V_{IL} = \frac{2V_{out} + V_{T0p} - V_{DD} + \left(\frac{K_n}{K_p}\right)\cdot V_{T0n}}{1+\left(\frac{K_n}{K_p}\right)}}$$

Substituting $K_R = \frac{K_n}{K_p}$.

$$V_{IL} = \frac{2V_{out} + V_{T0p} - V_{DD} + K_R \cdot V_{T0n}}{1+K_R}$$

Calculation of V_{IH} :-

When $V_{in} = V_{IH}$, V_{out} is low (i.e $\simeq V_{OL}$), $\frac{dV_{out}}{dV_{in}} = -1$.

nmos is in linear mode & pmos is in saturation.

Since $I_{D_n} = I_{D_p}$

$$\Rightarrow \frac{K_n}{2}\left[2\left(V_{GS_n} - V_{T0n}\right)V_{DS_n} - V_{DS_n}^2\right] = \frac{K_p}{2}\left[V_{GS_p} - V_{T0p}\right]^2$$

$$\Rightarrow \frac{K_n}{2}\left[2\left(V_{in} - V_{T0n}\right)\cdot V_{out} - V_{out}^2\right] = \frac{K_p}{2}\left[V_{in} - V_{DD} - V_{T0p}\right]^2$$

differentiate w.r.t V_{in} we get

$$\cancel{K_n\left[2(V_{in} - V_{T0n})\cdot\frac{dV_{out}}{dV_{in}} + 2V_{out} - 2V_{out}\frac{dV_{out}}{}\right.}$$

$$K_n\left[2(V_{in} - V_{T0n})\cdot\frac{dV_{out}}{dV_{in}} + 2V_{out} - 2V_{out}\frac{dV_{out}}{dV_{in}}\right]$$

$$= 2K_p\left[V_{in} - V_{DD} - V_{T0p}\right]$$

$$\Rightarrow K_n\left\{-V_{IH} + V_{T0n} + V_{out} + V_{out}\right\} = K_p\left\{V_{IH} - V_{DD} - V_{T0p}\right\}$$

$$\Rightarrow \frac{K_n}{K_p}\left[2V_{out} + V_{T0n}\right] - \frac{K_n}{K_p}\cdot V_{IH} = V_{IH} - V_{DD} - V_{T0p}$$

$$\Rightarrow V_{IH}\left[1 + \frac{K_n}{K_p}\right] = V_{DD} + V_{T0p} + \left(\frac{K_n}{K_p}\right)\left[2V_{out} + V_{T0n}\right]$$

$$\Rightarrow \boxed{V_{IH} = \frac{V_{DD} + V_{T0p} + K_R\left(2V_{out} + V_{T0n}\right)}{1+K_R}}$$

where $K_R = \frac{K_n}{K_p}$

Calculation of V_{th} :-

when $V_{in} = V_{out} = V_{th}$, nmos & pmos both are in Saturation mode. i.e

$$\frac{K_n}{2}(V_{GS_n} - V_{T0n})^2 = \frac{K_p}{2}(V_{GS_p} - V_{T0p})^2$$

$$\Rightarrow K_n(V_{th} - V_{T0n})^2 = K_p(V_{th} - V_{DD} - V_{T0p})^2$$

$$\Rightarrow \sqrt{K_n}(V_{th} - V_{T0n}) = \sqrt{K_p}(V_{th} - V_{DD} - V_{T0p})$$

$$\Rightarrow V_{th} - V_{T0n} = \frac{\sqrt{K_p}}{\sqrt{K_n}}(V_{DD} + V_{T0p} - V_{th})$$

$$\Rightarrow V_{th}\left[1 + \sqrt{\frac{K_p}{K_n}}\right] = V_{T0n} + \sqrt{\frac{K_p}{K_n}}(V_{DD} - V_{T0p})$$

$$\Rightarrow V_{th} = \frac{V_{T0n} + \sqrt{\frac{K_p}{K_n}}(V_{DD} - V_{T0p})}{\left(1 + \sqrt{\frac{K_p}{K_n}}\right)}$$

$$\Rightarrow \boxed{V_{th} = \frac{V_{T0n} + \sqrt{\frac{1}{K_R}}(V_{DD} - V_{T0p})}{\left(1 + \sqrt{\frac{1}{K_R}}\right)}}$$

Ex 5.4, page 210

—o—

VLSI Design

Topic:

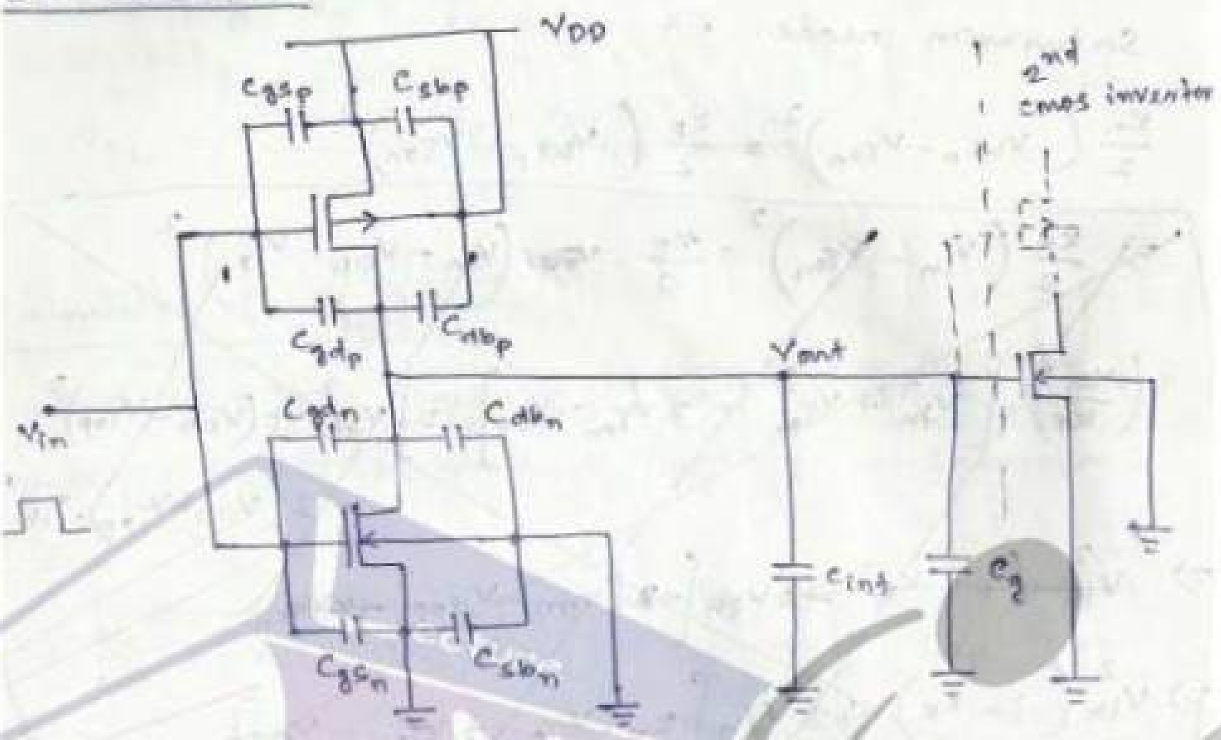
MOS Inverter-Switching Characteristics

Contributed By:

Verified Writer

MOS inverters - Switching characteristics & Interconnect Effects

Introduction



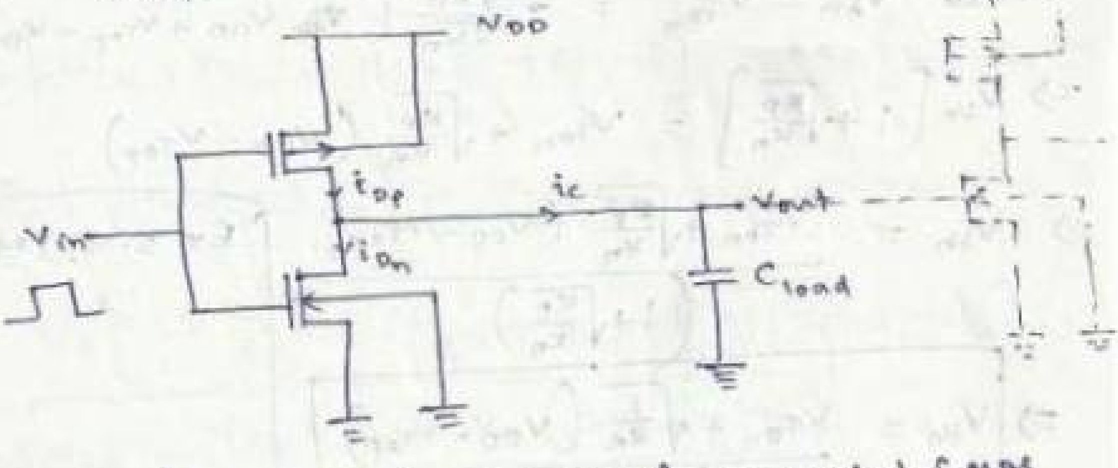
(cascaded cmos inverter stages)

Here C_{gd} & C_{gs} → gate overlap capacitance

C_{db} & C_{sb} → Junction capacitance

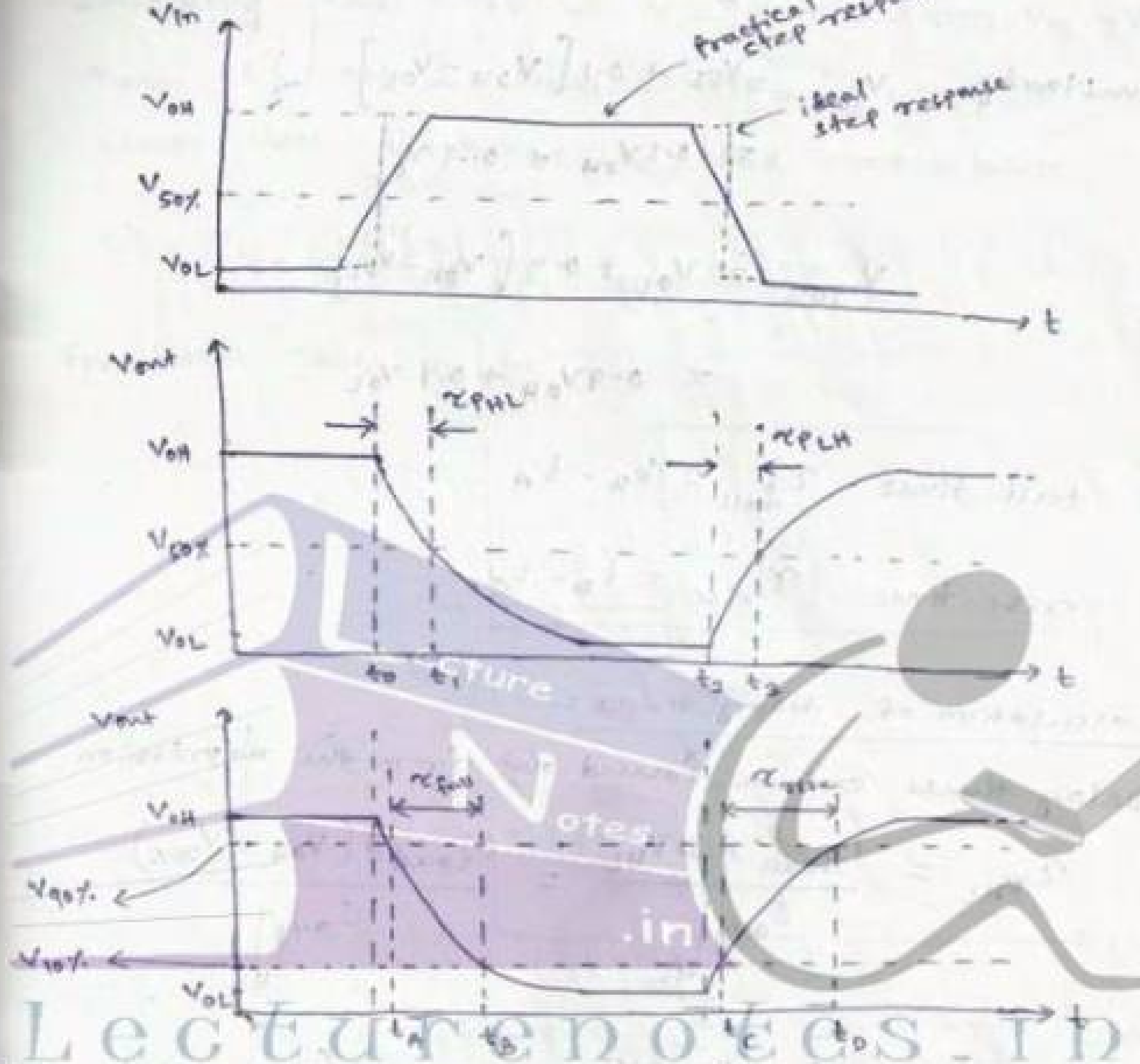
C_g → gate capacitance

C_{int} → Interconnect capacitance



(Equivalent circuit of cascaded cmos inverter with C_{load} as equivalent capacitance) where $C_{load} = C_{gd_n} + C_{gd_p} + C_{db_n} + C_{db_p} + C_{int} + C_g$.

Delay - time definitions :-



[input & output voltage waveforms showing diff. propagation delay times, rise & fall times.]

$$V_{50\%} = V_{OL} + \frac{1}{2}(V_{OH} - V_{OL}) = \frac{1}{2}(V_{OH} + V_{OL})$$

propagation delay time from High to low o/p logic (τ_{PHL}) & from low to high o/p logic (τ_{PLH}) are given by

$$\tau_{PHL} = t_1 - t_0$$

$$\tau_{PLH} = t_3 - t_2$$

Average propagation delay

$$\tau_p = \frac{\tau_{PHL} + \tau_{PLH}}{2}$$

Similarly $V_{10\%} = V_{OL} + 0.1 [V_{OH} - V_{OL}]$

$$= 0.1 V_{OH} + 0.9 V_{OL}$$

$$V_{90\%} = V_{OL} + 0.9 [V_{OH} - V_{OL}]$$

$$= 0.9 V_{OH} + 0.1 V_{OL}$$

& fall time $\tau_{fall} = t_B - t_A$

rise time $\tau_{rise} = t_D - t_C$

Calculation of delay times :-

Delay times can be found out from the expression

$$\tau_{PHL} = \frac{C_{load} \times \Delta V_{HL}}{I_{avg}\big|_{HL}} = \frac{C_{load} \times (V_{OH} - V_{50\%})}{I_{avg}\big|_{HL}}$$

& $$\tau_{PLH} = \frac{C_{load} \times \Delta V_{LH}}{I_{avg}\big|_{LH}} = \frac{C_{load} \times (V_{50\%} - V_{OL})}{I_{avg}\big|_{LH}}$$

where $I_{avg}\big|_{HL} = \frac{1}{2}\left[i_c(V_{in} = V_{out} = V_{OH}) + i_c(V_{in} = V_{OH};\ V_{out} = V_{50\%})\right]$

& $I_{avg}\big|_{LH} = \frac{1}{2}\left[i_c(V_{in} = V_{OL},\ V_{out} = V_{50\%}) + i_c(V_{in} = V_{out} = V_{OL})\right]$

⇒ This approach provides fairly inaccurate results.

Alternate Approach :-

$$C_{load} \cdot \frac{dV_{out}}{dt} = i_C = i_{D_p} - i_{D_n}$$

Assumptions:-

Initially $V_{out} = V_{OH}$ & V_{in} moves from V_{OL} to V_{OH}.

nmos is ON & pmos is off, i.e $i_{Dp} = 0$

Nmos will discharge the load capacitance.

$$\Rightarrow C_{load} \cdot \frac{dV_{out}}{dt} = -i_{D_n}$$

for ideal case :

$\Rightarrow$ Bet^n the time interval from t_0 to t_1, nmos is in saturation.

$$i_{D_n} = \frac{k_n}{2}\left[V_{in} - V_{Tn}\right]^2 = \frac{k_n}{2}\left[V_{OH} - V_{Tn}\right]^2 = -C_{load}\frac{dV_{out}}{dt}$$

for $V_{OH} - V_{Tn} < V_{out} \leq V_{OH}$

$$\Rightarrow dt = \frac{-2C_{load}\, dV_{out}}{k_n (V_{OH} - V_{Tn})^2}$$

Integrating dt from 't_0' to 't_1' & dV_{out} from V_{OH} to $(V_{OH} - V_{Tn})$, we get

$$\int_{t_0}^{t_1} dt = \frac{-2C_{load}}{k_n (V_{OH} - V_{Tn})^2} \int_{V_{OH}}^{V_{OH} - V_{Tn}} dV_{out}$$

$$\Rightarrow (t_1 - t_0) = \frac{-2C_{load}}{k_n (V_{OH} - V_{Tn})^2} \cdot \left[V_{OH} - V_{Tn} - V_{OH} \right]$$

$$\boxed{(t_1 - t_0) = \frac{2C_{load} \cdot V_{Tn}}{k_n (V_{OH} - V_{Tn})^2}}$$

$\Rightarrow$ Between the time interval from 't_1' to 't_2'. NMOS is in linear mode of operation.

$$i_{D_n} = \frac{k_n}{2} \left[2(V_{in} - V_{Tn}) V_{out} - V_{out}^2 \right]$$

$$= \frac{k_n}{2} \left[2(V_{OH} - V_{Tn}) V_{out} - V_{out}^2 \right]$$

for $V_{OH} - V_{Tn} < V_{out}$

$$\Rightarrow \frac{k_n}{2} \left[2(V_{OH} - V_{Tn}) V_{out} - V_{out}^2 \right] = -C_{load} \cdot \frac{dV_{out}}{dt}$$

$$\Rightarrow dt = \frac{-2C_{load}\, dV_{out}}{k_n \left[2(V_{OH} - V_{Tn}) V_{out} - V_{out}^2 \right]}$$

Integrating dt from 't_1' to 't_2' & dV_{out} from

$(V_{OH} - V_{Tn})$ to $V_{50\%}$, we get

$$\int_{t_1}^{t_2} dt = -\frac{2C_{load}}{k_n} \int_{V_{OH}-V_{Tn}}^{V_{50\%}} \frac{dV_{out}}{\left[2(V_{OH} - V_{Tn})V_{out} - V_{out}^2\right]}$$

$$\Rightarrow (t_2 - t_1) = -\frac{2C_{load}}{k_n} \left\{ \frac{1}{2(V_{OH}-V_{Tn})} \cdot \ln\left[\frac{\cdot}{2(V_{OH}-V_{Tn}) - V_{out}}\right]_{V_{OH}-V_{Tn}}^{V_{50\%}} \right\}$$

$$\boxed{(t_2 - t_1) = \frac{C_{load}}{k_n(V_{OH}-V_{Tn})} \cdot \ln\left[\frac{2(V_{OH}-V_{Tn}) - V_{50\%}}{V_{50\%}}\right]}$$

Then propagation delay from High to low

$$\tau_{PHL} = t\Big|_{V_{out} = V_{50\%}} - t\Big|_{V_{out} = V_{OH}}$$

$$= (t_2 - t_1) + (t_1 - t_0)$$

$$\boxed{\tau_{PHL} = \frac{C_{load}}{k_n(V_{OH}-V_{Tn})} \times \ln\left[\frac{2(V_{OH}-V_{Tn}) - V_{50\%}}{V_{50\%}}\right] + \frac{2C_{load} \cdot V_{Tn}}{k_n(V_{OH}-V_{Tn})^2}}$$

For a CMOS inverter

$V_{OH} = V_{DD}$ & $V_{OL} = 0$, $V_{50\%} = V_{DD}/2$

$$\boxed{\tau_{PHL} = \frac{C_{load}}{k_n(V_{DD}-V_{Tn})} \cdot \left[\frac{2 \cdot V_{Tn}}{(V_{DD}-V_{Tn})} + \ln\left(\frac{4(V_{DD}-V_{Tn})}{V_{DD}} - 1\right)\right]} \quad (1)$$

Ex 6.1, 6.2, page 225 to 228

In the similar way the propagation delay from low to high (τ_{PLH}) can be found out as

$$\boxed{\tau_{P_{LH}} = \frac{C_{load}}{k_p(V_{OH}-V_{OL}-|V_{Tp}|)} \times \left[\frac{2|V_{Tp}|}{(V_{OH}-V_{OL}-|V_{Tp}|)} + \ln\left(\frac{2(V_{OH}-V_{OL}-|V_{Tp}|)}{V_{OH}-V_{50\%}} - 1\right)\right]}$$

for a CMOS inverter, $V_{OH} = V_{DD}$, $V_{OL} = 0$, $V_{50\%} = V_{DD}/2$

$$\boxed{\tau_{P_{LH}} = \frac{C_{load}}{k_p(V_{DD}-|V_{Tp}|)}\left[\frac{2|V_{Tp}|}{(V_{DD}-|V_{Tp}|)} + \ln\left(\frac{4(V_{DD}-|V_{Tp}|)}{V_{DD}} - 1\right)\right]} \quad (2)$$

⇒ In case of ~~ideal~~ equal propagation delay i.e.

$\tau_{P_{LH}} = \tau_{P_{HL}}$ we have

$V_{Tn} = |V_{Tp}|$

$k_n = k_p \left(\text{or } \frac{W_p}{W_n} = \frac{\mu_n}{\mu_p} \right)$

⇒ for n-MOS depletion-load inverter type.

$$\tau_{P_{LH}} = \frac{C_{load}}{k_{n_{load}} \cdot |V_{T_{load}}|}\left[\frac{2(V_{DD}-|V_{T_{load}}|-V_{OL})}{|V_{T_{load}}|} + \ln\left(\frac{2|V_{T_{load}}|-(V_{DD}-V_{50\%})}{V_{DD}-V_{50\%}}\right)\right]$$

& $$\tau_{P_{HL}} \simeq \frac{C_{load}(V_{DD})}{2KW_n(V_{DD}-V_T)}$$

Inverter design with delay constraints:-

For a required value of τ^*_{PHL} and τ^*_{PLH}, the design parameter (W/L) ratio of the nmos and pmos transistor can be found as

$$\left(\frac{W_n}{L_n}\right) = \frac{C_{load}}{\tau^*_{PHL}\,\mu_n \cdot C_{ox}\,(V_{DD}-V_{Tn})}\left[\frac{2 \cdot V_{Tn}}{(V_{DD}-V_{Tn})} + \ln\left(\frac{4(V_{DD}-V_{Tn})}{V_{DD}} - 1\right)\right]$$

& $$\left(\frac{W_p}{L_p}\right) = \frac{C_{load}}{\tau^*_{PLH}\,\mu_p\,C_{ox}(V_{DD}-|V_{Tp}|)}\left[\frac{2|V_{Tp}|}{V_{DD}-|V_{Tp}|} + \ln\left(\frac{4(V_{DD}-|V_{Tp}|)}{V_{DD}} - 1\right)\right]$$

where $C_{load} = C_{gd_p} + C_{gd_n} + C_{db_p} + C_{db_n} + C_{int} + C_g$.

where 'C_{gd}' is the oxide-related capacitance and depends on the value of 'W'.

$$\begin{cases} C_{gd}\ (\text{cutoff}) = C_{ox} \cdot W \cdot L_D \\ C_{gd}\ (\text{Linear}) = C_{ox} \cdot W \cdot L_D + \frac{1}{2} C_{ox} \cdot W \cdot L \\ C_{gd}\ (\text{saturation}) = C_{ox} \cdot W \cdot L_D \end{cases}$$

& 'C_{db}' is the junction capacitance and also depends on the value of 'W'.

$C_{db} = A \cdot C_{j0} \cdot K_{eq}$, where $A = W \cdot Y = W \cdot D_{drain}$, where D_{drain} = Drain length

$$C_{j0} = \left[\frac{\epsilon_{si}\, e\, N}{2 V_{bi}}\right]^{1/2}, \quad K_{eq} = \frac{2\sqrt{V_{bi}}}{(V_1 - V_2)}\left\{\sqrt{V_{bi}-V_2} - \sqrt{V_{bi}-V_1}\right\}$$

Hence $C_{load} = C_{gd_p}(W_p) + C_{gd_n}(W_n) + C_{db_p}(W_p) + C_{db_n}(W_n) + C_{int} + C_g$

or $C_{load} = f(W_p, W_n)$

⇒ Since the drain regions of pmos & nmos have an adjacent n^+ & p^+ regions, then the equivalent large signal junction capacitance for drain region due to sidewall is

$$C_{sidewall} = \text{perimeter} \times C_{j_{sw}} \times K_{eq}(sw)$$

for pmos

$$C_{sidewall_p} = 2(W_p + D_{drain}) \cdot C_{j_{sw_p}} \cdot K_{eq_p}$$

& for Nmos

$$C_{sidewall_n} = 2(W_n + D_{drain}) \cdot C_{j_{sw_n}} \cdot K_{eq_n}$$

where $C_{j_{sw_n}} = x_{j_n} \cdot C_{jo_{sw_n}}$ → zero-bias sidewall junction capacitance

& $C_{j_{sw_p}} = x_{j_p} \cdot C_{jo_{sw_p}}$

x_j → channel depth

finally

$$C_{db_n} = W_n \cdot D_{drain} \cdot C_{jo_n} \cdot K_{eq_n} + 2(W_n + D_{drain}) C_{j_{sw_n}} \cdot K_{eq_n}$$

& $$C_{db_p} = W_p \cdot D_{drain} \cdot C_{jo_p} \cdot K_{eq_p} + 2(W_p + D_{drain}) C_{j_{sw_p}} \cdot K_{eq_p}$$

$$\Rightarrow C_{load} = (W_n \cdot C_{jo_n} \cdot K_{eq_n} + W_p \cdot C_{jo_p} \cdot K_{eq_p}) D_{drain}$$
$$+ 2(W_n + D_{drain}) C_{j_{sw_n}} \cdot K_{eq_n} + 2(W_p + D_{drain}) C_{j_{sw_p}} \cdot K_{eq_p}$$
$$+ C_{int} + C_g$$
$$\simeq \alpha_0 + \alpha_n W_n + \alpha_p \cdot W_p$$

where $\alpha_0 = 2D_{drain}(C_{j_{swn}} \cdot K_{eq_n} + C_{j_{swp}} \cdot K_{eq_p}) + C_{int} + C_g$

$$\alpha_n = K_{eq_n}(C_{j_{on}} \cdot D_{drain} + 2C_{j_{swn}})$$

$$\alpha_p = K_{eq_p}(C_{j_{op}} \cdot D_{drain} + 2C_{j_{swp}})$$

putting the value of C_{load} in τ_{PHL} & τ_{PLH} relations for cmos inverter we get

$$\tau_{PHL} = \left(\frac{\alpha_0 + \alpha_n W_n + \alpha_p W_p}{W_n}\right)\left(\frac{L_n}{\mu_n C_{ox}(V_{DD} - V_{Tn})}\right) \times$$

$$\left[\frac{2V_{Tn}}{V_{DD} - V_{Tn}} + \ln\left(\frac{4(V_{DD} - V_{Tn})}{V_{DD}} - 1\right)\right] = \Gamma_n\left(\frac{\alpha_0 + \alpha_n W_n + \alpha_p W_p}{W_n}\right)$$

&

$$\tau_{PLH} = \left(\frac{\alpha_0 + \alpha_n W_n + \alpha_p W_p}{W_p}\right)\left(\frac{L_p}{\mu_p C_{ox}(V_{DD} - |V_{Tp}|)}\right) \times$$

$$\left[\frac{2|V_{Tp}|}{V_{DD} - |V_{Tp}|} + \ln\left(\frac{4(V_{DD} - |V_{Tp}|)}{V_{DD}} - 1\right)\right]$$

$$= \Gamma_p\left(\frac{\alpha_0 + \alpha_n W_n + \alpha_p W_p}{W_p}\right)$$

Estimation of Interconnect Parasitics :- (fig 6.13, page 244)

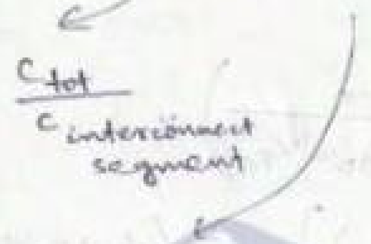
Interconnect Capacitance Estimation (fig 6.14, page 245.)

In large-scale IC, the parasitic interconnect capacitances are the most difficult parameters to estimate accurately since each interconnection line (wire) is a 3-D structure of metal/poly with different shape, thickness & vertical distance from ground plane (substrate).

figure 6.16 → fig. for parasitic resistance & capacitance estimation.

figure 6.17 ⇒ Influence of fringing electric field upon the parasitic wire capacitance.

figure 6.18 ⇒ variation of fringing-field factor ($FF = C_{tot}/C_{pp}$) with the interconnect geometry.



⇒ The influence of fringing field increases with decreasing $\left(\frac{W}{h}\right)$ ratio,

⇒ FF increases with increase in (t/h) & (t/h)

⇒ FF increases rapidly with the in $(\frac{t}{h})$ for higher value of (t/h).

⇒ Yuan & Trick in 1980s estimated the interconnect capacitance as

for $w \geq t/2$

$$C = \varepsilon\left[\frac{(w - t/2)}{h} + \frac{2\pi}{\ln\left(1 + \frac{2h}{t} + \sqrt{\frac{2h}{t}\left(\frac{2h}{t} + 2\right)}\right)}\right]$$

& for $w < t/2$

$$C = \varepsilon\left[\frac{w}{h} + \frac{\pi\left(1 - 0.0543 \cdot \frac{t}{2h}\right)}{\ln\left(1 + \frac{2h}{t} + \sqrt{\frac{2h}{t}\left(\frac{2h}{t} + 2\right)}\right)} + 1.47\right]$$

fig-6.20 - capacitive coupling components., page 249)

Interconnect Resistance Estimation

The parasitic resistance of a metal/poly line can also have a significant influence of the signal propagation delay over that line.

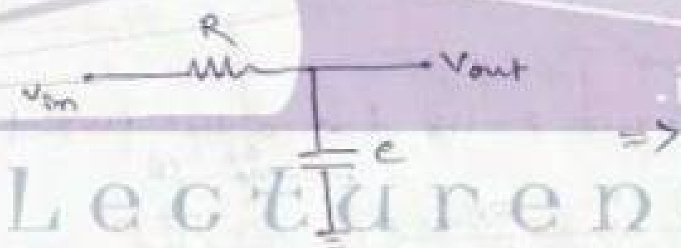
for a ~~sepa~~ single interconnect as fig 6.16, page 246, the wire resistance can be written as

$$R_{wire} = \rho \cdot \frac{l}{w \cdot t} = R_{sheet}\left(\frac{l}{w}\right)$$

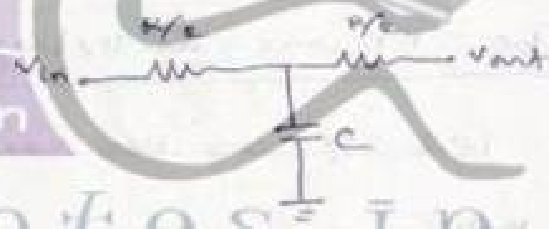
where $R_{sheet} = \frac{\rho}{t}$, $\rho \rightarrow$ ~~the~~ resistivity

Calculation of Interconnect Delay

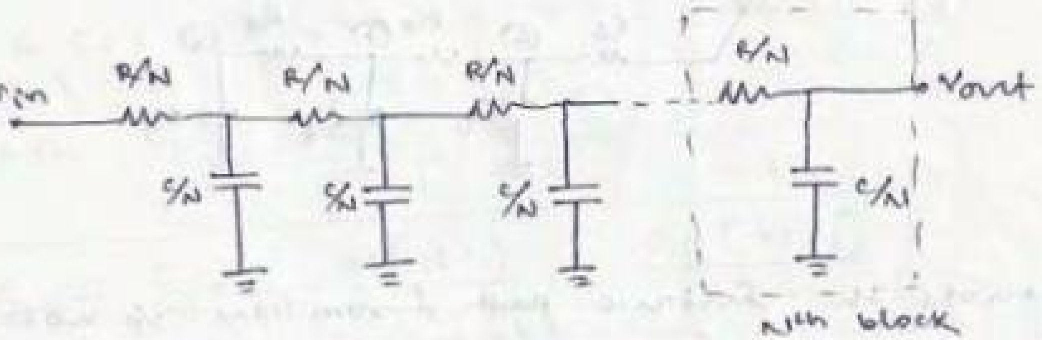
RC delay models :-



(simple lumped RC model of interconnect line)



(T-model of same ~~RC~~ RC model of interconnect line)



(Distributed RC ladder n/w model having N equal segments)

$$v_{out}(t) = v_{in}\left(1 - \exp\left(-\frac{t}{RC}\right)\right)$$

at $t = 0$

$$v_{out}(t) = V_{DD}\left(1 - \exp\left(-\frac{t}{RC}\right)\right)$$

at $t = \tau_{PLH}$, the o/p voltage reaches 50% of the initial voltage.

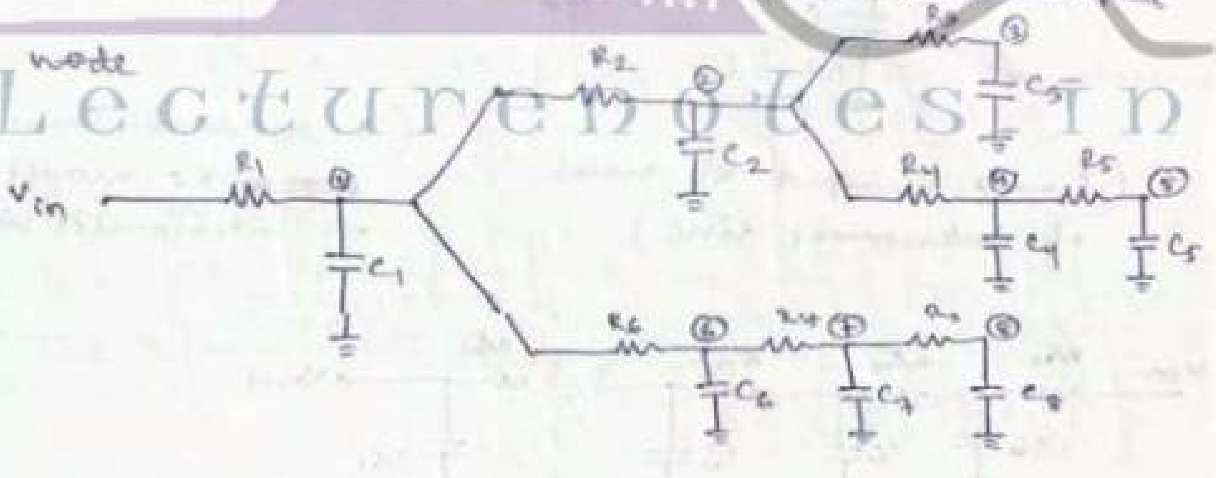
$$\Rightarrow V_{50\%} = V_{DD}\left[1 - \exp\left(-\frac{\tau_{PLH}}{RC}\right)\right]$$

$$\Rightarrow \frac{V_{DD}}{2} = V_{DD}\left[1 - \exp\left(-\frac{\tau_{PLH}}{RC}\right)\right]$$

$$\Rightarrow \boxed{\tau_{PLH} \simeq 0.69\,RC}$$

The Elmore ~~ww~~ Delay:

~~let P_i be the unique path from the input node~~



let P_i denote the unique path from the i/p node to node i, $i = 1, 2, 3, \ldots N$.

let $P_{ij} = P_i \cap P_j$ denote the portion of the path betn the i/p & the node i, which is common to the path betn the i/p & node j.

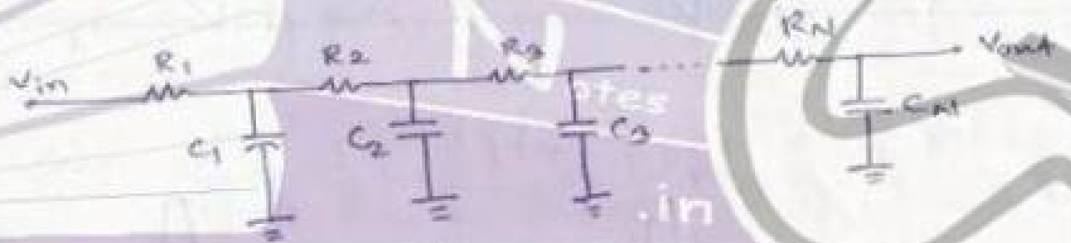
Assuming the i/p signal to be step i/p at t=0, then the Elmore delay at node 'i' of the RC n/w tree

$$\tau_{D_i} = \sum_{j=1}^{N} C_j \sum_{\forall k \in P_{ij}} R_k$$

Here $\tau_{D_7} = R_1C_1 + R_1C_2 + R_1C_3 + R_1C_4 + R_1C_5 + (R_1+R_6)C_6 + (R_1+R_6+R_7)C_7 + (R_1+R_6+R_7)C_8$

Similarly

$\tau_{D_5} = R_1C_1 + R_1C_6 + R_1C_7 + R_1C_8 + (R_1+R_2)C_2 + (R_1+R_2)C_3 + (R_1+R_2+R_4)C_4 + (R_1+R_2+R_4+R_5)C_5$



$$\tau_{DN} = \sum_{j=1}^{N} C_j \sum_{k=1}^{j} R_k$$

if $R_1 = R_2 = R_3 = \cdots = R_N = R/N$

& $C_1 = C_2 = C_3 = \cdots = C_N = C/N$

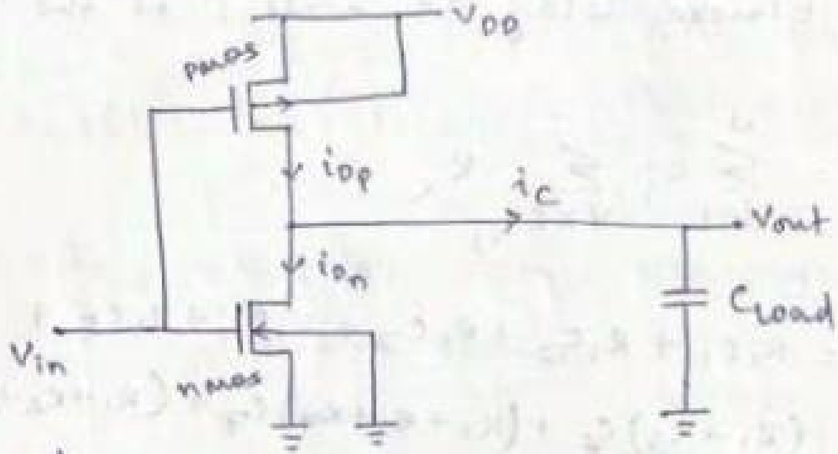
then $\tau_{DN} = \sum_{j=1}^{N} \frac{C}{N} \sum_{k=1}^{j} \frac{R}{N}$

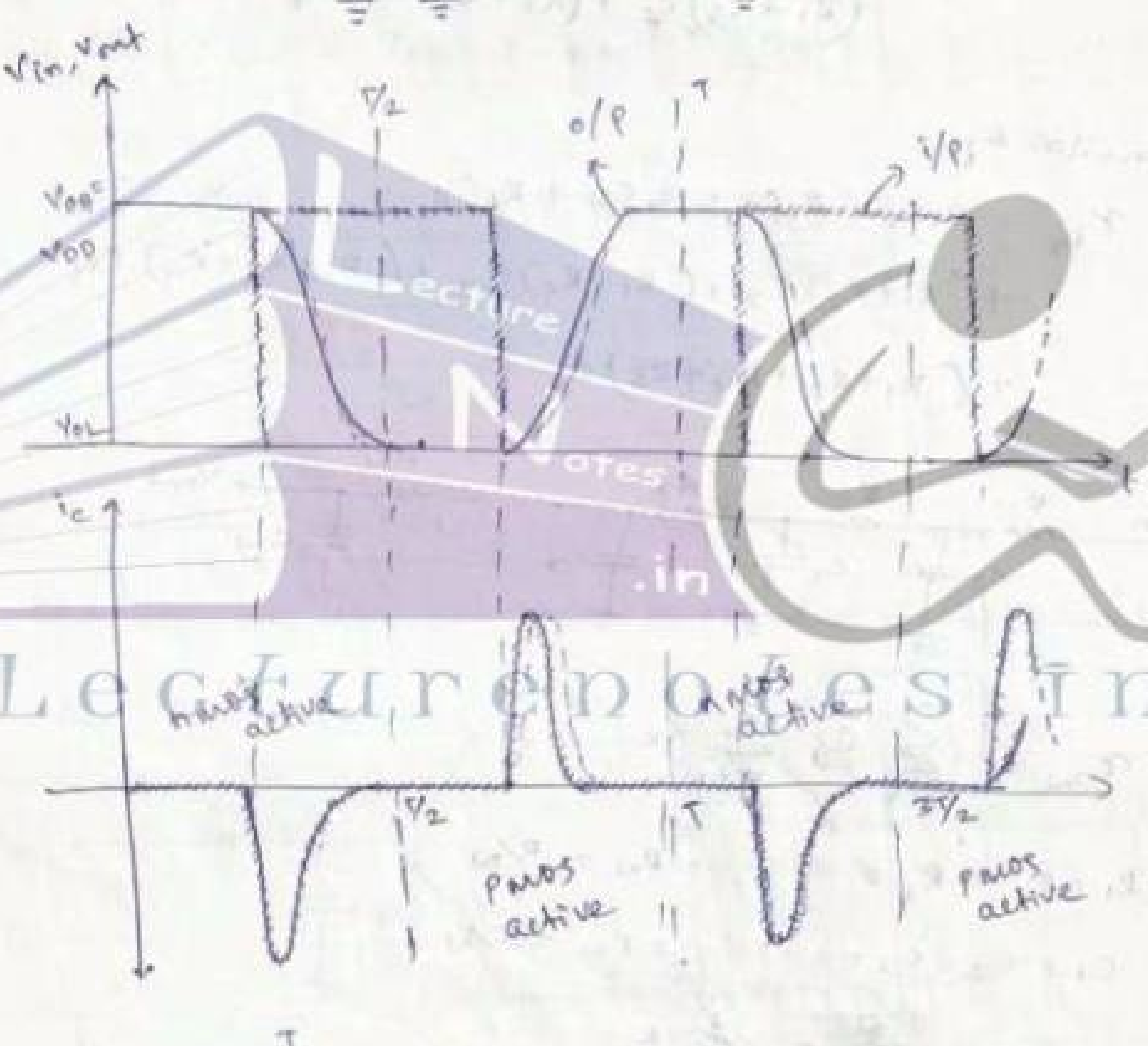
$$\Rightarrow \boxed{\tau_{DN} = \frac{C}{N} \times \frac{R}{N} \cdot \frac{N(N+1)}{2} = \frac{RC(N+1)}{2N}}$$

for large distributed RC network where 'N' is very large

$$\boxed{\tau_{DN}\Big|_{N\to\infty} = \frac{RC}{2}}$$

Switching power dissipation of CMOS inverter:-





$$P_{avg} = \frac{1}{T}\int_0^T v(t).\,i(t).\,dt$$

$$= \frac{1}{T}\left[\int_0^{T/2} V_{out} \times \left(-C_{Load}\frac{dV_{out}}{dt}\right).\,dt + \int_{T/2}^{T}(V_{DD}-V_{out})\left(C_{Load}\frac{dV_{out}}{dt}\right).\,d\right.$$

$$\Rightarrow P_{avg} = \frac{C_{load}}{T}\left[\int_0^{V_{DD}} V_{out}\cdot(dV_{out}) + \int_0^{V_{DD}} (V_{DD}-V_{out})\, dV_{out}\right]$$

$$= \frac{C_{load}}{T}\left[\frac{V_{out}^2}{2}\bigg|_0^{V_{DD}} + V_{DD}\cdot V_{out}\bigg|_0^{V_{DD}} - \frac{V_{out}^2}{2}\bigg|_0^{V_{DD}}\right]$$

$$= \frac{C_{load}}{T}\cdot\left[\frac{V_{DD}^2}{2} + V_{DD}^2 - \frac{V_{DD}^2}{2}\right] = \frac{C_{load}}{T}\cdot V_{DD}^2$$

$$\Rightarrow \boxed{P_{avg} = \frac{1}{T}\cdot C_{load}\cdot V_{DD}^2 = C_{load}\cdot V_{DD}^2\cdot f}$$

Power - Delay Product :-

if P^*_{avg} is the average switching power dissipation at maximum operating frequency "f_{max}" & 'τ_p' is the average propagation delay.

then $P^*_{avg} = C_{load}\cdot V_{DD}^2 \times f_{max}$

where $f_{max} = \frac{1}{\tau_{max}} = \frac{1}{\tau_{PHL}+\tau_{PLH}}$

then power-delay product is written as

$$PDP = 2P^*_{avg}\cdot\tau_p$$

$$= 2\left[C_{load}\cdot V_{DD}^2\left(\frac{1}{\tau_{PHL}+\tau_{PLH}}\right)\right]\left(\frac{\tau_{PHL}+\tau_{PLH}}{2}\right)$$

$$\left\{\because \tau_p = \frac{\tau_{PLH}+\tau_{PHL}}{2}\right\}$$

i.e $\boxed{PDP = C_{load}\cdot V_{DD}^2}$ for cmos ~~inverter~~ logic gates

—o—

VLSI Design

Topic:

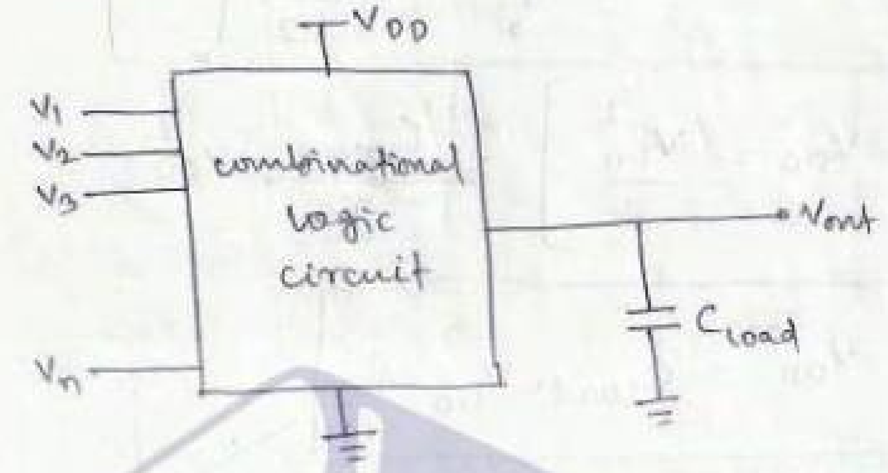
Combinational MOS Logic Circuits

Contributed By:

Verified Writer

ch-06 COMBINATIONAL MOS Logic Circuits

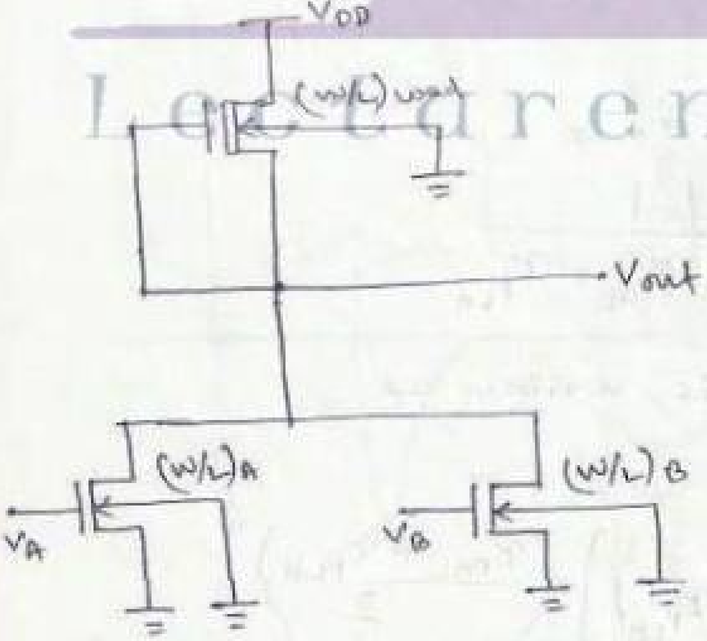
Introduction :- Here we will study the static & dynamic characteristics of various combinational mos logic circuits.



(Generic combinational logic circuit)

MOS logic circuits with depletion nMOS loads :-

2-input NOR gate :-



(2-i/p NOR gate with depletion nmos load)

(logic symbol)

V_A	V_B	V_{out}
Low	Low	High
Low	High	Low
High	Low	Low
High	High	Low

(Truth Table)

(with reference to the derivations for CMOS inverter)

Calculation of V_{OH} :-

when both V_A & V_B (input voltages) are lower than driver threshold voltage, then the driver transistors are turned off. i.e $I_{D_{driver}} = 0$

$$\Rightarrow I_{D_{load}} = \frac{k_{n_{load}}}{2}\left[2|V_{T_{load}}|\cdot(V_{DD}-V_{OH})-(V_{DD}-V_{OH})^2\right]=0$$

$$\Rightarrow \boxed{V_{OH} = V_{DD}}$$

Calculation of V_{OL} :-

The o/p voltage will be low (i.e V_{OL}) if atleast one of the input is high i.e

(1) $V_A = V_{OH}$, $V_B = V_{OL}$

(2) $V_A = V_{OL}$, $V_B = V_{OH}$

(3) $V_A = V_{OH}$, $V_B = V_{OH}$

for case ① & ② since one of the driver transistor become off, the analysis is reduced to simple nmos depletion-load inverter.

Assuming $V_{T_{0A}} = V_{T_{0B}} = V_{T_0}$,

for case ①

$$k_R = \frac{k_{driver_A}}{k_{load}} = \frac{k'_{n_{driver}}\cdot\left(\frac{W}{L}\right)_A}{k'_{n_{load}}\times\left(\frac{W}{L}\right)_{load}}$$

since (A) is on.

for case ②

$$k_R = \frac{k_{driver_B}}{k_{load}} = \frac{k'_{n_{driver}}\left(\frac{W}{L}\right)_B}{k'_{n_{load}}\left(\frac{W}{L}\right)_{Load}}$$

if $\left(\frac{W}{L}\right)_A = \left(\frac{W}{L}\right)_B$, then $k_R\Big|_{case①} = k_R\Big|_{case②}$

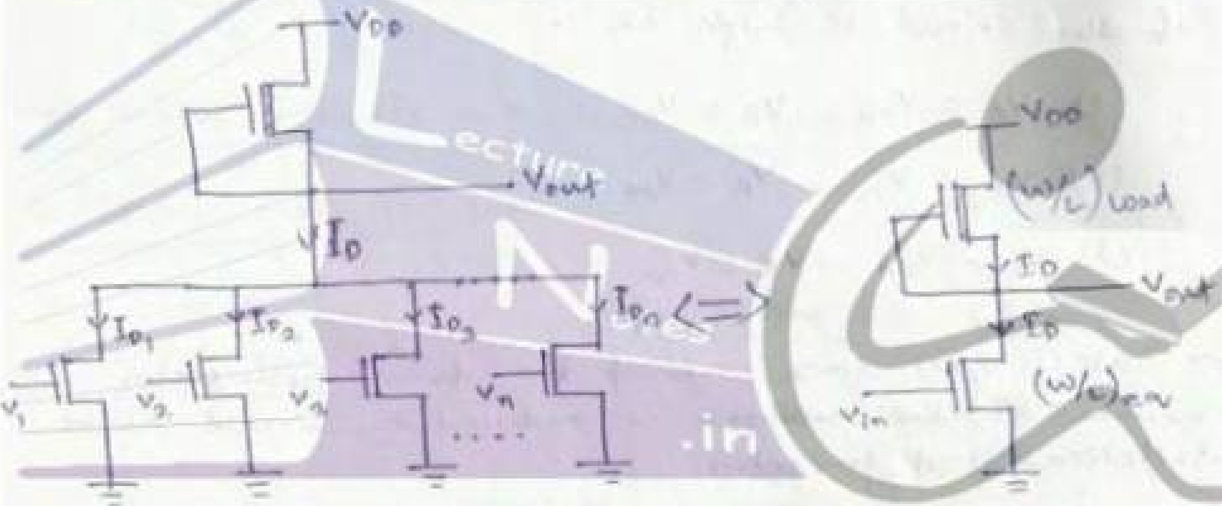
& for case (3) both driver transistors are operating.

$$K_R = \frac{K_{driver_A} + K_{driver_B}}{K_{load}} = \frac{k'_{n\,driver}\left[\left(\frac{W}{L}\right)_A + \left(\frac{W}{L}\right)_B\right]}{k'_{n\,load}\left(\frac{W}{L}\right)_{load}}$$

&
$$\boxed{V_{OL} = (V_{DD} - V_{T0}) - \sqrt{(V_{DD} - V_{T0})^2 - \frac{[V_{T\,load}]^2}{K_R}}}$$

where K_R has different values for different conditions.

Generalized NOR gate with multiple inputs :-



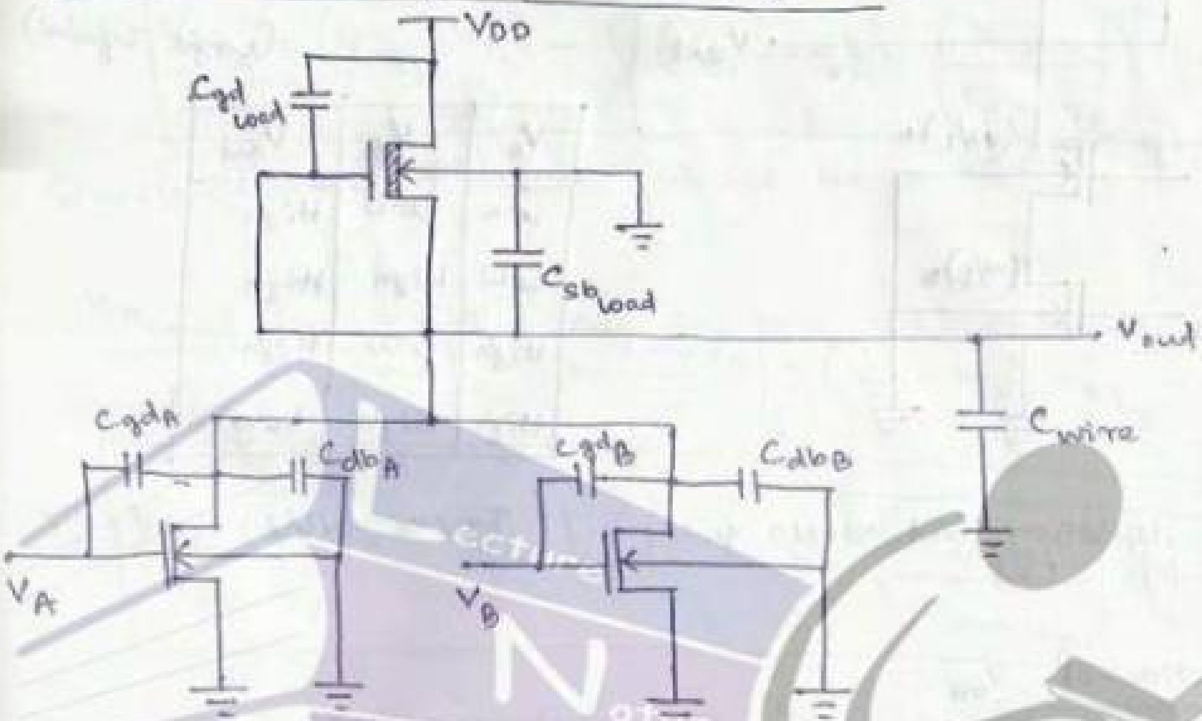
$$I_D = \sum_{K=1}^{n} I_{D_K} = \begin{cases} \sum_{K=1}^{n} \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)_K \left[2(V_{GS_K} - V_{T0})V_{out} - V_{out}^2\right] & \text{(for linear)} \\ \sum_{K=1}^{n} \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)_K \cdot (V_{GS_K} - V_{T0})^2 & \text{(for saturation)} \end{cases}$$

if $V_{GS_K} = V_{GS}$ for $K = 1, 2, \ldots n$

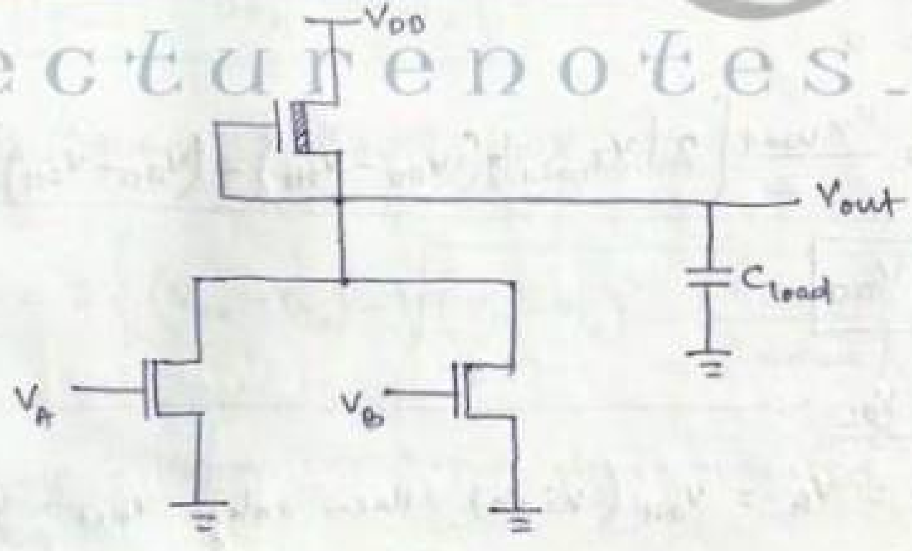
then
$$\boxed{I_D = \begin{cases} \frac{\mu_n \cdot C_{ox}}{2} \cdot \left[2(V_{GS} - V_{T0})V_{out} - V_{out}^2\right] \times \left\{\sum_{K=1}^{n}\left(\frac{W}{L}\right)_K\right\} & \text{(for linear)} \\ \frac{\mu_n C_{ox}}{2} \cdot (V_{GS} - V_{T0})^2 \left\{\sum_{K=1}^{n}\left(\frac{W}{L}\right)_K\right\} & \text{(for saturation)} \end{cases}}$$

or $$\boxed{\left(\frac{W}{L}\right)_{equivalent} = \sum_{K=1}^{n}\left(\frac{W}{L}\right)_K}$$

Transient analysis of NOR gate :-

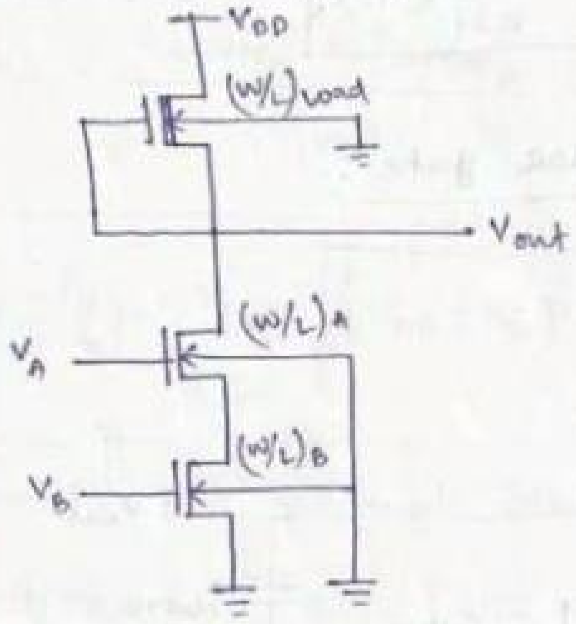


⇕

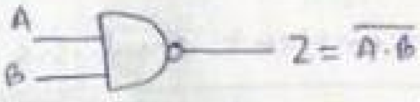


where $$\boxed{C_{load} = C_{gd_A} + C_{gd_B} + C_{db_A} + C_{db_B} + C_{gd_{load}} + C_{sb_{load}} + C_{wire}}$$

2-Input NAND gate :-



(2-i/p depletion load NAND gate) ckt.



(Logic symbol)

V_A	V_B	V_{out}
Low	Low	High
Low	High	High
High	Low	High
High	High	Low

(Truth table)

Calculation of V_{OH} :-

when both V_A & V_B are less than driver threshold voltage, then the driver transistors are off

i.e. $I_{D_{driver}} = 0$

$$\Rightarrow I_{D_{load}} = \frac{K_{n_{load}}}{2}\left[2|V_{T_{load}}|\cdot(V_{DD}-V_{OH}) - (V_{DD}-V_{OH})^2\right] = 0$$

$$\Rightarrow \boxed{V_{OH} = V_{DD}}$$

Calculation of V_{OL} :-

when $V_A = V_B = V_{OH}$ (high), then only $V_{out} = V_{OL}$ (low)

& for this condition (load in saturation & drivers are linear)

$$I_{D_{load}} = I_{D_{driver_A}} = I_{D_{driver_B}}$$

$\boxed{V_{GS_{load}} = 0}$

$$\Rightarrow \frac{K_{n_{load}}}{2}|V_{T_{load}}|^2 = \frac{K_{n_{driverA}}}{2}\left[2(V_{GS_A}-V_{T_0})\cdot V_{DS_A} - V_{DS_A}^2\right]$$

$$= \frac{K_{n_{driverB}}}{2}\left[2(V_{GS_B}-V_{T_0})\cdot V_{DS_B} - V_{DS_B}^2\right]$$

Assuming
~~Putting~~ $V_{GS_A} = V_{GS_B} = V_{OH}$ (input high), we get

$$\frac{K_{n\,load}}{2}|V_{T\,load}|^2 = \frac{K_{n\,driver_A}}{2}\left[2(V_{OH}-V_{T_0})\cdot V_{DS_A} - V_{DS_A}^2\right]$$

$$\Rightarrow \boxed{V_{DS_A} = (V_{OH}-V_{T_0}) - \sqrt{(V_{OH}-V_{T_0})^2 - \left(\frac{K_{load}}{K_{driver_A}}\right)\cdot|V_{T\,load}|^2}}$$

Similarly the 2^{nd} eq^n would be

$$\frac{K_{n\,load}}{2}\cdot|V_{T\,load}|^2 = \frac{K_{n\,driver_B}}{2}\left[2(V_{OH}-V_{T_0})\cdot V_{DS_B} - V_{DS_B}^2\right]$$

$$\Rightarrow \boxed{V_{DS_B} = (V_{OH}-V_{T_0}) - \sqrt{(V_{OH}-V_{T_0})^2 - \left(\frac{K_{load}}{K_{driver_B}}\right)\cdot|V_{T\,load}|^2}}$$

Then the corresponding o/p V_{OL} can be written as

$$\boxed{V_{OL} = V_{DS_A} + V_{DS_B}}$$

for $K_{driver_A} = K_{driver_B} = K_{driver}$

$$\boxed{V_{OL} = 2\left\{(V_{OH}-V_{T_0}) - \sqrt{(V_{OH}-V_{T_0})^2 - \left(\frac{K_{load}}{K_{driver}}\right)\cdot|V_{T\,load}|^2}\right\}}$$

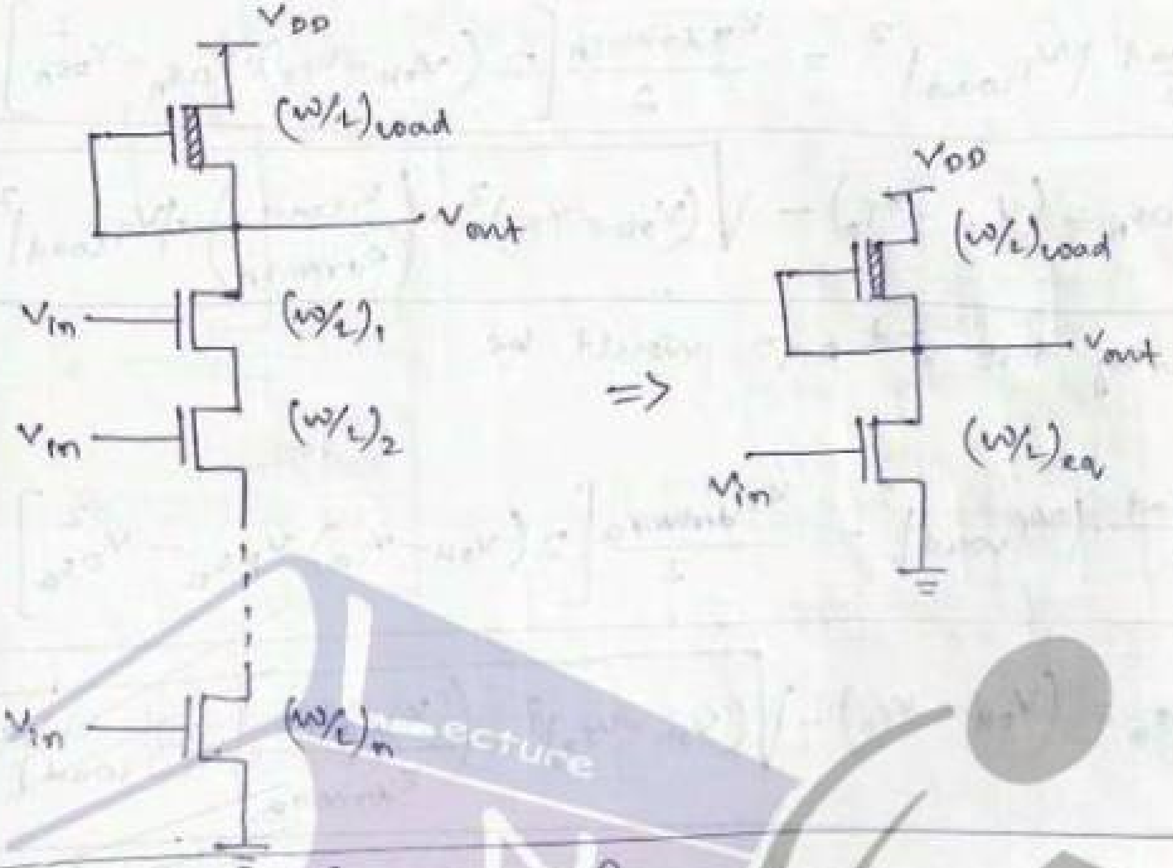
$\Rightarrow$ Since both driver transistors operated in linear region.

$$I_{D_A} = \frac{K_{n\,driver}}{2}\left[2(V_{GS_A}-V_{T_0})V_{DS_A} - V_{DS_A}^2\right]$$

$$I_{D_B} = \frac{K_{n\,driver}}{2}\left[2(V_{GS_B}-V_{T_0})V_{DS_B} - V_{DS_B}^2\right]$$

Since $I_{D_A} = I_{D_B} = I_D \Rightarrow I_D = \frac{1}{2}[I_{D_A}+I_{D_B}]$

Generalized NAND structure with multiple inputs:-



$$I_D = \begin{cases} \dfrac{\mu_n C_{ox}}{2} \cdot \left[\dfrac{1}{\sum_{K=1}^{n} \frac{1}{\left(\frac{W}{L}\right)_K}}\right] \times \left[2(V_{in}-V_{T0})\, V_{out} - V_{out}^2\right], \text{ linear} \\ \dfrac{\mu_n C_{ox}}{2} \left[\dfrac{1}{\sum_{K=1}^{n} \frac{1}{\left(\frac{W}{L}\right)_K}}\right] \times \left[(V_{in}-V_{T0})^2\right], \text{ Saturation} \end{cases}$$

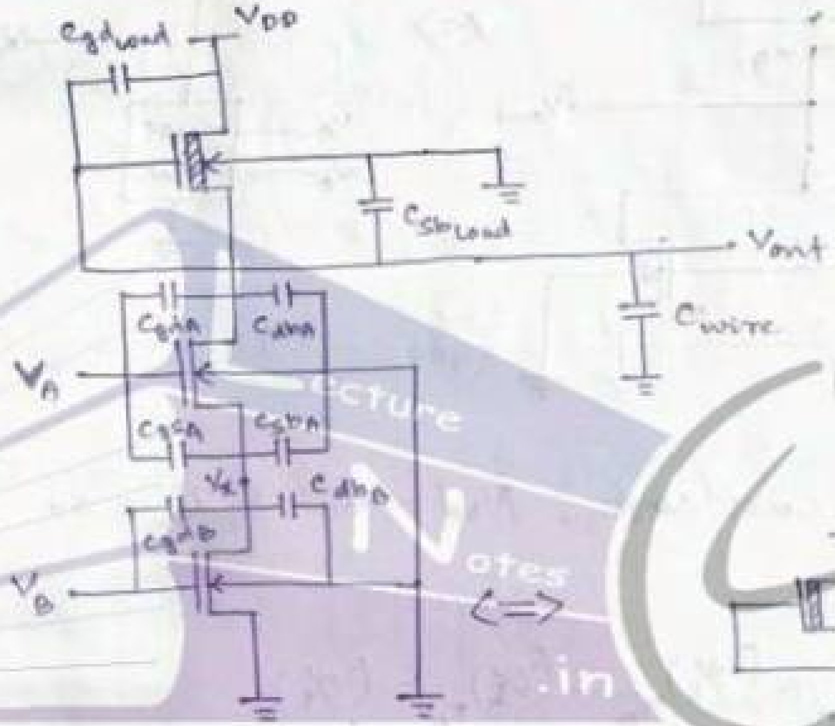
or $I_D = \dfrac{\mu_n C_{ox}}{2} \cdot \left(\dfrac{W}{L}\right)_{equivalent} \times \begin{cases} \left[2(V_{in}-V_{T0})V_{out} - V_{out}^2\right], \text{ linear} \\ \left[(V_{in}-V_{T0})^2\right], \text{ Saturation} \end{cases}$

where $\boxed{\left(\dfrac{W}{L}\right)_{equivalent} = \left[\dfrac{1}{\sum_{K=1}^{n} \frac{1}{\left(\frac{W}{L}\right)_K}}\right]}$

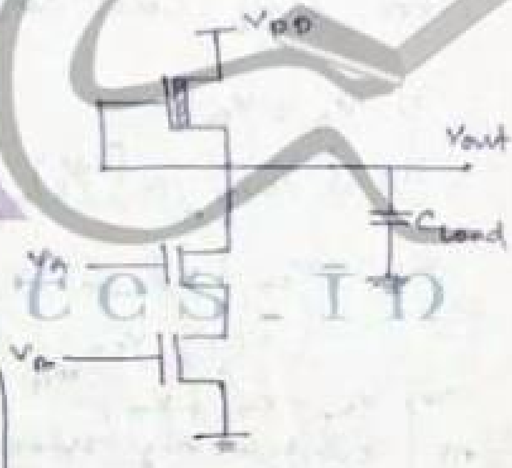
if $\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = \cdots = \left(\frac{W}{L}\right)_n = \left(\frac{W}{L}\right)$

$$\Rightarrow \boxed{\left(\frac{W}{L}\right)_{\text{equivalent}} = \frac{1}{\frac{n}{\left(\frac{W}{L}\right)}} = \frac{1}{n} \times \left(\frac{W}{L}\right)}$$

Transient analysis of 2 i/p NAND gate :-



$\Longleftrightarrow$



Then

$$\boxed{\begin{aligned} C_{load} = {} & C_{gd_{load}} + C_{sb_{load}} \\ & + C_{gd_A} + C_{db_A} + C_{gs_A} + C_{sb_A} \\ & + C_{gd_B} + C_{db_B} + C_{wire} \end{aligned}}$$

C_{gs_A}, C_{sb_A}, C_{gd_B} & C_{db_B} are also included because the internal node voltage 'V_x' also changes with V_{out} during transition.

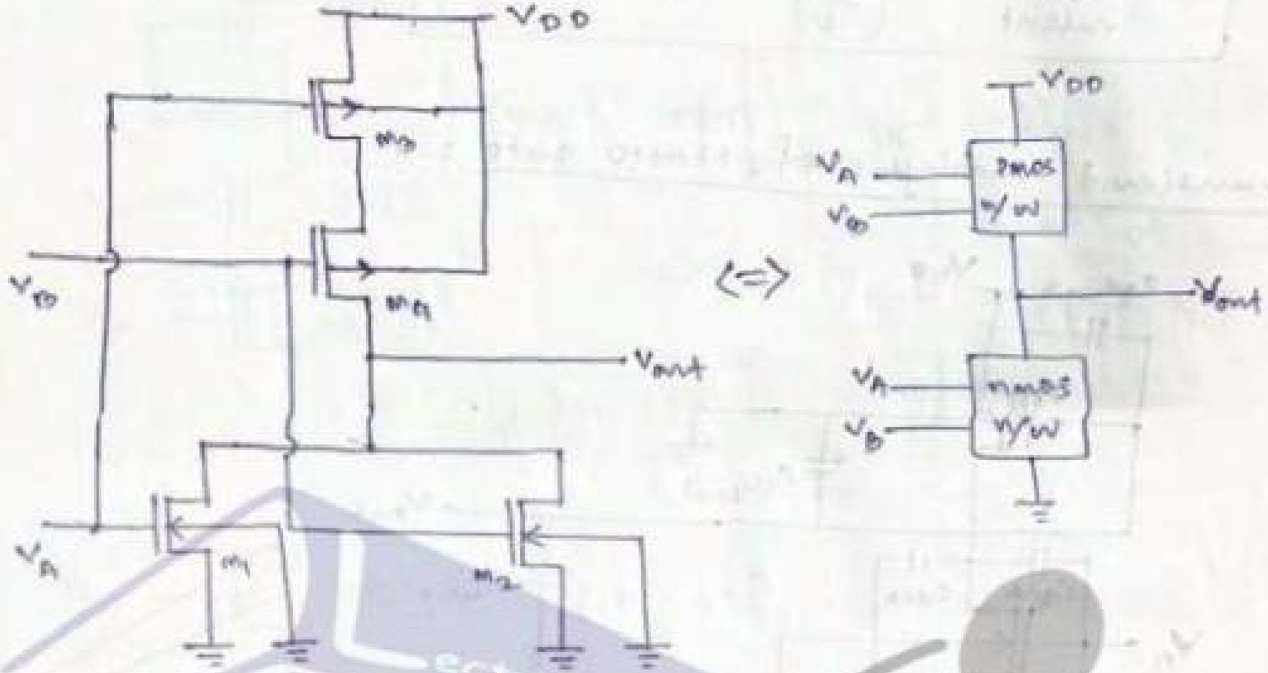
if $V_A = V_{OH}$ & V_B changes from V_{OH} to V_{OL}

$$\boxed{C_{load} = C_{gd_{load}} + C_{sb_{load}} + C_{gd_A} + C_{db_A} + C_{wire}}$$

when $V_B = V_{OH}$ & V_A changes from V_{OH} to V_{OL}

CMOS logic circuits :-

CMOS NOR2 (2 input NOR gate) :-



Assumption

for any CMOS logic, $V_{OH} = V_{DD}$ & $V_{OL} = 0V$

i) $V_A = V_B$

ii) $(W/L)_{n_A} = (W/L)_{n_B} = (W/L)_{P_A} = (W/L)_{P_B}$

iii) V_{SB} (for PMOS) is neglected

i.e $V_{SB} \simeq 0V$

iv) $K_{n_1} = K_{n_2} = K_n$, $K_{P_3} = K_{P_4} = K_P$

At switching threshold V_{th}.

$$V_A = V_B = V_{out} = V_{th}$$

Here M_1, M_2 & M_4 are saturated whereas M_3 is in linear region of operation.

$$I_{D_1} = I_{D_2} = \frac{K_n}{2}(V_{th} - V_{Tn})^2$$

$$I_D = I_{D_1} + I_{D_2} = K_n(V_{th} - V_{Tn})^2$$

$$\Rightarrow \boxed{\sqrt{I_D} = \sqrt{K_n}\left(V_{th} - V_{Tn}\right)} \longrightarrow (1)$$

$$I_{D_3} = \frac{K_p}{2}\left[2\left(V_{DD} - V_{th} - |V_{Tp}|\right)V_{SD_3} - V_{SD_3}^2\right]$$

$$\& \; I_{D_4} = \frac{K_p}{2}\left[V_{DD} - V_{th} - |V_{Tp}| - V_{SD_3}\right]^2$$

let $\left(V_{DD} - V_{th} - |V_{Tp}|\right) = x$ & $I_{D_3} = I_{D_4} = I_D$

$$\Rightarrow I_{D} = \frac{K_p}{2}\left[2x \cdot V_{SD_3} - V_{SD_3}^2\right] \longrightarrow (2)$$

$$\& \; I_{D} = \frac{K_p}{2}\left(x - V_{SD_3}\right)^2 \longrightarrow (3)$$

$$eq (3) \Rightarrow \left(x - V_{SD_3}\right) = \sqrt{\frac{2I_D}{K_p}}$$

$$\Rightarrow V_{SD_3} = \left(x - \sqrt{\frac{2I_D}{K_p}}\right)$$

Putting the value of V_{SD_3} in eq (2) we get

$$2x\left(x - \sqrt{\frac{2I_D}{K_p}}\right) - \left(x - \sqrt{\frac{2I_D}{K_p}}\right)^2 = \frac{2I_D}{K_p}$$

$$\Rightarrow 2x^2 - x\sqrt{\frac{8I_D}{K_p}} - \left[x^2 + \frac{2I_D}{K_p} - x\sqrt{\frac{8I_D}{K_p}}\right] = \frac{2I_D}{K_p}$$

$$\Rightarrow 2x^2 - \cancel{x\sqrt{\frac{8I_D}{K_p}}} - x^2 - \frac{2I_D}{K_p} + \cancel{x\sqrt{\frac{8I_D}{K_p}}} - \frac{2I_D}{K_p} = 0$$

$$\Rightarrow x^2 - \frac{4I_D}{K_p} = 0 \Rightarrow x = 2\sqrt{\frac{I_D}{K_p}}$$

$$\Rightarrow \boxed{V_{DD} - V_{th} - |V_{Tp}| = 2\sqrt{\frac{I_D}{K_p}}} \longrightarrow (4)$$

Putting the value of $\sqrt{I_D}$ from eqn ① in eqn ④

$$V_{DD} - V_{th} - |V_{Tp}| = \frac{2}{\sqrt{k_p}}\left[\sqrt{k_n}\left(V_{th} - V_{Tn}\right)\right]$$

$$\Rightarrow V_{DD} - V_{th} - |V_{Tp}| = 2\sqrt{\frac{k_n}{k_p}}\, V_{th} - 2\sqrt{\frac{k_n}{k_p}}\, V_{Tn}$$

$$\Rightarrow V_{th}\left[1 + 2\sqrt{\frac{k_n}{k_p}}\right] = \left(V_{DD} - |V_{Tp}|\right) + 2\sqrt{\frac{k_n}{k_p}}\, V_{Tn}$$

$$\Rightarrow V_{th} = \frac{V_{DD} - |V_{Tp}| + 2\sqrt{\frac{k_n}{k_p}}\, V_{Tn}}{1 + 2\sqrt{\frac{k_n}{k_p}}}$$

$$\boxed{V_{th\,(NOR2)} = \frac{V_{Tn} + \frac{1}{2}\sqrt{\frac{k_p}{k_n}}\left(V_{DD} - |V_{Tp}|\right)}{1 + \frac{1}{2}\sqrt{\frac{k_p}{k_n}}}}$$

⇒ Equivalent inverter approach

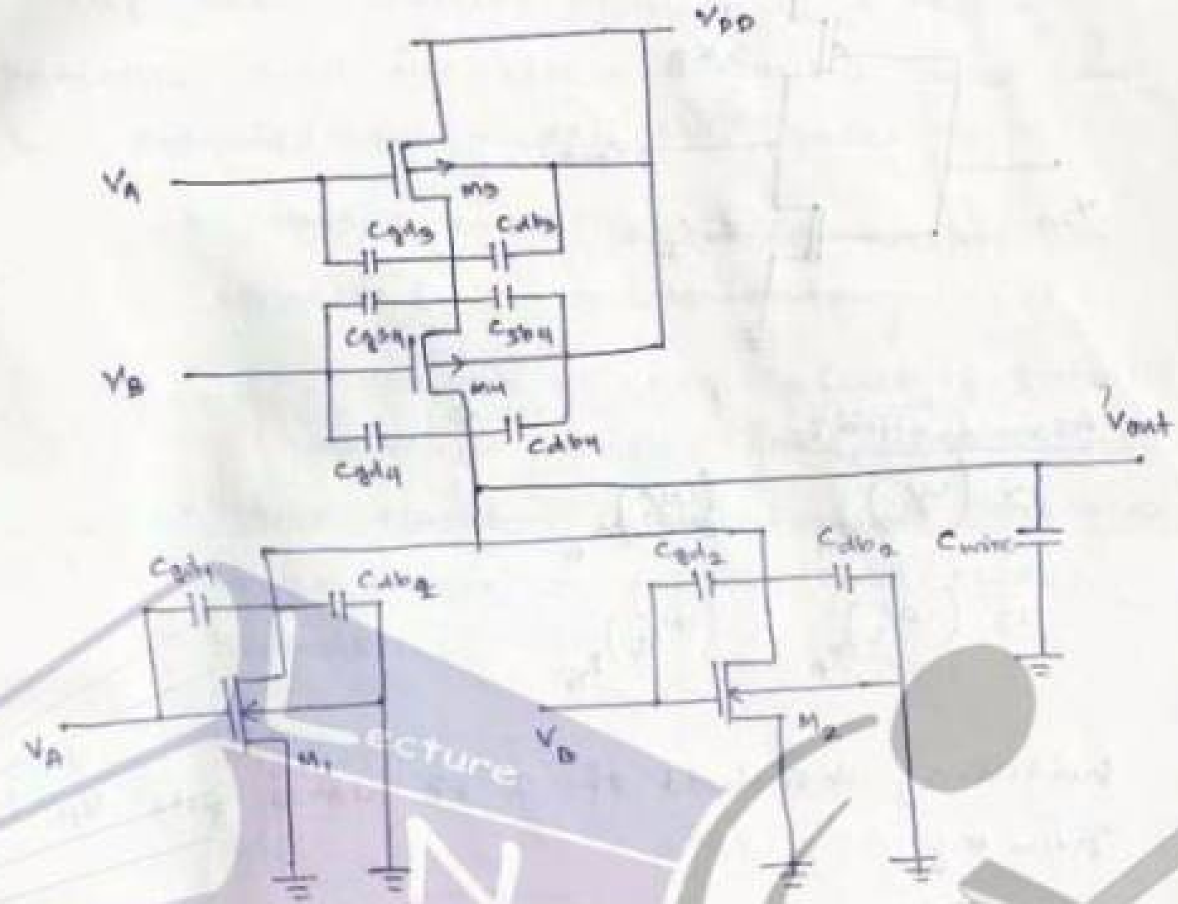
when inputs are identical,

⇒ 2 parallel nmos transistors with k_n is equivalent to a single nmos with $k_n' = 2k_n$.

⇒ 2 series pmos transistors with k_p is equivalent to a single pmos with $k_p' = k_p/2$

$$\boxed{V_{th\,(inverter)} = \frac{V_{Tn} + \sqrt{\frac{k_p}{k_n}} \cdot \left(V_{DD} - |V_{Tp}|\right)}{1 + \sqrt{\frac{k_p}{k_n}}}}$$

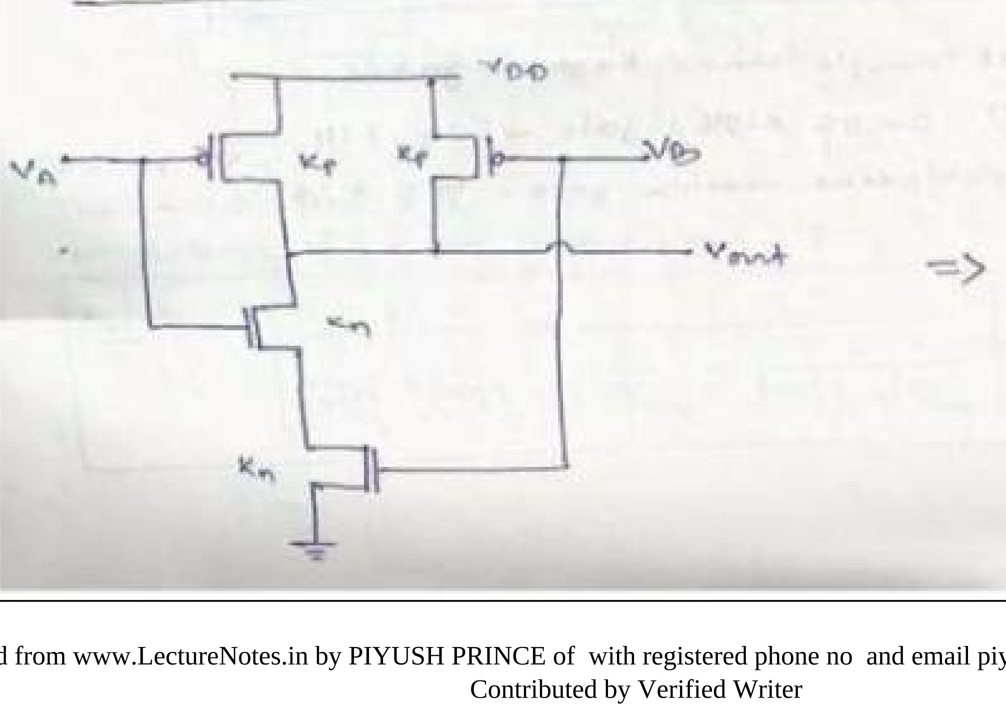
Transient analysis of 2 i/p NOR gate :-

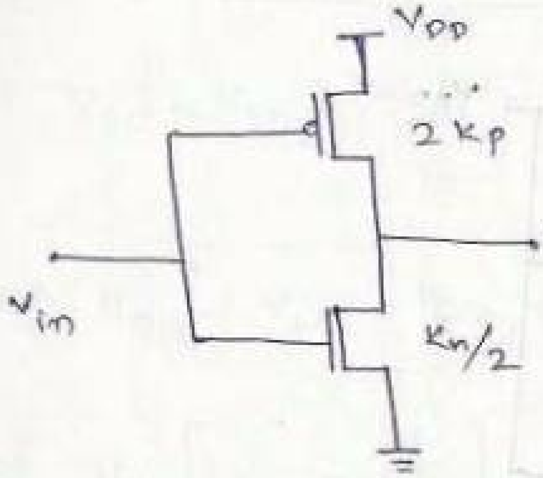


$$C_{load} = C_{gd_3} + C_{db_4} + C_{gd_1} + C_{db_1} + C_{gd_2} + C_{db_2}$$

$$+ C_{gd_3} + C_{db_3} + C_{gs_4} + C_{gs_3} \, C_{sb_4} + C_{wire}$$

CMOS NAND2 (2 i/p NAND gate) :-





Assumption :-

1> $\left(\frac{W}{L}\right)_{n_A} = \left(\frac{W}{L}\right)_{n_B}$

2> $\left(\frac{W}{L}\right)_{P_A} = \left(\frac{W}{L}\right)_{P_B}$

Switching threshold of 2 i/p NAND gate V_{th} is given by

$$V_{th}(NAND2) = \frac{V_{Tn} + \sqrt{\frac{2K_p}{K_n/2}}\left(V_{DD} - |V_{Tp}|\right)}{1 + \sqrt{\frac{2K_p}{K_n/2}}}$$

$$\boxed{V_{th}(NAND2) = \frac{V_{Tn} + 2\sqrt{K_p/K_n}\left(V_{DD} - |V_{Tp}|\right)}{1 + 2\sqrt{K_p/K_n}}}$$

layout of simple cmos logic gates.

1> cmos NOR2 gate - fig 7.14
2> cmos NAND2 gate - fig 7.15 } page 293

complex logic circuits :-

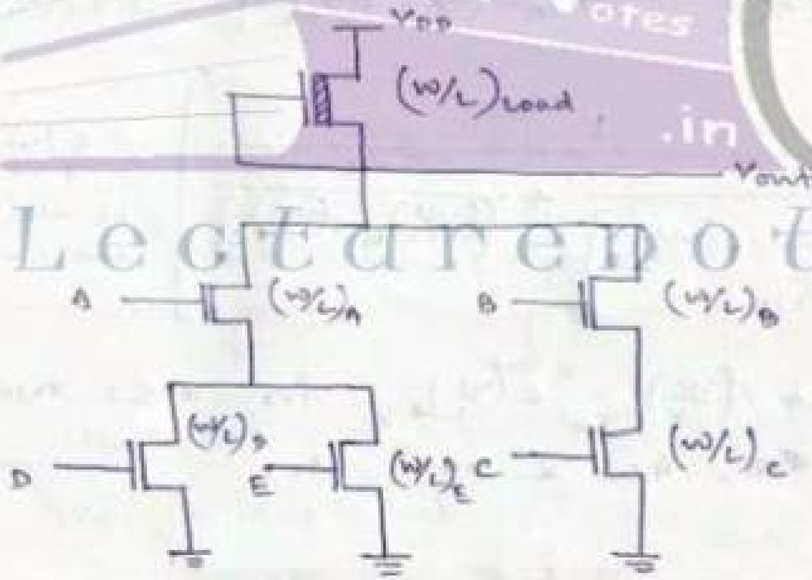
The basic circuit structures & design principles used for NOR 2 & NAND 2 gates can be extended to complex logic gates.

* NMOS (Driver) & PMOS (Load) transistors are connected in complementary pairs.

* OR operation is equivalent to parallel driver (NMOS) & series load (PMOS) transistors

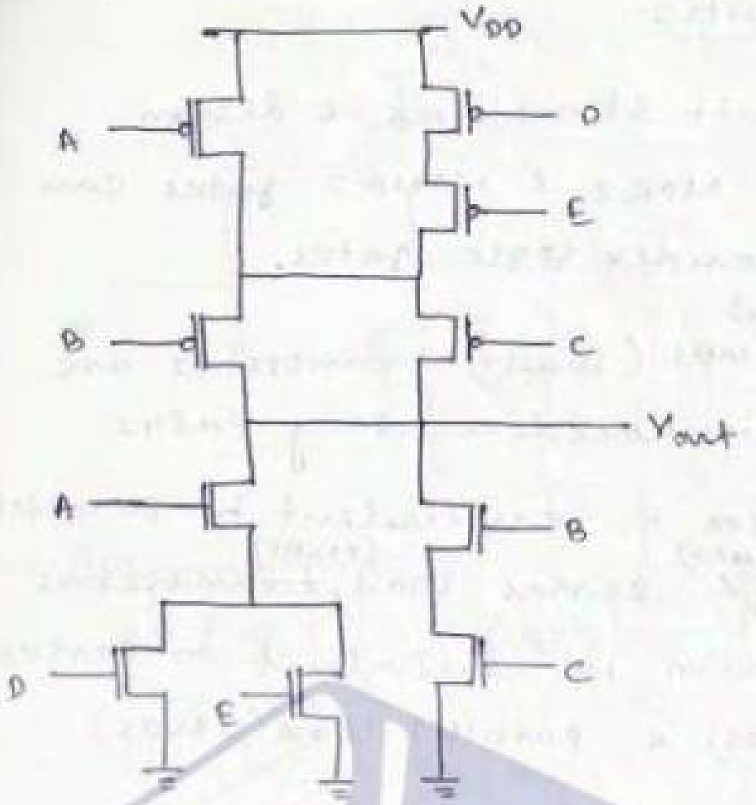
* AND operation is equivalent to series driver (NMOS) & parallel load (PMOS) transistors.

$$Z = \overline{A(D+E)+B\cdot C}$$



The equivalent (W/L) ratio of the pull-down network consisting of NMOS transistors is

$$\boxed{\left(\frac{W}{L}\right)_{eq} = \left[\frac{1}{\frac{1}{(W/L)_B}+\frac{1}{(W/L)_C}} + \frac{1}{\frac{1}{(W/L)_A}+\frac{1}{(W/L)_D+(W/L)_E}}\right]}$$



$$\left(\frac{W}{L}\right)_{eq_n} = \left[\frac{1}{\frac{1}{(W/L)_B}+\frac{1}{(W/L)_C}} + \frac{1}{\frac{1}{(W/L)_A}+\frac{1}{(W/L)_D+(W/L)_E}}\right] \simeq \frac{7}{6}\left(\frac{W}{L}\right)$$

if all (W/L) are same

$$\left(\frac{W}{L}\right)_{eq_p} = \left[\frac{1}{\frac{1}{\frac{1}{\frac{1}{(W/L)_D}+\frac{1}{(W/L)_E}}+(W/L)_A} + \frac{1}{\left[(W/L)_B+(W/L)_C\right]}}\right] \simeq \frac{6}{7}\left(\frac{W}{L}\right)$$

if all (W/L) are same.

Similarly findout $\left(\frac{W}{L}\right)_{eq_n}$ & $\left(\frac{W}{L}\right)_{eq_p}$ for XOR2, XNOR2 NOR3, NAND3, AOI & OAI etc.

⇒ Ex 7.2, page 302

$(W/L)_{eq_n}$ & $(W/L)_{eq_p}$ & transistor-level schematic (page 305).

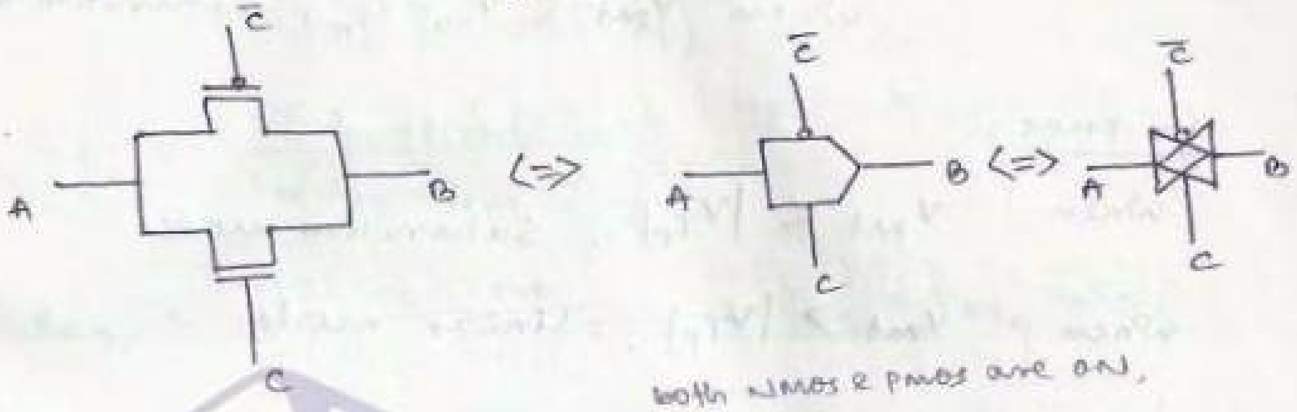
for 1-bit full adder.

Sum = $A \oplus B \oplus C$

Carry = $AB + AC + BC$

CMOS transmission gate (pass gate) :-

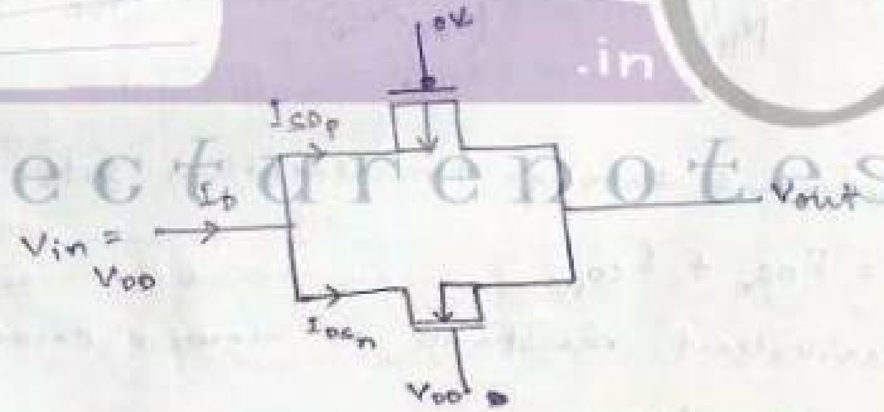
A CMOS transmission gate (CMOS TG) operates as a bidirectional switch between node 'A' & node 'B' which is controlled by signal 'C' & its complement '$\bar{C}$'.



(a) When C is high (say V_{DD}), both NMOS & PMOS are ON, A & B are connected

(b) When C is low (say 0), both NMOS & PMOS are off, so A & B are open circuit (high impedance):

Here ~~[illegible]~~ $B = AC + A\bar{C} = A$ (logic-wise)



For NMOS:

$$V_{DS_n} = V_{DD} - V_{out}$$

$$V_{GS_n} = V_{DD} - V_{out}$$

For PMOS :-

$$V_{DS_P} = V_{out} - V_{DD}$$

$$V_{GS_P} = -V_{DD}$$

consequently the operating regions of NMOS & PMOS can be written as :

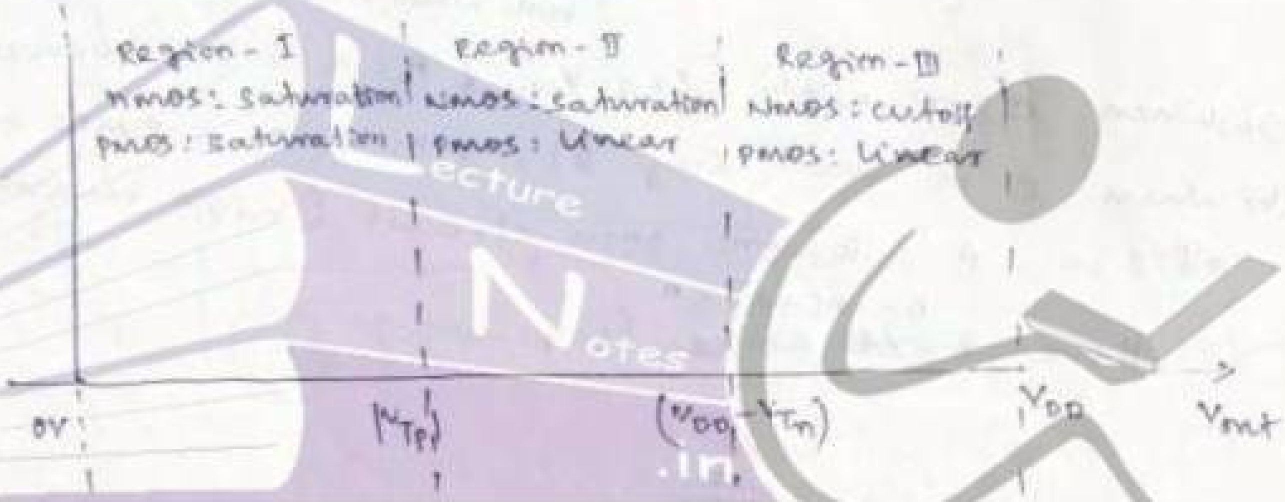
<u>for NMOS</u> :- when $V_{out} > V_{DD} - V_{Tn}$, cutoff mode.

when $V_{out} < V_{DD} - V_{Tn}$, Saturation mode.

<u>for PMOS</u> :-

when $V_{out} < |V_{Tp}|$, Saturation mode.

when $V_{out} > |V_{Tp}|$, linear mode.



The total current through transmission gate is

$$I_D = I_{DS_n} + I_{SD_p}$$

and the equivalent resistance of NMOS & PMOS are

$$R_{eq_n} = \frac{V_{DD} - V_{out}}{I_{DS_n}}$$

$$R_{eq_p} = \frac{V_{DD} - V_{out}}{I_{SD_p}}$$

& $R_{eq} = (R_{eq_n} \parallel R_{eq_p})$

Region - I :-

NMOS & PMOS both in Saturation

$$I_{DS_n} = \frac{K_n}{2}(V_{DD} - V_{out} - V_{Tn})^2$$

& $$I_{SD_p} = \frac{K_p}{2}(V_{DD} - |V_{Tp}|)^2$$

$$R_{eq_n} = \frac{2(V_{DD} - V_{out})}{K_n(V_{DD} - V_{out} - V_{Tn})^2}$$

$$R_{eq_p} = \frac{2(V_{DD} - V_{out})}{K_p(V_{DD} - |V_{Tp}|)^2}, \quad R_{eq} = R_{eq_n} \parallel R_{eq_p}$$

Here V_{SB} (PMOS) = 0

but V_{SB} (NMOS) = V_{out}

so $V_{Tp} = V_{T0p}$

but $V_{Tn} = V_{T0n} + \gamma\left(\sqrt{|2\phi_F| + V_{out}} - \sqrt{|2\phi_F|}\right)$

Region - II :-

NMOS : Saturation

PMOS : Linear

$$I_{DS_n} = \frac{K_n}{2}(V_{DD} - V_{out} - V_{Tn})^2$$

$$I_{SD_p} = \frac{K_p}{2}\left[2(V_{DD} - |V_{Tp}|)(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2\right]$$

$$= \frac{K_p(V_{DD} - V_{out})}{2}\left[2(V_{DD} - |V_{Tp}|) - (V_{DD} - V_{out})\right]$$

$$R_{eq_n} = \frac{2(V_{DD} - V_{out})}{K_n(V_{DD} - V_{out} - V_{Tn})^2}$$

$$R_{eq_p} = \frac{2}{K_p[2(V_{DD} - |V_{Tp}|) - (V_{DD} - V_{out})]}$$

$$R_{eq} = R_{eq_n} \parallel R_{eq_p}$$

Region - III :-

nmos : cut-off
pmos : linear

$$I_{DS_n} = 0$$

$$I_{SD_p} = \frac{k_p}{2}(V_{DD} - V_{out})\left[2(V_{DD} - |V_{Tp}|) - (V_{DD} - V_{out})\right]$$

$$R_{eq_n} = \frac{V_{DD} - V_{out}}{I_{DS_n}} = \infty$$

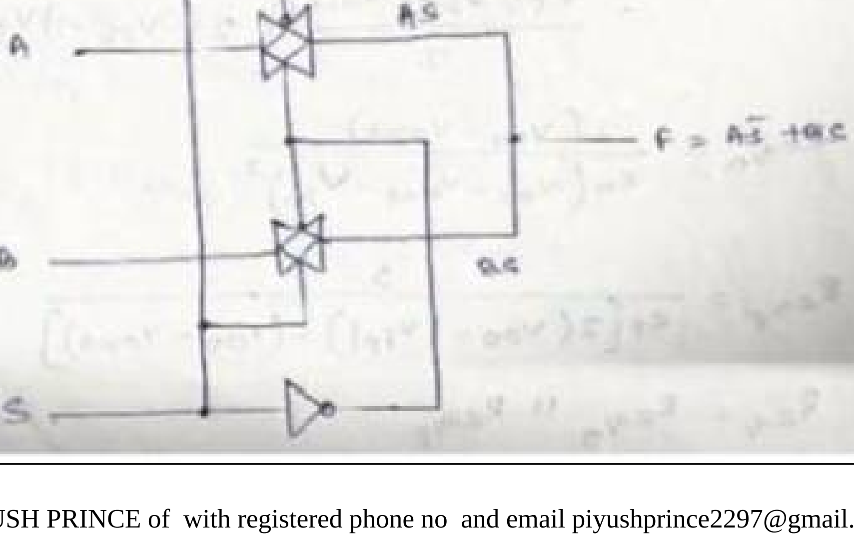
$$R_{eq_p} = \frac{2}{k_p\left[2(V_{DD} - |V_{Tp}|) - (V_{DD} - V_{out})\right]}$$

$$R_{eq} = R_{eq_n} \parallel R_{eq_p} = R_{eq_p}$$

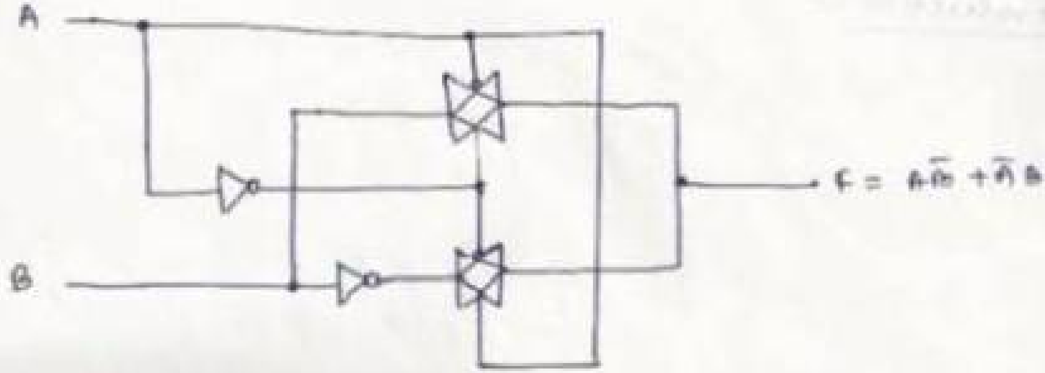
figure 7.35 : Equivalent resistance R_{eq_n}, R_{eq_p} & R_{eq} of CMOS TG plotted as a function of V_{out}.
Page 310

Design & Implementation of various logic structure using CMOS TG :-

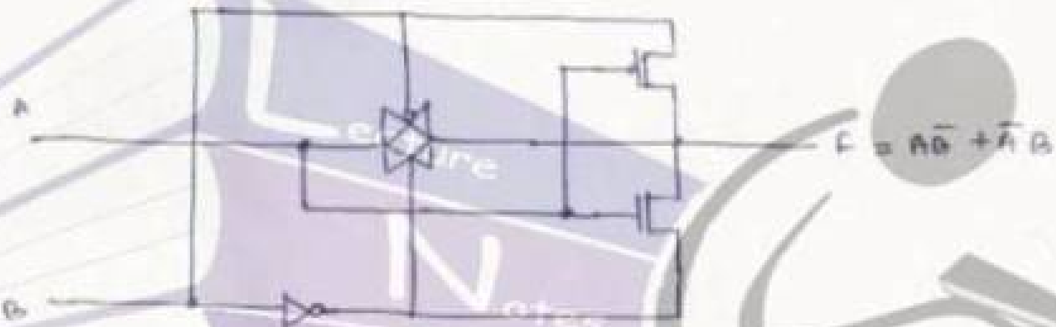
i) $F = A\bar{S} + BS$ (2 I/P MUX)



ii) $F = A\bar{B} + \bar{A}B$ (2 i/p XOR)



or



iii) $F = AB + \bar{A}\bar{C} + A\bar{B}C$

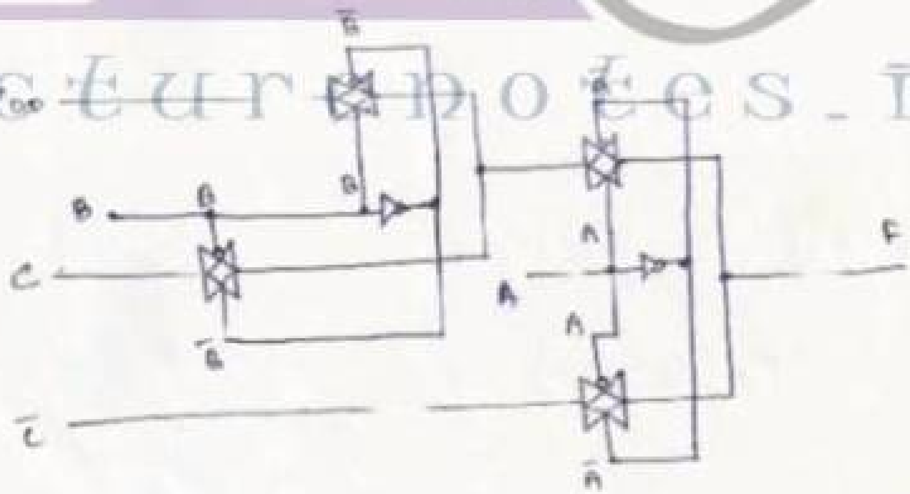


fig 7.40(b), page 313

<u>complementary pass-transistor Logic (CPL) :-</u>

Figure 7.42 (a) & (b), page 315

Figure 7.43 & 7.44, page 315 & 316

— o —

VLSI Design

Topic:

Sequential MOS Logic Circuits

Contributed By:

Verified Writer

MODULE - 3

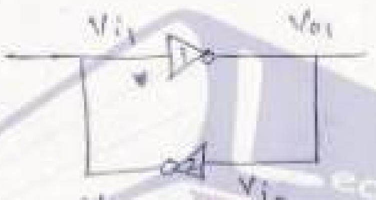
Sequential MOS logic circuits

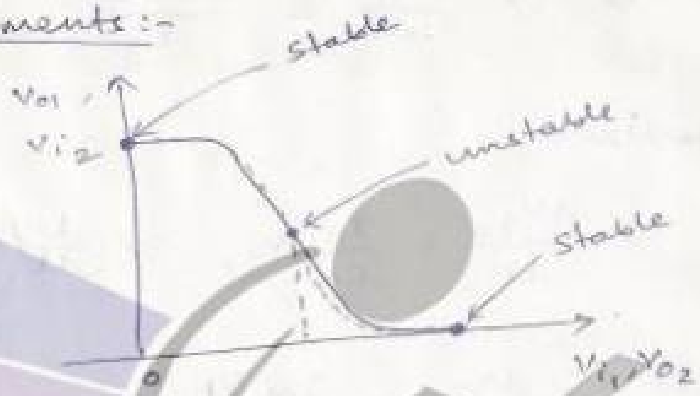
Introduction :-

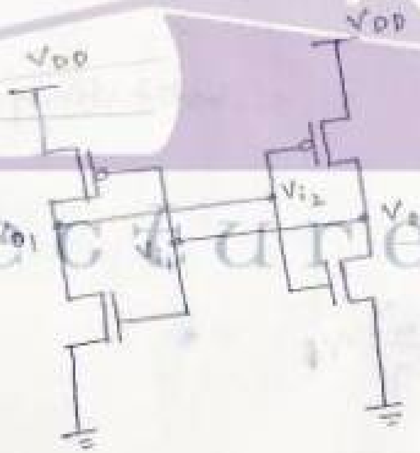
The main components of sequential system are 3 types :-

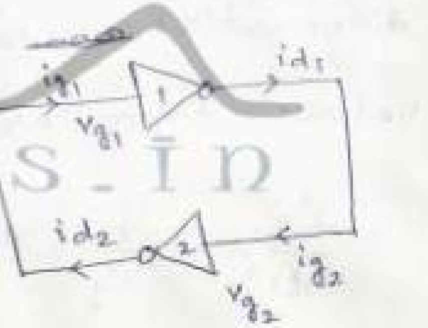
1) Bistable circuits
2) Monostable circuits
3) Astable circuits.

Behavior of Bistable Elements :-









(small signal analysis of inverter bistable circuit

$$\left.\begin{aligned} i_{g_1} &= i_{d_2} = g_m v_{g_2} = g_m q_2/C_g \\ i_{g_2} &= i_{d_1} = g_m v_{g_1} = g_m q_1/C_g \end{aligned}\right\} \text{———} (1)$$

again $i_{g_1} = C_g \frac{dv_{g_1}}{dt}$ & $i_{g_2} = C_g \frac{dv_{g_2}}{dt}$ ——— (2)

$\Rightarrow$

combining eq^n ① & ② we get

$$\left.\begin{aligned} g_m v_{g_2} &= C_g \cdot \frac{dv_{g_1}}{dt} \\ g_m v_{g_1} &= C_g \frac{dv_{g_2}}{dt} \end{aligned}\right\} \quad \text{—— (3)}$$

$$\Rightarrow \frac{dv_{g_1}}{dt} = \frac{g_m}{C_g}(v_{g_2}) \quad \text{or} \quad \frac{d}{dt}\left(\frac{q_1}{C_g}\right) = \frac{g_m}{C_g}\frac{q_2}{C_g}$$

$$\Rightarrow \frac{dq_1}{dt} = \frac{g_m q_2}{C_g} \quad \text{—— (4.1)}$$

Similarly

$$\frac{dv_{g_2}}{dt} = \frac{g_m}{C_g} v_{g_1} \Rightarrow \frac{d}{dt}\left(\frac{q_2}{C_g}\right) = \frac{g_m}{C_g} \cdot \frac{q_1}{C_g}$$

$$\Rightarrow \frac{dq_2}{dt} = \frac{g_m q_1}{C_g} \quad \text{—— (4.2)}$$

differentiate eq^n (4.1) wrt 't' & substituting the values from eq^n 4.2 we get

$$\frac{d^2 q_1}{dt^2} = \frac{g_m}{C_g} \cdot \frac{dq_2}{dt} = \left(\frac{g_m}{C_g} \times \frac{g_m}{C_g}\right) q_1$$

$$\left.\begin{aligned} &\Rightarrow \frac{d^2 q_1}{dt^2} = \left(\frac{g_m}{C_g}\right)^2 \cdot q_1 \quad \& \\ &\text{Similarly} \quad \frac{d^2 q_2}{dt^2} = \left(\frac{g_m}{C_g}\right)^2 q_2 \end{aligned}\right\} \quad \text{—— (5)}$$

eqⁿ (5.1) $\Rightarrow$

$$\frac{d^2 q_1(t)}{dt^2} = \left(\frac{g_m}{C_g}\right)^2 \cdot q_1(t) = \frac{1}{\tau_0^{\,2}}\, q_1 \qquad \text{where } \tau_0 = \frac{C_g}{g_m}.$$

$$\Rightarrow q_1(t) = \frac{1}{2}\left[q_1(0) - \tau_0\, q_1'(0)\right] e^{-t/\tau_0} + \frac{1}{2}\left[q_1(0) + \tau_0\, q_1'(0)\right] e^{t/\tau_0}$$

$$\& \; q_2(t) = \frac{1}{2}\left[q_2(0) - \tau_0\, q_2'(0)\right] e^{-t/\tau_0} + \frac{1}{2}\left[q_2(0) + \tau_0\, q_2'(0)\right] e^{t/\tau_0}$$

again $q_1(t) = C_g \cdot v_{g_1}(t) = C_g \cdot v_{o_2}(t)$

$\& \; q_2(t) = C_g\, v_{g_2}(t) = C_g\, v_{o_1}(t)$

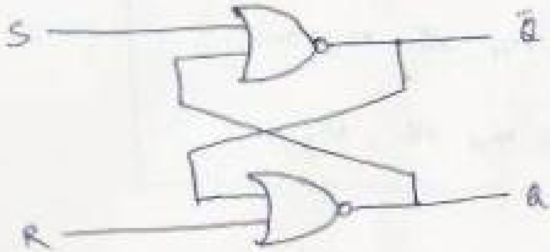
$$\Rightarrow \boxed{\begin{aligned} v_{o_2}(t) &= \frac{1}{2}\left[v_{o_2}(0) - \tau_0\, v_{o_2}'(0)\right] e^{-t/\tau_0} + \frac{1}{2}\left[v_{o_2}(0) + \tau_0\, v_{o_2}'(0)\right] e^{t/\tau_0} \\ v_{o_1}(t) &= \frac{1}{2}\left[v_{o_1}(0) - \tau_0\, v_{o_1}'(0)\right] e^{-t/\tau_0} + \frac{1}{2}\left[v_{o_1}(0) + \tau_0\, v_{o_1}'(0)\right] e^{t/\tau_0} \end{aligned}}$$

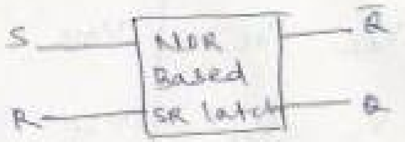
↳ Time-domain analysis of v_{o_1} & v_{o_2}

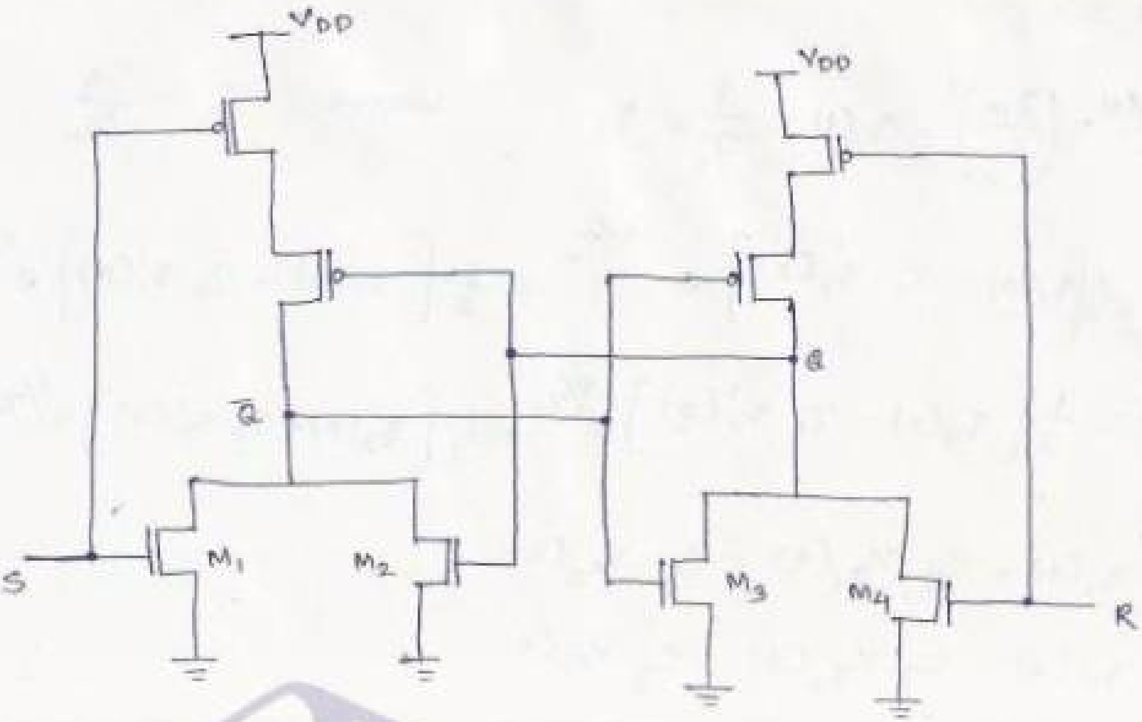
for $t \gg 1$, $e^{-t/\tau_0} \simeq 0$

$$\boxed{\begin{aligned} v_{o_2}(t) &= \frac{1}{2}\left[v_{o_2}(0) + \tau_0\, v_{o_2}'(0)\right] e^{t/\tau_0} \\ v_{o_1}(t) &= \frac{1}{2}\left[v_{o_1}(0) + \tau_0\, v_{o_1}'(0)\right] e^{t/\tau_0} \end{aligned}}$$

SR latch circuit :- NOR Based :







Truth Table

S	R	Q_{n+1}	$\bar{Q}_{n+1}$	operation
0	0	Q_n	$\bar{Q}_n$	Hold/No change
0	1	1	0	Set
1	0	0	1	Reset
1	1	0	0	Not allowed/forbidden

operating modes of transistors in NOR-based CMOS SR latch circuit :-

S	R	Q_{n+1}	$\bar{Q}_{n+1}$	operation
V_{OH}	V_{OL}	V_{OH}	V_{OL}	M_1 & M_2 ON, M_3 & M_4 off
V_{OL}	V_{OH}	V_{OL}	V_{OH}	M_1 & M_2 off, M_3 & M_4 ON
V_{OL}	V_{OL}	V_{OH}	V_{OL}	M_1 & M_4 off, M_2 ON
or V_{OL}	V_{OL}	V_{OL}	V_{OH}	M_1 & M_4 off, M_3 ON

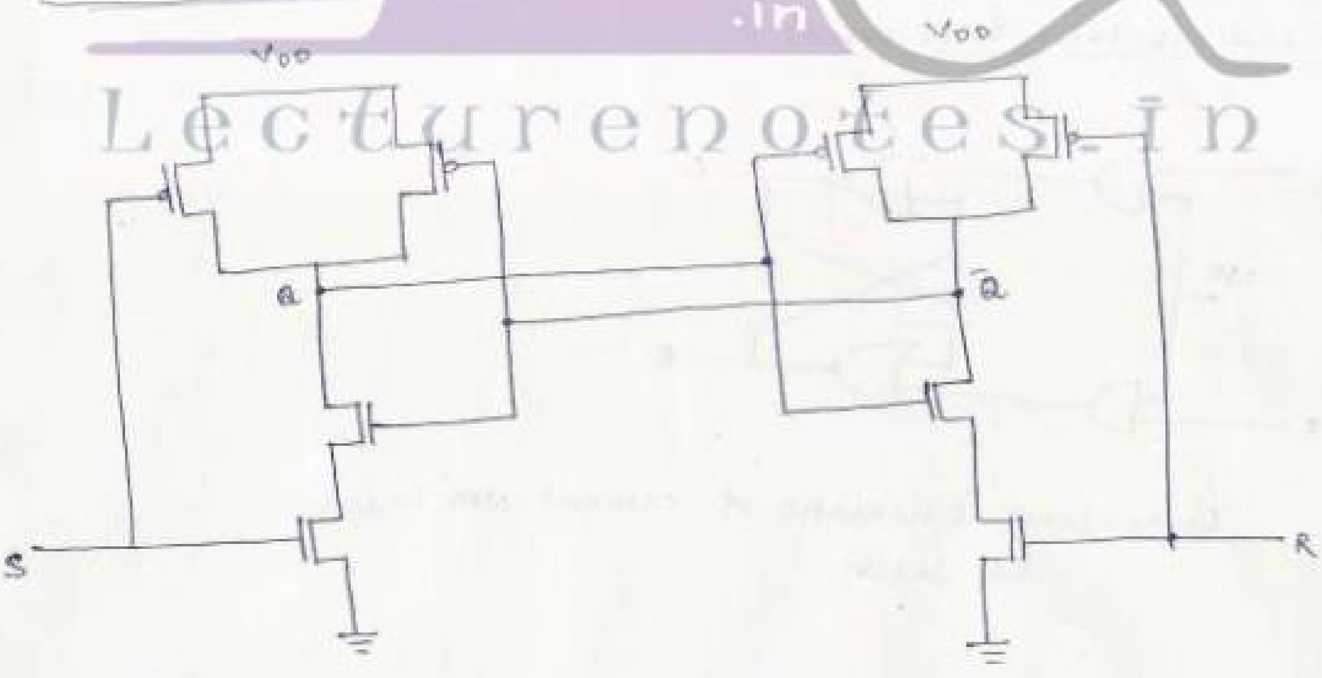
(figure 8.9, page 333, circuit diagram of CMOS SR latch from transient analysis.)

$$C_Q = C_{gb_2}, C_{gb_5} + C_{db_3} + C_{db_4} + C_{db_7} + C_{sb_7} + C_{db_8}$$

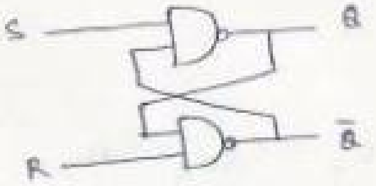
$$C_{\bar{Q}} = C_{gb_3} + C_{gb_7} + C_{db_1} + C_{db_2} + C_{db_5} + C_{sb_5} + C_{db_6}$$

$$\tau_{rise_Q}(\text{SR latch}) = \tau_{rise_Q}(\text{NOR2}) + \tau_{fall_{\bar{Q}}}(\text{NOR2})$$

NAND Based SR latch:-



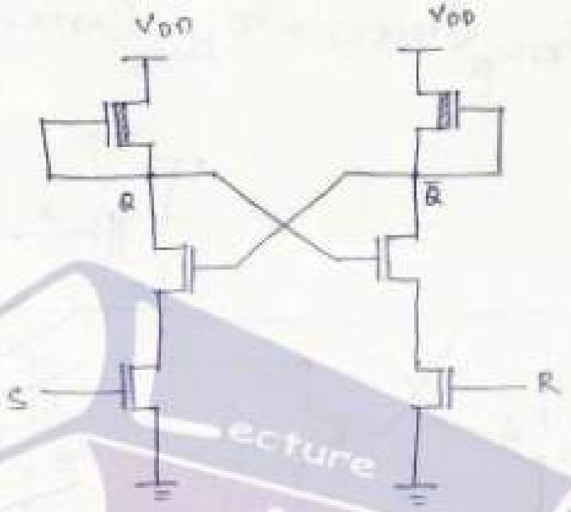
(CMOS SR latch circuit using NAND2 gates)



(Gate level schematic of NAND-based SR latch)

Truth Table :-

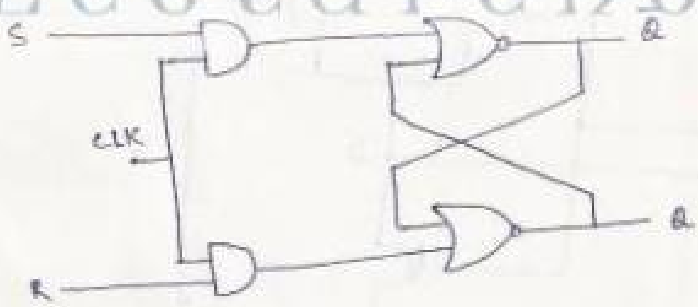
S	R	Q_{n+1}	$\bar{Q}_{n+1}$	operation
0	0	1	1	not allowed
0	1	1	0	set
1	0	0	1	Reset
1	1	Q_n	$\bar{Q}_n$	hold/no change



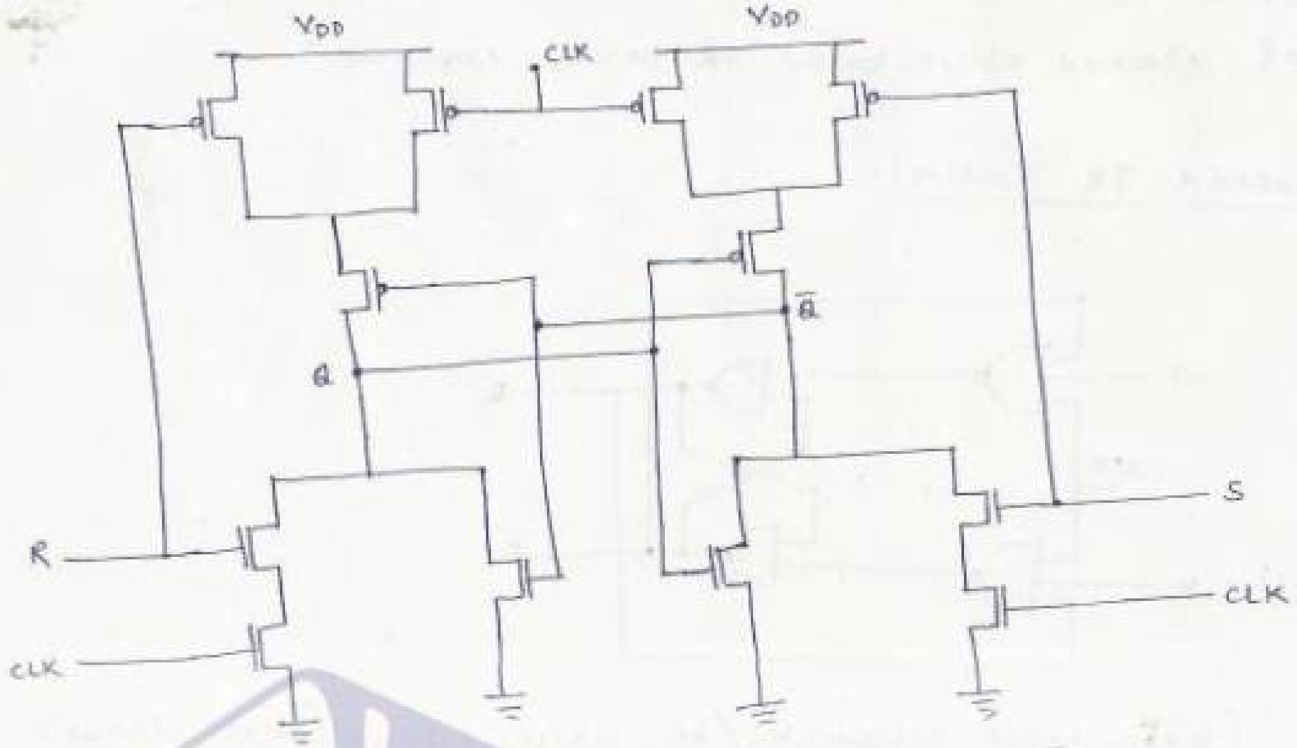
[Depletion-load NMOS NAND based SR latch ckt]

clocked latch and flip-flop circuits :-

clocked SR latch (NOR based) :-



Grate-level schematic of clocked NOR based SR latch



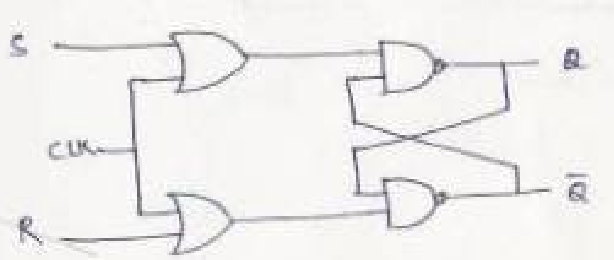
(CMOS implementation of clocked NOR based SR latch circuit)

Truth Table / State Table :-

CLK	S	R	Q_{n+1}	$\bar{Q}_{n+1}$	operation
0	0	0	Q_n	$\bar{Q}_n$	Hold
0	0	1	Q_n	$\bar{Q}_n$	Hold
0	1	0	Q_n	$\bar{Q}_n$	Hold
0	1	1	Q_n	$\bar{Q}_n$	Hold
1	0	0	Q_n	$\bar{Q}_n$	Hold
1	0	1	0	1	Reset
1	1	0	1	0	Set
1	1	1	0	0	Not Allowed

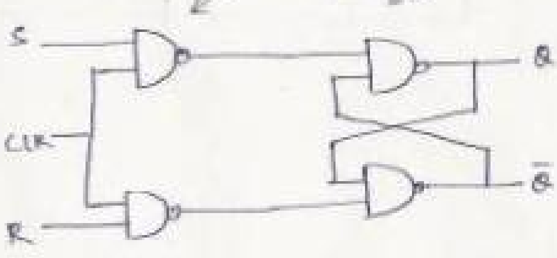
(figure 8.15: verify the waveform acc to the state table)

NAND-based clocked SR latch:-



or

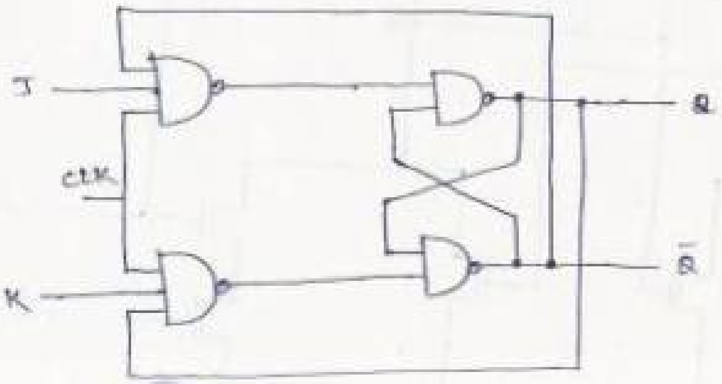
(All NAND implementation of clocked SR latch)



(Gate-level schematic of clocked NAND based SR latch)

* Draw the CMOS and depletion load circuit implementation of clocked NAND-based SR latch circuit.

Clocked JK Latch:-

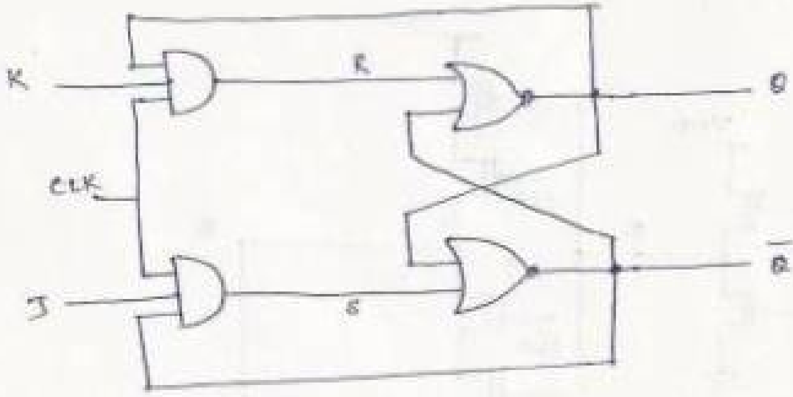


(Gate-level schematic/All-NAND schematic of clocked JK latch circuit)

* Draw the CMOS and depletion load circuit implementation of clocked NAND-based JK latch circuit.

Truth Table/State Table:-

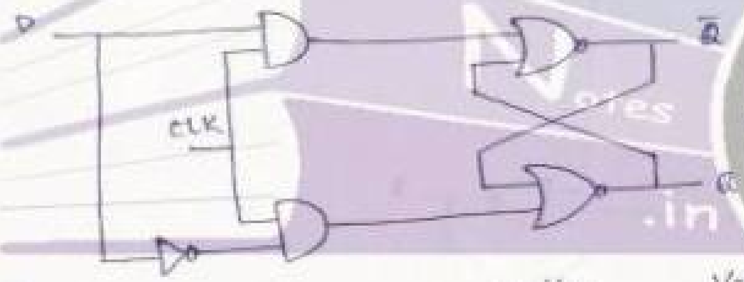
CLK	J	K	Q_{n+1}	$\bar{Q}_{n+1}$	operation
0	0	0	Q_n	$\bar{Q}_n$	Hold
0	0	1	Q_n	$\bar{Q}_n$	Hold
0	1	0	Q_n	$\bar{Q}_n$	Hold
0	1	1	Q_n	$\bar{Q}_n$	Hold
1	0	0	Q_n	$\bar{Q}_n$	Hold
1	0	1	0	1	~~Set~~ Reset
1	1	0	1	0	Set
1	1	1	$\bar{Q}_n$	Q_n	Toggle



(Gate-level schematic of clocked NOR-based JK latch circuit)

(figure 8.21, page 340, CMOS implementation of All-NAND-base clocked JK latch)

<u>CMOS D-latch and Edge-triggered flip-flop :-</u>



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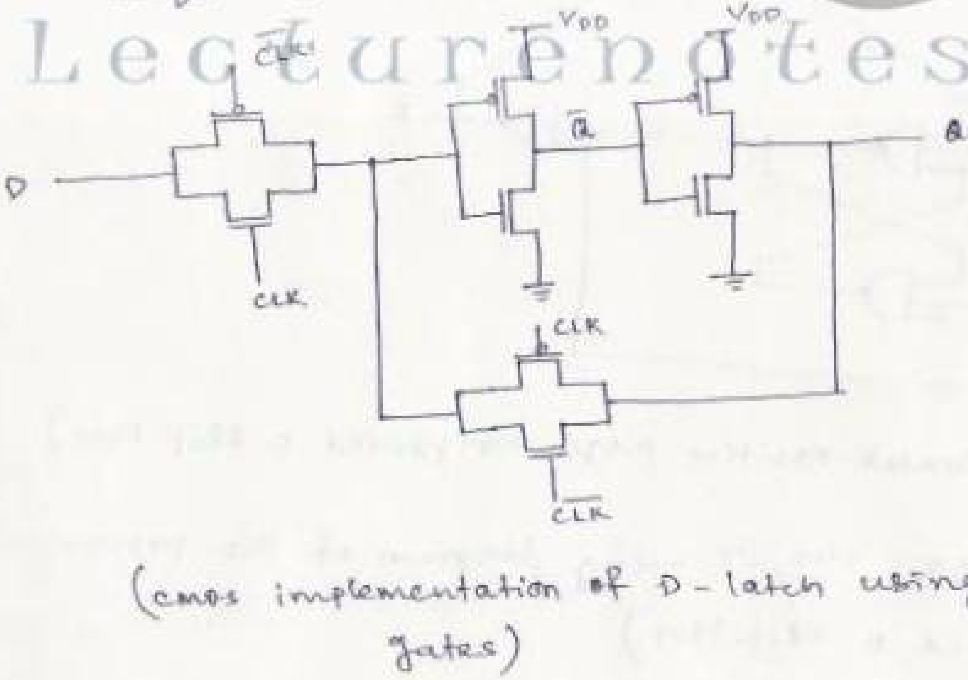
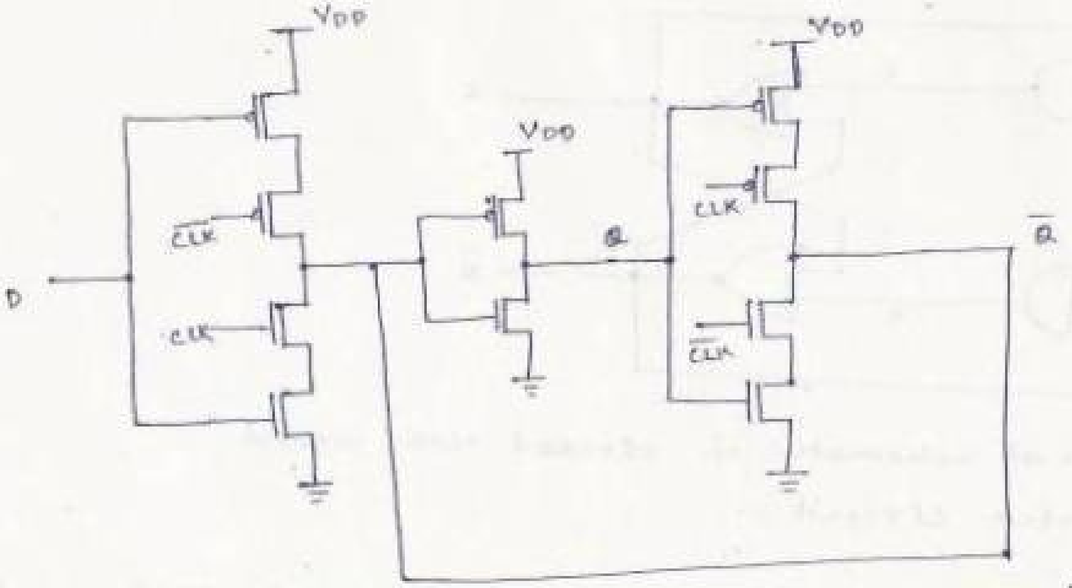
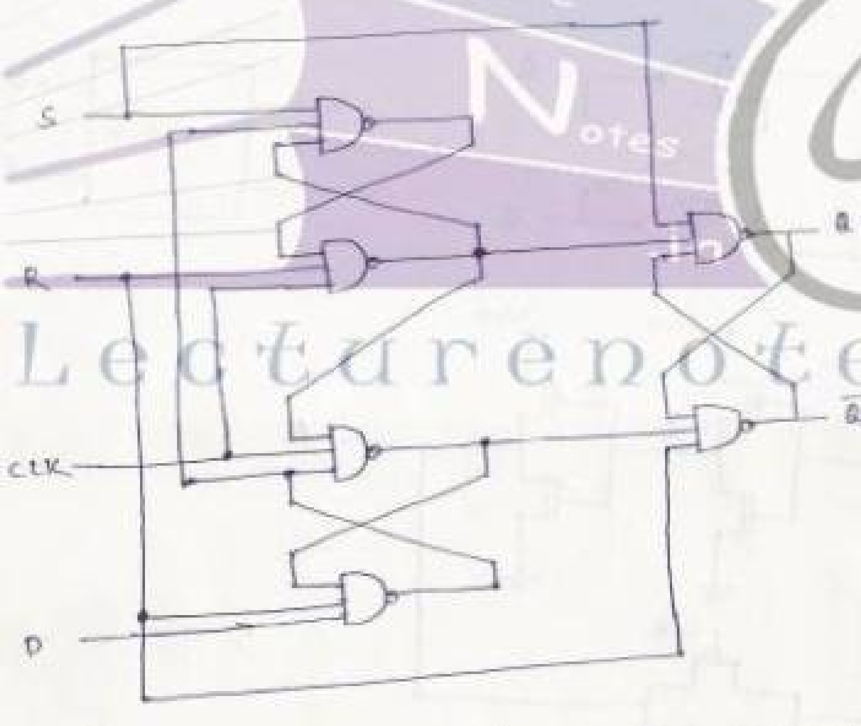


(CMOS implementation of D-latch using 2 transmission gates)



(Alternative CMOS implementation of D-latch)

(figure 8.30, CMOS negative (falling) edge-triggered master-slave D flip-flop, page 346.)



(NAND3-based positive edge-triggered D flip-flop)

(Figure 8.35, page 349, Timing diagram of the positive edge-triggered D flip-flop)

—o—

LECTURENOTES.IN

VLSI Design

Topic:

Dynamic MOS Logic Circuits

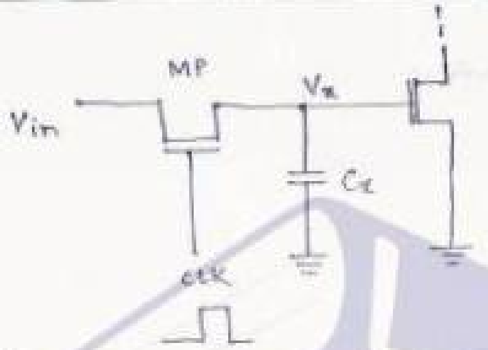
Contributed By:

Verified Writer

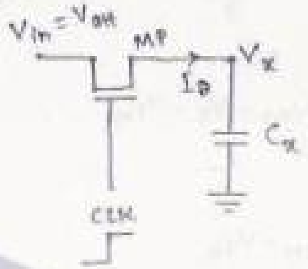
ch-8: Dynamic logic circuits

Introduction:- The static combinational & sequential logic circuit requires a large number of transistors and hence causes more time delay. So in high-density & high performance logic implementation, dynamic logic circuits offers minimum circuit delay and silicon area

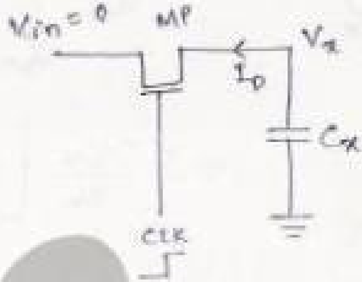
Basic principles of pass Transistor circuits:-



(basic building block for nmos dynamic logic, where nmos pass transistor driving the gate of another nmos transistor)



(Equivalent circuit for the logic '1' transfer)



(Equivalent cir for the logic '0' transfer)

Logic '1' Transfer:- let $V_x(\text{at } t=0) = 0V$

$V_{in} = V_{OH} = V_{DD}$

& clk goes from 0 to V_{DD} at $t=0$

Then $V_{DS} = V_{DD} - V_x \simeq V_{DD}$ at $t=0$

$V_{GS} = V_{DD} - V_x \simeq V_{DD}$ at $t=0$

$\Rightarrow V_{DS} > V_{GS} - V_{Tn}$, Transistor MP operates in Saturation r

Then $I_D = C_x \cdot \frac{d(V_x - 0)}{dt} = \frac{k_n}{2}\left[V_{GS} - V_{Tn}\right]^2 = \frac{k_n}{2}\left[V_{DD} - V_x - V_{Tn}\right]^2$

$$\Rightarrow dt = \frac{2C_x}{k_n} \cdot \frac{dV_x}{(V_{DD} - V_x - V_{Tn})^2}$$

integrating both the sides

$$\int_0^t dt = \frac{2C_x}{K_n}\int_0^{V_x} \frac{dV_x}{(V_{DD}-V_x-V_{Tn})^2}$$

let $V_{DD}-V_x-V_{Tn} = y$

$dy = -dV_x$

$$\int_0^t dt = \frac{2C_x}{K_n}\int_{V_{DD}-V_{Tn}}^{V_{DD}-V_x-V_{Tn}} -\frac{dy}{y^2} = \cancel{\frac{2C_x}{K_n}\int_{V_{DD}-V_{Tn}}^{V_{DD}-V_x-V_{Tn}}}$$

$$= -\frac{2C_x}{K_n}\left[\frac{y^{-1}}{-1}\right]_{V_{DD}-V_{Tn}}^{V_{DD}-V_x-V_{Tn}}$$

$$\Rightarrow t = \frac{2C_x}{K_n}\left[\frac{1}{V_{DD}-V_x-V_{Tn}}\right]_0^{V_x}$$

$$\boxed{t = \frac{2C_x}{K_n}\left[\left(\frac{1}{V_{DD}-V_x-V_{Tn}}\right)-\left(\frac{1}{V_{DD}-V_{Tn}}\right)\right]}$$

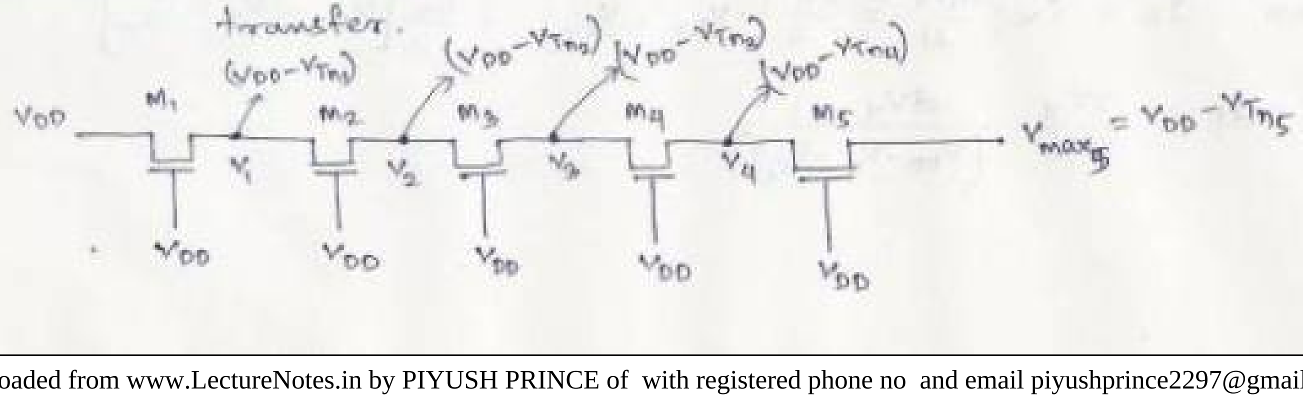
Solving this expression for $V_x(t)$ we get

$$\boxed{V_x(t) = (V_{DD}-V_{Tn})\frac{\left\{\frac{K_n(V_{DD}-V_{Tn})}{2C_x}\right\}t}{1+\left\{\frac{K_n(V_{DD}-V_{Tn})}{2C_x}\right\}t}}$$

The maximum voltage for $V_x(t)$ can be obtained when $t \to \infty$,

$$V_{max} = V_x(t)\Big|_{t\to\infty} = V_{DD}-V_{Tn} = V_{DD}-V_{T0n}-\gamma\left\{\sqrt{|2\phi_F|+V_{max}}-\sqrt{|2\phi_F|}\right.$$

<u>case-I</u> :- Node voltages in pass-transistor chain during logic '1' transfer.



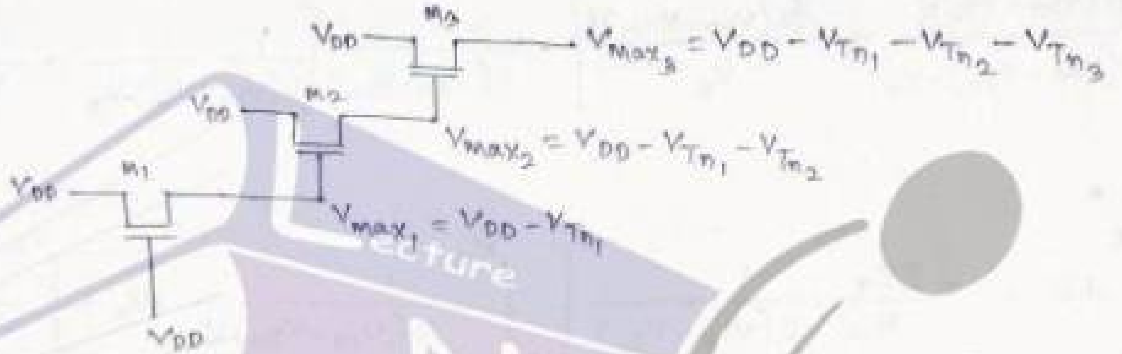
Here $V_{max_1} = V_{DD} - V_{Tn_1}$, $V_{Tn_1} = V_{Ton} - \gamma\left[\sqrt{|2\phi_F| + V_{max_1}} - \sqrt{|2\phi_F|}\right]$

$V_{max_2} = V_{DD} - V_{Tn_2}$, $V_{Tn_2} = V_{Ton} - \gamma\left[\sqrt{|2\phi_F| + V_{max_2}} - \sqrt{|2\phi_F|}\right]$

$V_{max_3} = V_{DD} - V_{Tn_3}$, $V_{Tn_3} = V_{Ton} - \gamma\left[\sqrt{|2\phi_F| + V_{max_3}} - \sqrt{|2\phi_F|}\right]$

$\vdots \qquad\qquad \vdots$

Case II :- Node voltages during logic '1' transfer, when each pass transistor is driving another pass transistor.



logic '0' Transfer :- V_x (at $t=0$) $= V_{max} = V_{DD} - V_{Tn}$

$V_{in} = 0V$

& clk goes from 0 to V_{DD} at $t=0$

$V_{DS_{max}} = V_{max} - 0 = V_{max} = V_{DD} - V_{Tn}$, $V_{DS} = V_x - 0$ i.e.

$V_{DS} \le V_{DD} - V_{Tn}$

& $V_{GS} = V_{DD} - 0 = V_{DD}$ } at $t = 0$

$\Rightarrow$ $V_{DS} < V_{GS} - V_{Tn}$, Transistor MP is operating in Linear region

Then $I_D = C_x \cdot \frac{d(0 - V_x)}{dt} = \frac{K_n}{2}\left[2(V_{DD} - V_{Tn})V_x - V_x^2\right]$ at time 't'

$\Rightarrow dt = -\frac{2C_x}{K_n} \cdot \frac{dV_x}{2(V_{DD} - V_{Tn})V_x - V_x^2}$

integrating both sides we get

$$\int_0^t dt = -\frac{2C_x}{K_n}\int_{V_{DD}-V_{Tn}}^{V_x} \frac{dV_x}{2(V_{DD}-V_{Tn})V_x - V_x^2} \quad \text{--- (1)}$$

again $$\frac{1}{2(V_{DD}-V_{Tn})V_x - V_x^2} = \frac{2(V_{DD}-V_{Tn})/2(V_{DD}-V_{Tn})}{[2(V_{DD}-V_{Tn})\cancel{V_x} - V_x^{\cancel{2}}]V_x}$$

$$= \frac{[V_x + 2(V_{DD}-V_{Tn}) - V_x]/2(V_{DD}-V_{Tn})}{[2(V_{DD}-V_{Tn})\cancel{V_x} - V_x^{\cancel{2}}]V_x} = \frac{V_x/2(V_{DD}-V_{Tn}) + \{2(V_{DD}-V_{Tn})-V_x\}/2(V_{DD}-V_{Tn})}{[2(V_{DD}-V_{Tn}) - V_x]V_x}$$

$$= \left[\frac{1/\{2(V_{DD}-V_{Tn})\}}{2(V_{DD}-V_{Tn})-V_x} + \frac{1/\{2(V_{DD}-V_{Tn})\}}{V_x}\right] = \frac{1}{2(V_{DD}-V_{Tn})}\left[\frac{1}{2(V_{DD}-V_{Tn})-V_x} + \frac{1}{V_x}\right]$$

$\Rightarrow$ eq^n (1)

$$\int_0^t dt = -\frac{2C_x}{K_n}\cdot\frac{1}{2(V_{DD}-V_{Tn})}\left[\int_{V_{DD}-V_{Tn}}^{V_x}\frac{1}{2(V_{DD}-V_{Tn})-V_x}dV_x + \int_{V_{DD}-V_{Tn}}^{V_x}\frac{1}{V_x}dV_x\right]$$

let $2(V_{DD}-V_{Tn}) - V_x = Y$, $\frac{dY}{dV_x} = -1 \Rightarrow dY = -dV_x$

$$\Rightarrow t = \frac{-2C_x}{2K_n(V_{DD}-V_{Tn})}\left[\int\frac{1}{V_x}dV_x - \int\frac{1}{Y}dY\right]$$

$$= \frac{-2C_x}{2K_n(V_{DD}-V_{Tn})}\left[\ln V_x\Big|_{V_{DD}-V_{Tn}}^{V_x} - \ln Y\Big|_{V_{DD}-V_{Tn}}^{2(V_{DD}-V_{Tn})-V_x}\right]$$

$$= \frac{2C_x}{2K_n(V_{DD}-V_{Tn})}\left[\ln\frac{2(V_{DD}-V_{Tn})-V_x}{V_x}\right]_{V_{DD}-V_{Tn}}^{V_x}$$

$$= \frac{C_x}{K_n(V_{DD}-V_{Tn})}\left[\ln\frac{2(V_{DD}-V_{Tn})-V_x}{V_x} - \ln\frac{2(V_{DD}-V_{Tn})-(V_{DD}-V_{Tn})}{(V_{DD}-V_{Tn})}\right]$$

$$\Rightarrow t = \frac{C_x}{k_n(V_{DD}-V_{Tn})}\left[\ln\frac{2(V_{DD}-V_{Tn})-V_x}{V_x} - \cancelto{0}{\ln\frac{(V_{DD}-V_{Tn})}{(V_{DD}-V_{Tn})}}\right]$$

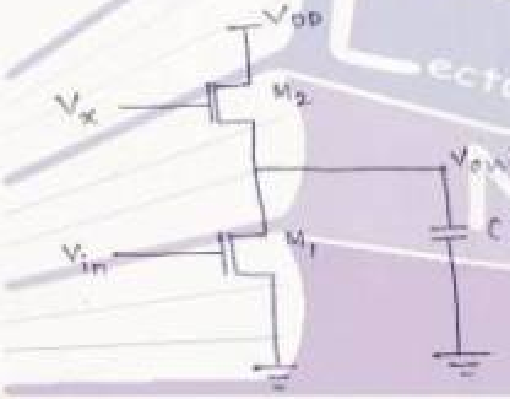
$$\Rightarrow \boxed{t = \frac{C_x}{k_n(V_{DD}-V_{Tn})}\cdot\ln\frac{2(V_{DD}-V_{Tn})-V_x}{V_x}}$$

Alternate Approach :-

$$\int\frac{1}{x^2-a^2}\,dx = \begin{cases}\frac{1}{2a}\ln\frac{a-x}{a+x}, & |x|<|a|\\ \frac{1}{2a}\ln\frac{x-a}{x+a}, & |x|>|a|\end{cases}$$

Voltage Bootstrapping :-

It is a technique for overcoming threshold voltage drops in digital circuits.

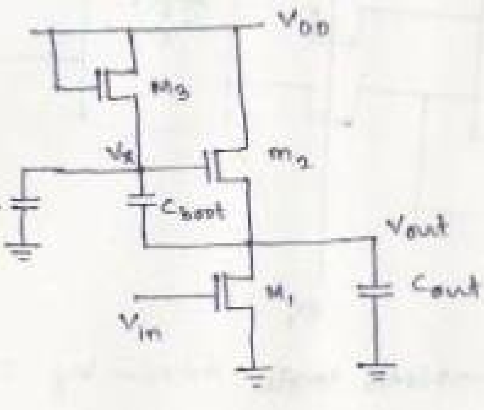


(Enhancement-type circuit where output node is weakly driven)

Here $V_x \leq V_{DD}$ for which M_2 operates in saturation region.

When the input voltage V_{in} is low, the maximum value of outp

$$V_{out}(max) = V_x - V_{T_2}(V_{out})$$



$$V_x \geqslant V_{DD} + V_{T_2}(V_{out})$$
$$\& \; V_x = V_{DD} - V_{T_3}(V_x)$$

Assuming $i_{C_S} \simeq i_{C_{boot}}$

$$\Rightarrow C_S \cdot \frac{dV_x}{dt} \simeq C_{boot} \cdot \frac{d(V_{out} - V_x)}{dt}$$

$$\Rightarrow (C_S + C_{boot}) \frac{dV_x}{dt} \simeq C_{boot} \frac{dV_{out}}{dt}$$

$$\Rightarrow \frac{dV_x}{dt} = \frac{C_{boot}}{C_S + C_{boot}} \frac{dV_{out}}{dt} \Rightarrow dV_x = \frac{C_{boot}}{C_S + C_{boot}} \cdot dV_{out}$$

integrating V_x from $(V_{DD} - V_{T3})$ to V_x and integrating V_{out} from V_{OL} to V_{DD} we get

$$\int_{V_{DD}-V_{T3}}^{V_x} dV_x = \left[\frac{C_{boot}}{C_S + C_{boot}}\right] \int_{V_{OL}}^{V_{DD}} dV_{out}$$

$$\Rightarrow V_x - (V_{DD} - V_{T3}) = \frac{C_{boot}}{C_S + C_{boot}} (V_{DD} - V_{OL})$$

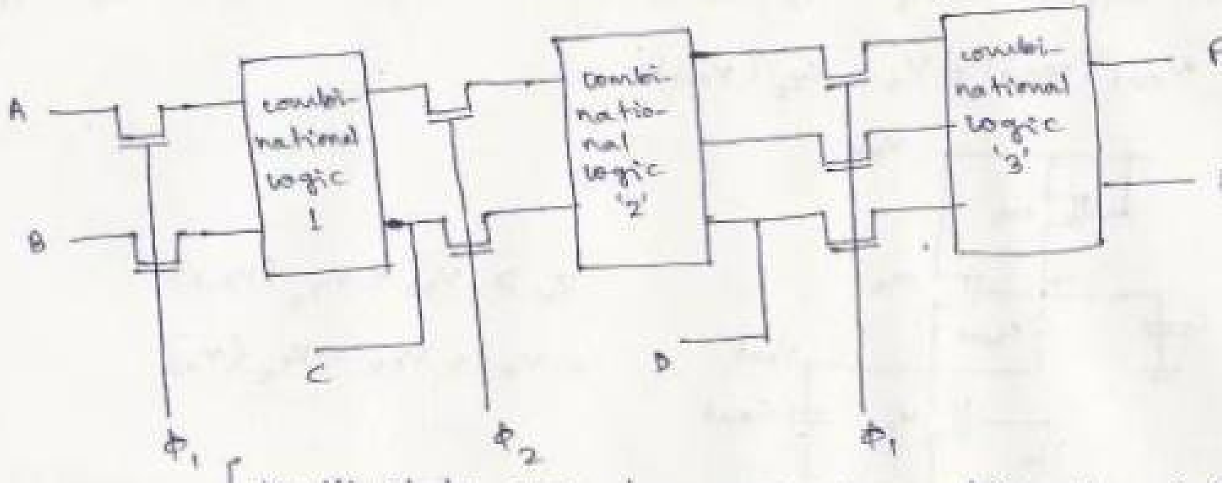
$$\Rightarrow \boxed{V_x = (V_{DD} - V_{T3}) + \left[\frac{C_{boot}}{C_S + C_{boot}}\right] \cdot (V_{DD} - V_{OL})}$$

if $C_{boot} >> C_S$, then

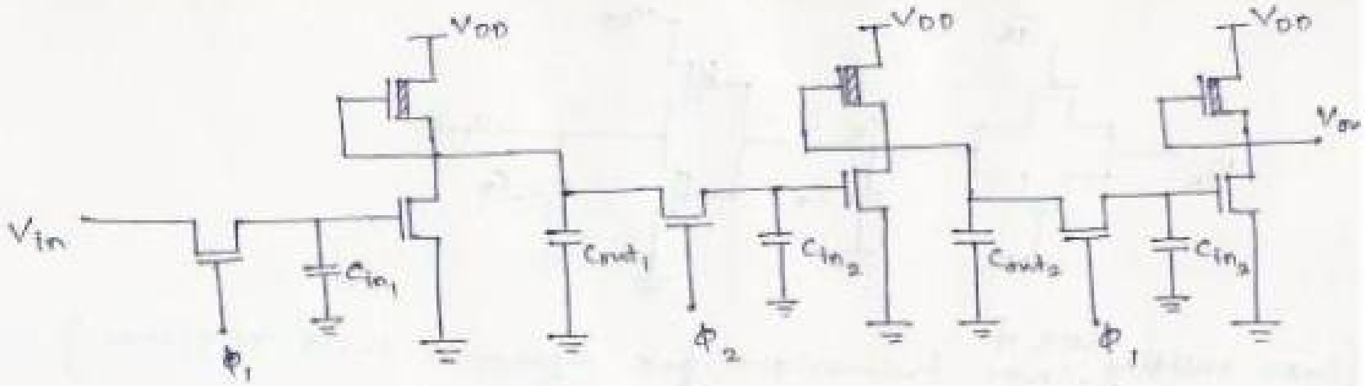
$V_x(max) = V_{DD} - V_{T3} + V_{DD} - V_{OL} = 2V_{DD} - V_{T3} - V_{OL}$

which can be compared with previous $V_{x_{max}} = V_{DD} - V_{Tn}$

Synchronous Dynamic circuit Techniques :-



[multi-state pass transistor logic driven by 2 non-overlapping clocks.]



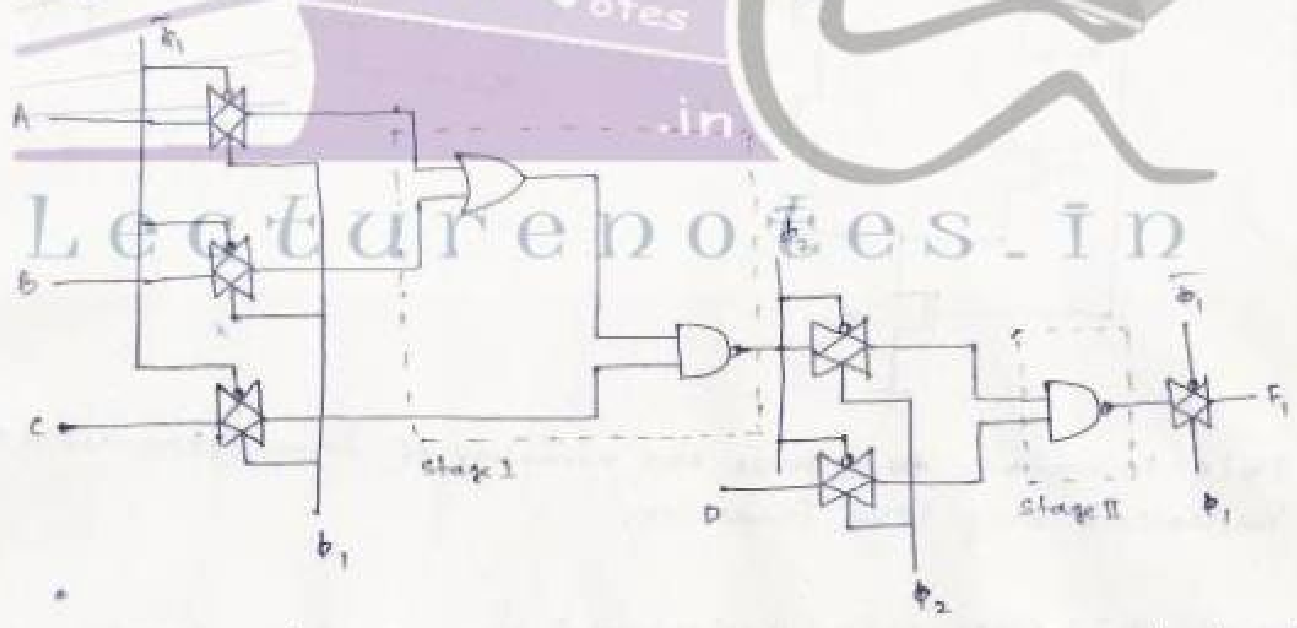
(3-stage depletion-load nMOS dynamic shift register driven with 2- clock signals ϕ_1 & ϕ_2)

figure 9.17, 9.18 & 9.19, Page 376, 377.

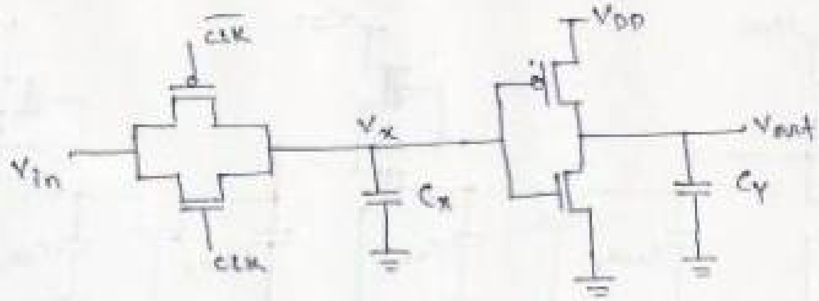
Dynamic cmos circuit techniques :-

cmos transmission gate logic :-

In this design, the static cmos gates are used for implementing the logic blocks whereas the cmos transmission gates are used for transferring the output levels of one stage to the inputs of the next stage.



(Example of Dynamic cmos transmission gate logic)

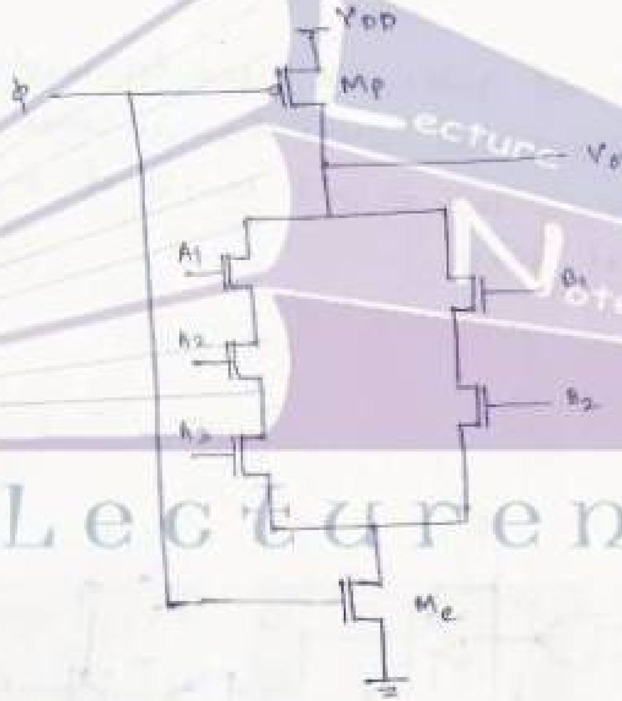


[Basic building block of CMOS transmission gate dynamic shift register.]

(Figure 9.25, Page 381.)

Dynamic CMOS logic (Precharge - Evaluate logic) :-

(figure 9.26, Page 381, Dynamic CMOS logic gate implementing a complex Boolean function.)



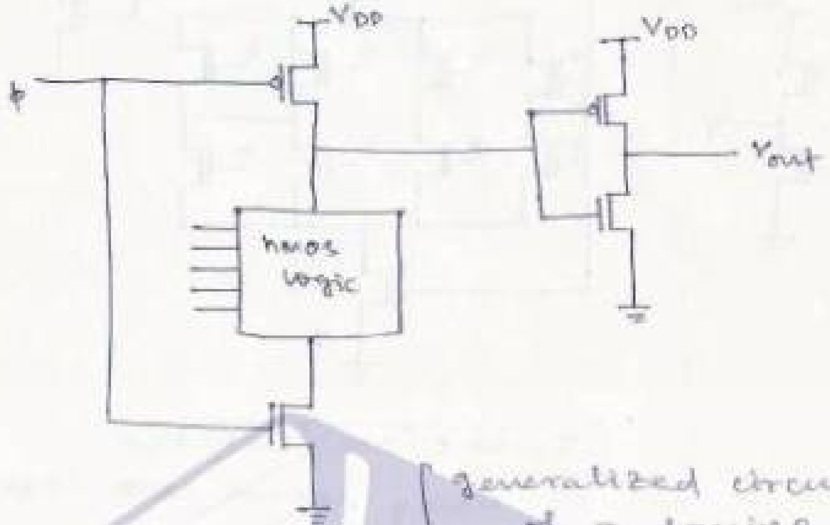
This technique reduces the number of transistors used to implement any logic function.

(Figure 9.27, Page 382: Illustration of the cascading problem in dynamic CMOS logic.)

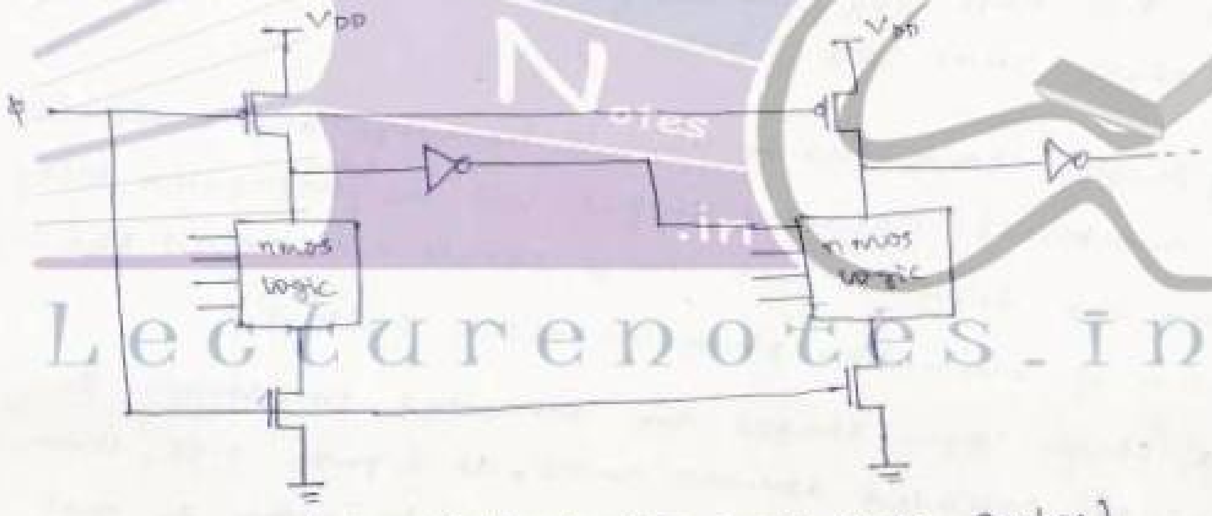
→ This figure illustrates that the dynamic CMOS logic stages driven by the same clock signal cannot be cascaded directly. So alternative clocking schemes/circuit structure must be developed to overcome this problem.

High performance Dynamic CMOS circuits:-

Domino CMOS logic:- It is the cascaded form of a dynamic CMOS logic circuit with a static CMOS ~~and~~ inverter circuit



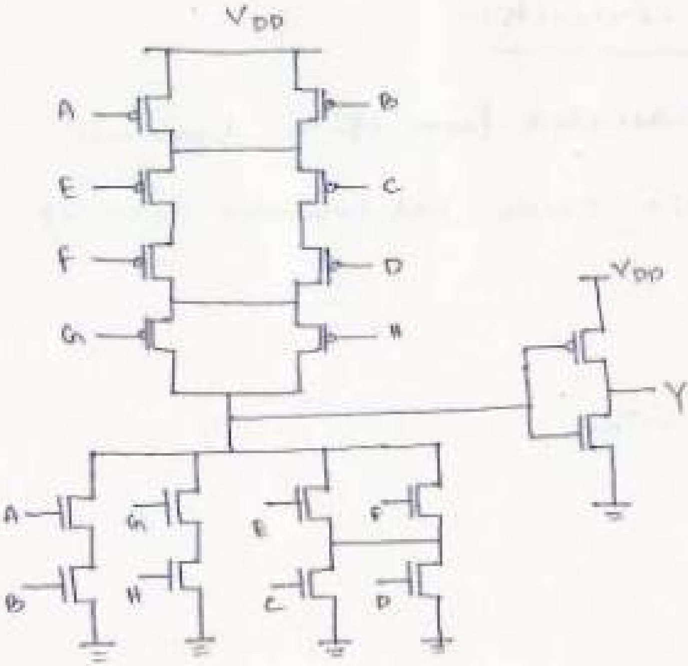
[generalized circuit diagram of a domino cmos logic circuit]

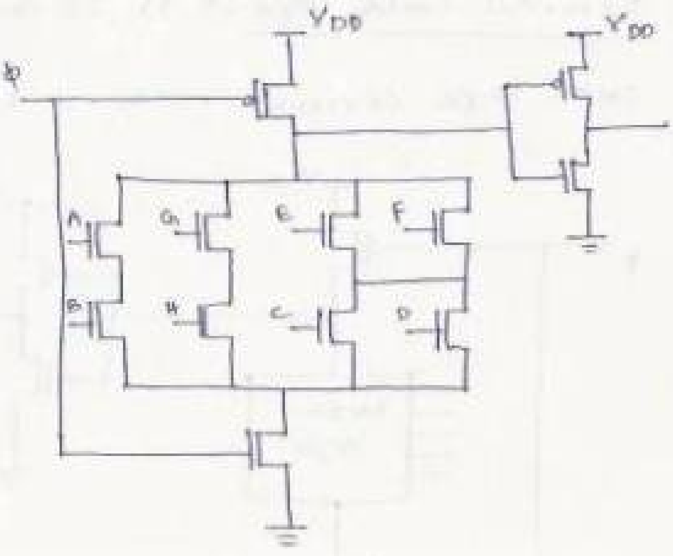


(cascaded domino cmos logic gates)

Domino cmos logic implementation for the expression

$$Y = AB + (C+D)(E+F) + GH$$





($Y = \overline{AB + (C+D)(E+F) + GH}$ using cmos logic)

($Y = \overline{AB + (C+D)(E+F) + GH}$ using domino cmos logic)

→ (figure 9.31, page 386, cascading domino cmos logic gates with static cmos logic gates.)

→ A ~~seg~~ single clock can be used to precharge and evaluate any number of cascaded stages, but the total propagation delay between the first & last stage should not exceed the duration of evaluation phase (in clock pulse ϕ).

→ when static logic stages are connected in between the stages of cascaded domino cmos, as figure 9.31, then the number of static stages should be even so that the input of the next domino cmos stage experience only '0' to '1' transitions during evaluation stage.

(figure 9.35, page 389, Example of a multiple-output domino cmos gate realizing 4 functions.)

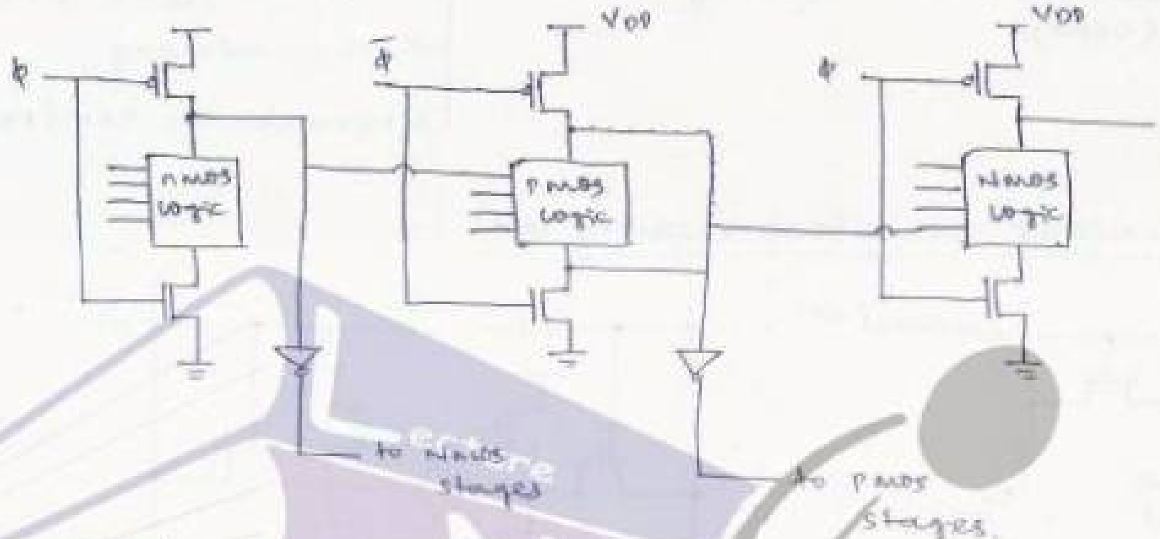
$$C_1 = G_1 + P_1 C_0$$
$$C_2 = G_2 + P_2 G_1 + P_2 P_1 C_0$$
$$C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_0$$
$$C_4 = G_4 + P_4 G_3 + P_4 P_3 G_2 + P_4 P_3 P_2 G_1 + P_4 P_3 P_2 P_1 G_0$$

$G_i = A_i \cdot$
$P_i = A_i \oplus$

NORA CMOS Logic (NP-Domino Logic) :-

In domino CMOS logic circuits, all the logic operations are performed by the NMOS transistors in pull-down phase, whereas the PMOS transistor is only used to precharge the dynamic nodes.

Alternatively in NORA CMOS logic circuits we have alternate NMOS & PMOS logic networks for evaluation phases.



Advantages of NORA CMOS circuit :-

1) It donot require a static CMOS inverter after every stage as in ~~dynam~~ domino CMOS logic.

2) It allows pipelined system architecture.

figure 9.39 : page 395 - NORA CMOS logic circuit example.

Disadvantage: - (NORA logic)

1) It suffers from charge sharing problem & leakage problem

Zipper CMOS Circuits :-

Similar to NORA CMOS logic with different clock signal arrangement.

figure 9.40, page 397 : circuit diagram & clock signal for zipper CMOS circuit.

True singl-phase clock (TSPC) Dynamic CMOS : figure 9.41, Page :

VLSI Design

Topic:

Semiconductor Memories

Contributed By:

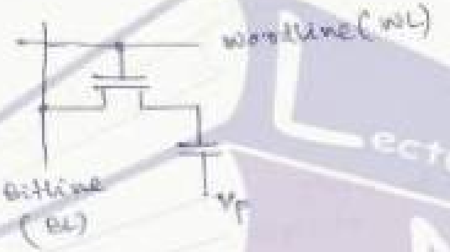
Verified Writer

chapter-9

Semiconductor memories

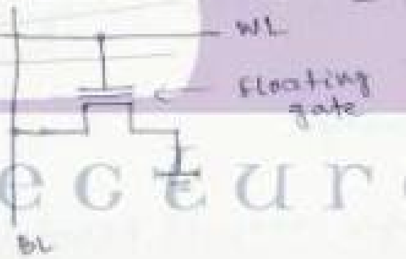
- R/W memory or RAM (Random Access memory)
  - Dynamic RAM (DRAM)
  - Static RAM (SRAM)
- ROM (Read-only memory)
  - Mask (fuse) ROM
  - Programmable ROM (PROM)
    - Erasable PROM (EPROM)
    - Electrically Erasable PROM (EEPROM)
  - Flash memory
  - Ferroelectric RAM (FRAM)

Equivalent circuits of memory cells :-

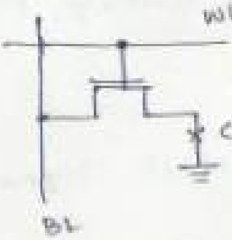


(DRAM circuit)

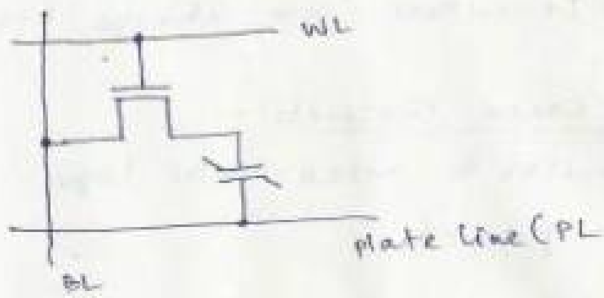
(SRAM circuit)



(EEPROM circuit)



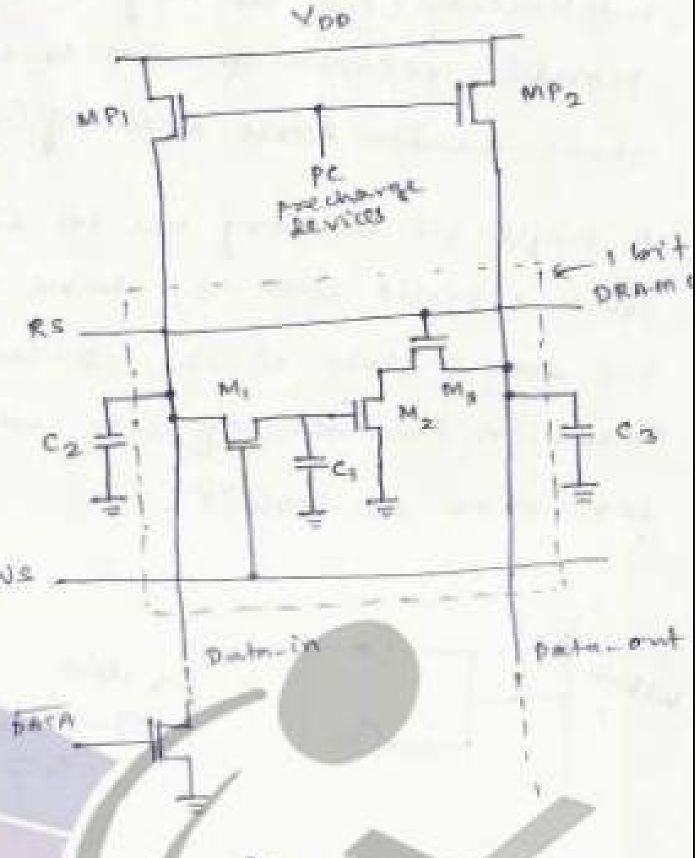
(mask (fuse) ROM circuit)



(FRAM circuit)

Dynamic Random Access memory (DRAM):-

operation of 3-transistor DRAM cell :-



(3-transistor DRAM cell with pull-up and R/W circuitry)

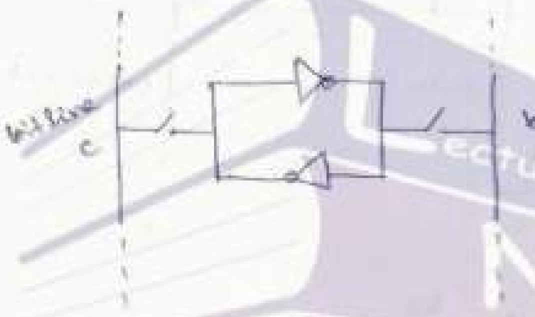
* The binary information is stored in the form of charge in the parasitic node capacitance 'C_1'.
* The storage transistor 'M_2' is turned on or off depending on the charge stored in 'C_1'. Whereas the pass transistors M_1 & M_3 acts as access switches for data read and data write operations.
* The precharge event is driven by clock signal 'ϕ_1' and the "read" and "write" events are driven by clock signal 'ϕ_2'.
* Every "data read" and "data write" operation is preceded by a precharge cycle which activates the column pull-up transistors MP_1 & MP_2 and the corresponding capacitance C_2 & C_3 are charged up to logic-high level.
* All the "data read" and "data write" operations are performed during clock signal 'ϕ_2' phase as high.

(Figure 10.9: Page 418, voltage waveforms associated with 3-transistor DRAM.)

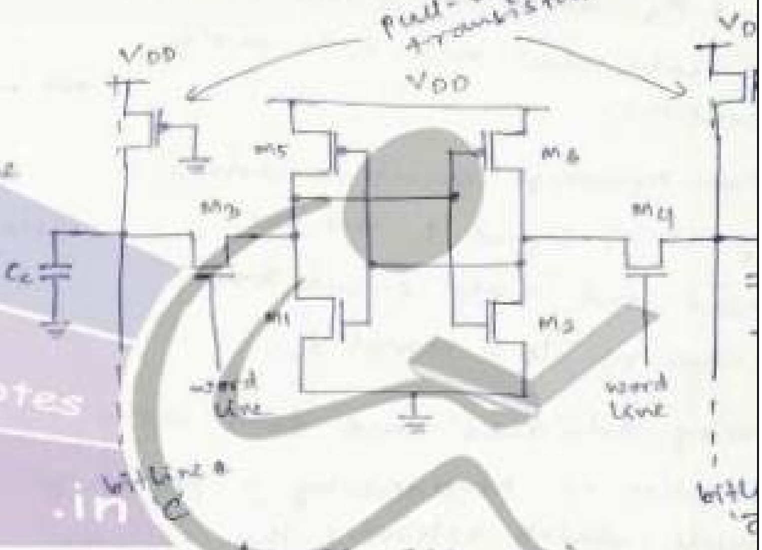
- Static Random Access memory (SRAM):-

If the stored data bits are retained by the memory cir indefinitely (i.e for any period of time) without any need fo periodic refresh operation to the RAM circuit, then it is calle Static Random Access memory (SRAM).

A single-bit memory cell in static RAM consists of a simple latch circuit with 2 stable operating states. depending on the stored state of the 2-inverter latch circuit, the data stored in the memory cell will be interpreted either as logic '0' or as logic '1'.



(1 bit SRAM cell)



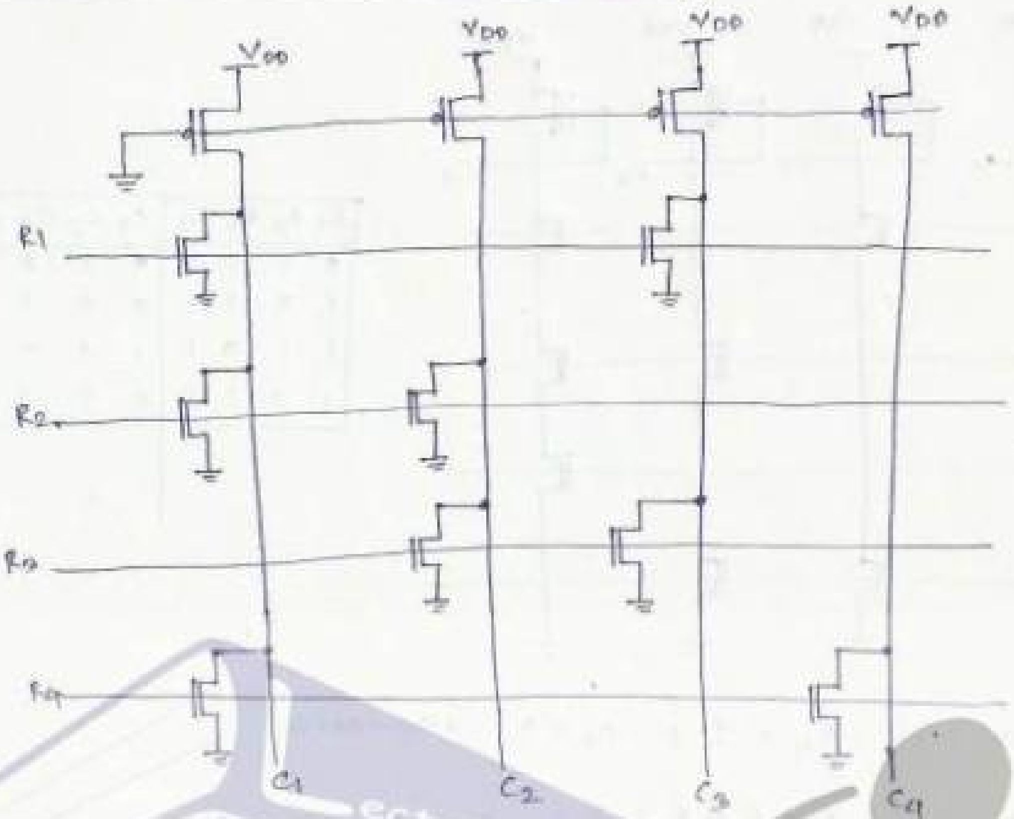
(CMOS SRAM cell)

(figure 10.33: page 443, voltage levels for SRAM cell for read operation)

(figure 10.34: page 444, voltage levels for SRAM cell for write operation)

Nonvolatile memory:- These are the memory which has an advantage of storing the data in the absence of power supply which is one of the disadvantage of MOS memory structures like DRAM and SRAM. flash memory based on floating-gate concept is most popular nonvolatile memory due to its small cell size and better functionality.

4 bit x 4 bit NOR Based Rom Array :-



R_1	R_2	R_3	R_4	C_1	C_2	C_3	C_4
1	0	0	0	0	1	0	1
0	1	0	0	0	0	1	1
0	0	1	0	1	0	0	1
0	0	0	1	0	1	1	0

$C_1 = 1$ (V_{DD}), when R_1, R_2 & R_4 are '0' i.e off.

$C_1 = 0$ (ground), when atleast one of R_1, R_2 & R_4 is '1'.

$C_2 = 1$ (V_{DD}), when R_2 & R_3 are '0' i.e corresponding transistors are off.

$C_2 = 0$ (ground), when atleast one of R_2 and R_3 is '1'.

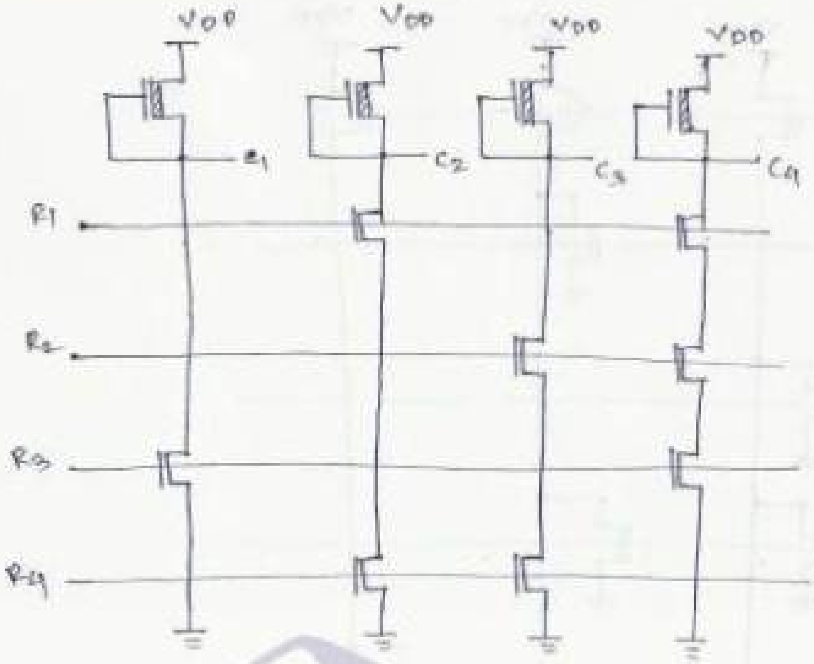
$C_3 = 1$ (V_{DD}), when R_1 & R_3 are '0' i.e off.

$C_3 = 0$ (gnd), when atleast one of R_1 & R_3 is '1'.

$C_4 = 1$ (V_{DD}), when R_4 is '0'.

$C_4 = 0$ (gnd), when R_4 is '1'.

4-bit × 4-bit NAND-based ROM array



R_1	R_2	R_3	R_4	C_1	C_2	C_3	C_4
0	1	1	1	0	1	0	1
1	0	1	1	0	0	1	1
1	1	0	1	1	0	0	1
1	1	1	0	0	1	1	0

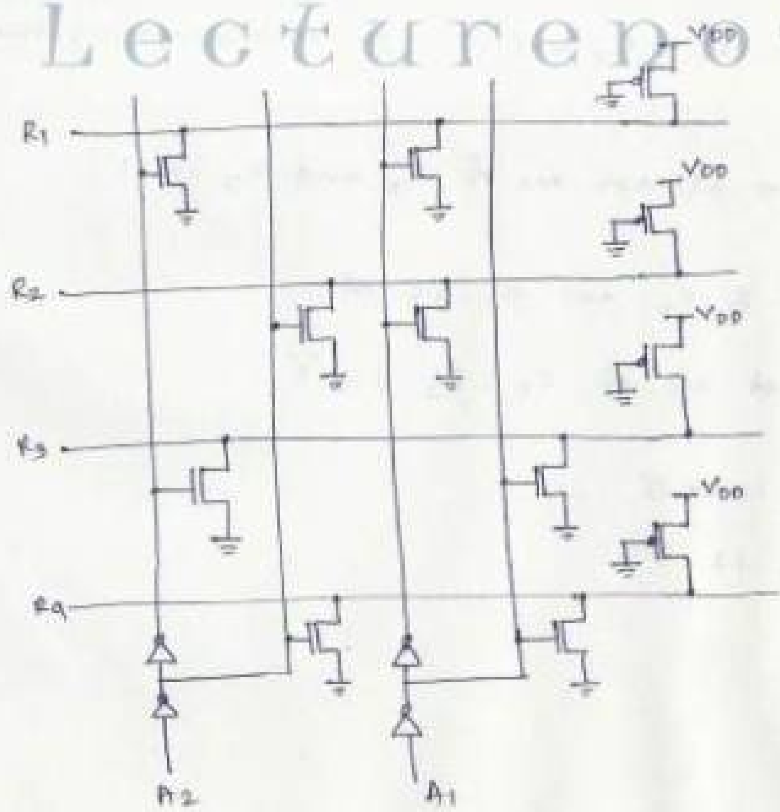
a) $C_1 = 0$ if $R_3 = 1$; $C_1 = 1$, otherwise

b) $C_2 = 0$ if $R_1 = 1$ & $R_4 = 1$; $C_2 = 1$, otherwise

c) $C_3 = 0$ if $R_2 = 1$ & $R_4 = 1$; $C_3 = 1$ otherwise

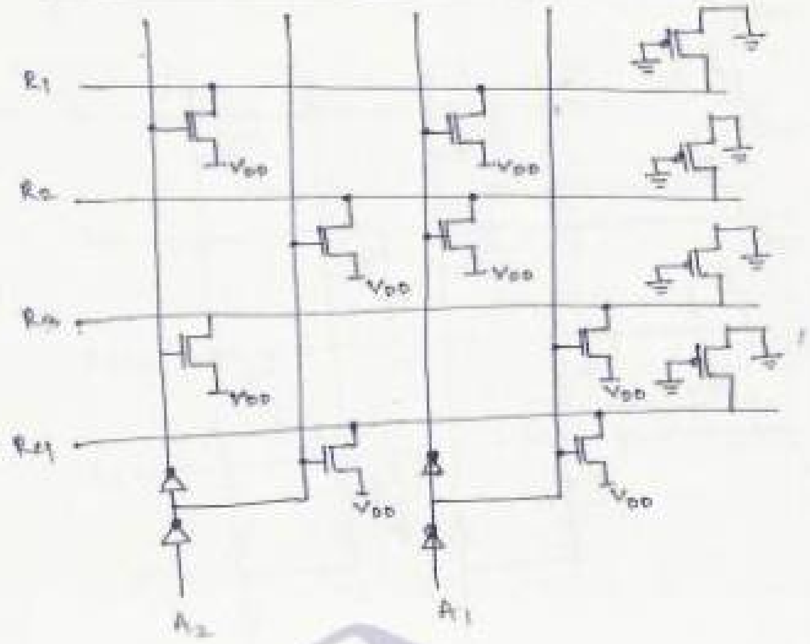
d) $C_4 = 0$ if $R_1 = 1$ & $R_2 = 1$ & $R_3 = 1$; $C_4 = 1$, otherwise.

Row Address Decoder NOR-Based:



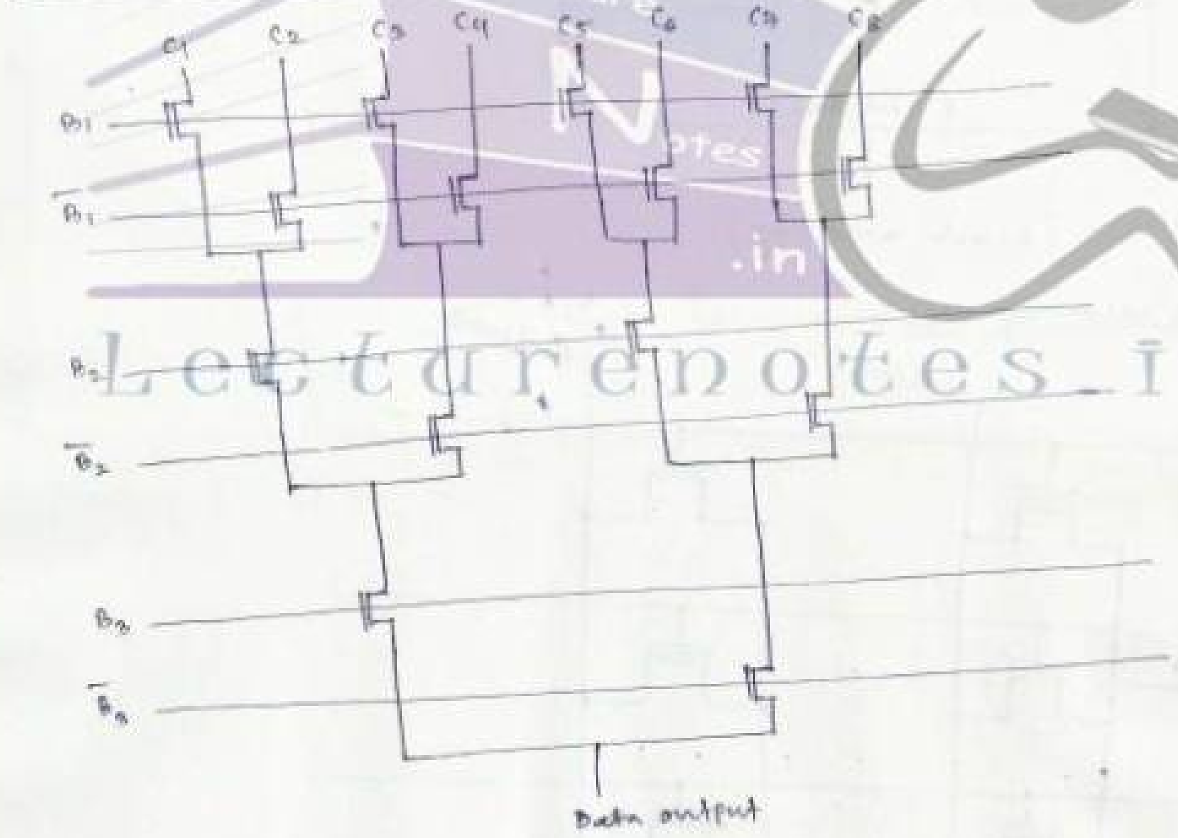
A_1	A_2	R_1	R_2	R_3	R_4
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

NAND-based row-address decoder



A_1	A_2	R_1	R_2	R_3	R_4
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

nmos pass transistor based column-address decoder

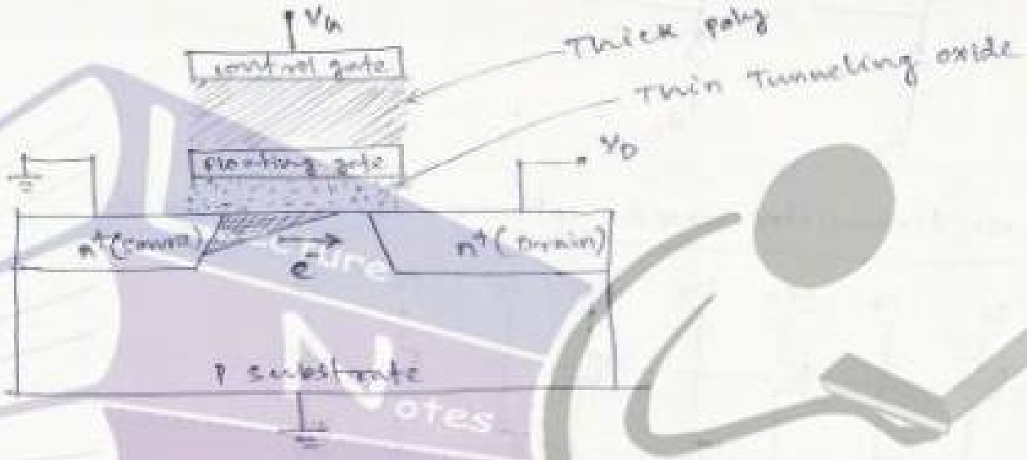


Flash memory :-

Flash memory cell consists of one transistor with a floating gate whose threshold voltage can be changed by applying an electric field to its gate.

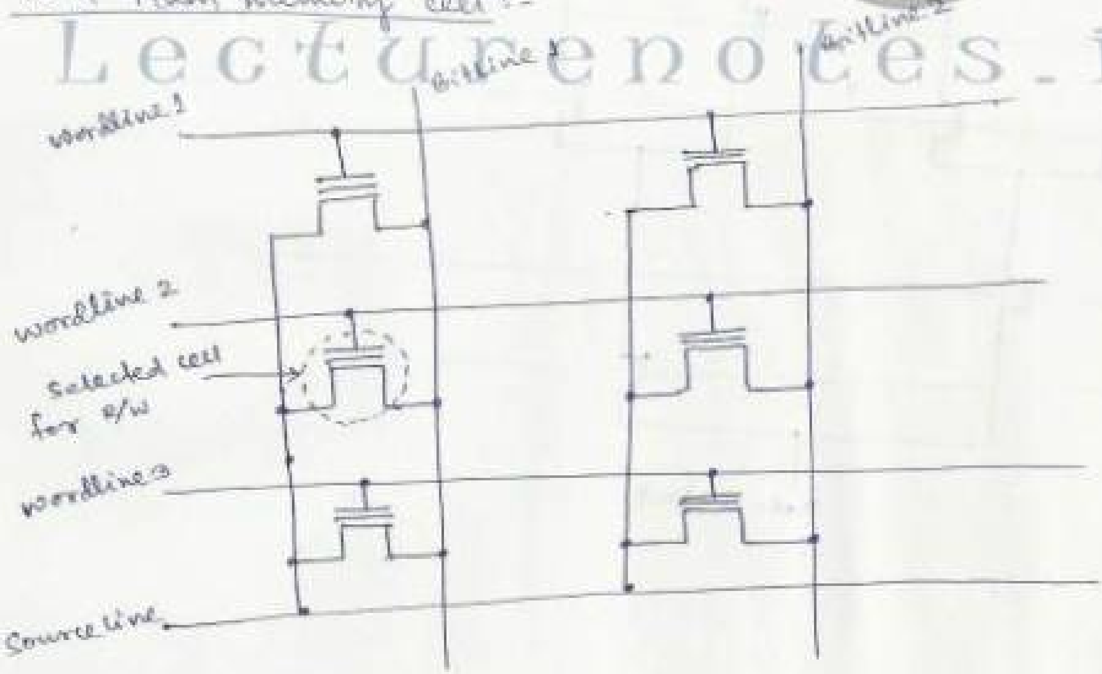
→ when electrons are accumulated at the floating gate, the threshold voltage of the memory cell becomes higher and the memory cell is considered to be in '1' state.

→ The threshold voltage of the memory cell can be lowered by removing electrons from the floating gate and the memory cell is regarded to be in '0' state.



(Flash memory)

NOR Flash memory cell :-



→ During "erase" operation, 0V and high voltage are simultaneously applied to all the control gates (wordlines) and the sources of the cells respectively. So any electrons present in the floating gate will move to the source & hence the data in all the cells are erased.

→ During "write" operation, high voltages are applied to the control gate and drain of the selected cell (12V to control gate and 8V to the drain terminal). Due to this, hot electrons are generated at drain and are injected into the floating gate which causes a high threshold voltage at the gate terminal (logic '1'). The remaining transistor will show logic '0'.

→ During 'read' operation, a moderate voltage is applied to the control gate (say V_{DD}) and a small voltage (say 1V) is applied to drain.

* When the cell data is '0' (low threshold value) & the transistor will turn on and current flows through the cell transistor.
* When cell data is '1' (high threshold voltage), the transistor will be turn-off & no current flows through the cell transistor.

NAND flash memory cell :-

(figure 10.63, Page 469)

Ferroelectric Random Access memory (FRAM)

Ferroelectric RAM uses a ferroelectric capacitor where the dielectric material of the typical capacitor is replaced by a ferroelectric material such as $Pb(Zr_xTi_{1-x})O_3$ and $SbBi_2Ta_2O_9$.

The total charge of the ferroelectric capacitor varies as a function of the applied voltage and it exists in the absence of electric field also. This is because of the spontaneous polarization in the crystal structure. The net change in the direction of ~~pop~~ polarization happens when the applied electric field is larger than V_c.

(Figure 10.67, page 474: structure of FRAM)

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ch-10

Design for Testability

Introduction:- It is a technique of determining the faults present in a fabricated chip. Since the number of transistors integrated into a single chip increases day by day, the task of chip testing to ensure the correct functionality becomes more ~~xxx~~ complex and time-consuming. It is seen that complexity and time-consumption for testing a chip is tenfolds from that of a circuit level test and that of system level is tenfolds as that of chip level testing, it is highly required to test the design in the early stage.

Fault Types and models:-

Faults caused by physical defects:-

1> Defects in silicon substrate
2> photolithographic defects
3> mask contamination and scratches
4> process variations & abnormalities
5> oxide defects.

The physical defects can cause electrical as well as logical faults.

Electrical faults:-

1> shorts (Bridging faults)
2> opens
3> Transistor stuck-on, & stuck-open.
4> Resistive shorts and opens.
5> Excessive change in threshold voltage
6> Excessive steady-state currents.

VLSI Design

Topic:

Design For Testability

Contributed By:

Verified Writer

direction of ~~pop~~ polarization happens when the applied electric field is larger than V_c.

(Figure 10.67, page 474: structure of FRAM)

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ch-10 **Design for Testability**

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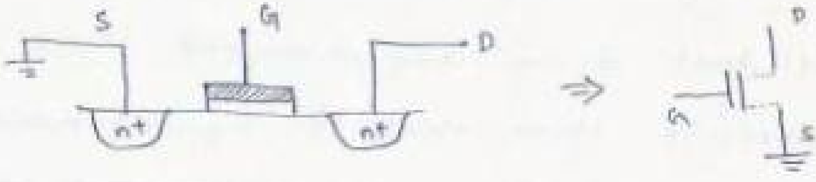
The physical defects can cause electrical as well as logical faults.

Electrical faults:-

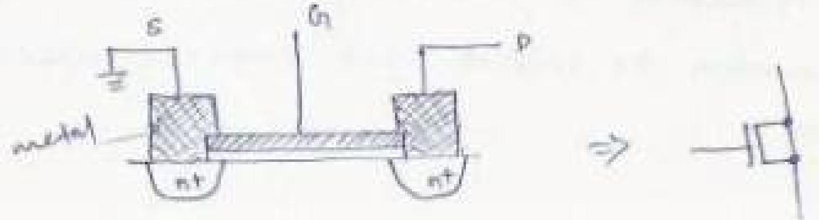
1> shorts (Bridging faults)
2> opens
3> Transistor stuck-on, & stuck-open.
4> Resistive shorts and opens.
5> Excessive change in threshold voltage
6> Excessive steady-state currents.

Logical faults:-

1) Logical stuck-at-0 or stuck-at-1
2) Slower transition (Delay fault)
3) AND-Bridging, OR Bridging



Stuck-open or stuck-off fault.



stuck-on or stuck-short fault

Ad Hoc Testable Design Techniques:-

The testability of a chip can be increased by inserting more access circuits to the original design. Some of the techniques are:

1) Partition-and-Mux Technique:-

A design with many serial gates, functional blocks or large circuits are difficult to test, such circuits are partitioned and several such small blocks or partitions are multiplexed.

Such type of techniques are very useful in fault testing with higher number of accessible nodes.

2) Initialize Sequential Circuit:-

When the sequential circuit is powered up, its initial state can be random unknown state. So it is not possible to start the test sequence correctly. The sta of a sequential circuit can be brought to a known state through "initialization".

3> Disable internal oscillators & clocks:-

To avoid synchronization problems during testing, intern oscillators and clocks should be disabled.

4> Avoid Asynchronous logic and Redundant logic:-

The design and test of an asynchronous logic circuit are more difficult than that of synchronous logic circuits. The operation of an asynchronous logic circuit is more sensitive to input test patterns which causes race condition.

Sometimes designed-in logic redundancy is used to mask a static hazard condition for reliability but these redundant nodes can not be observed since the primary output value can not be made dependent on the value of the redundant node.

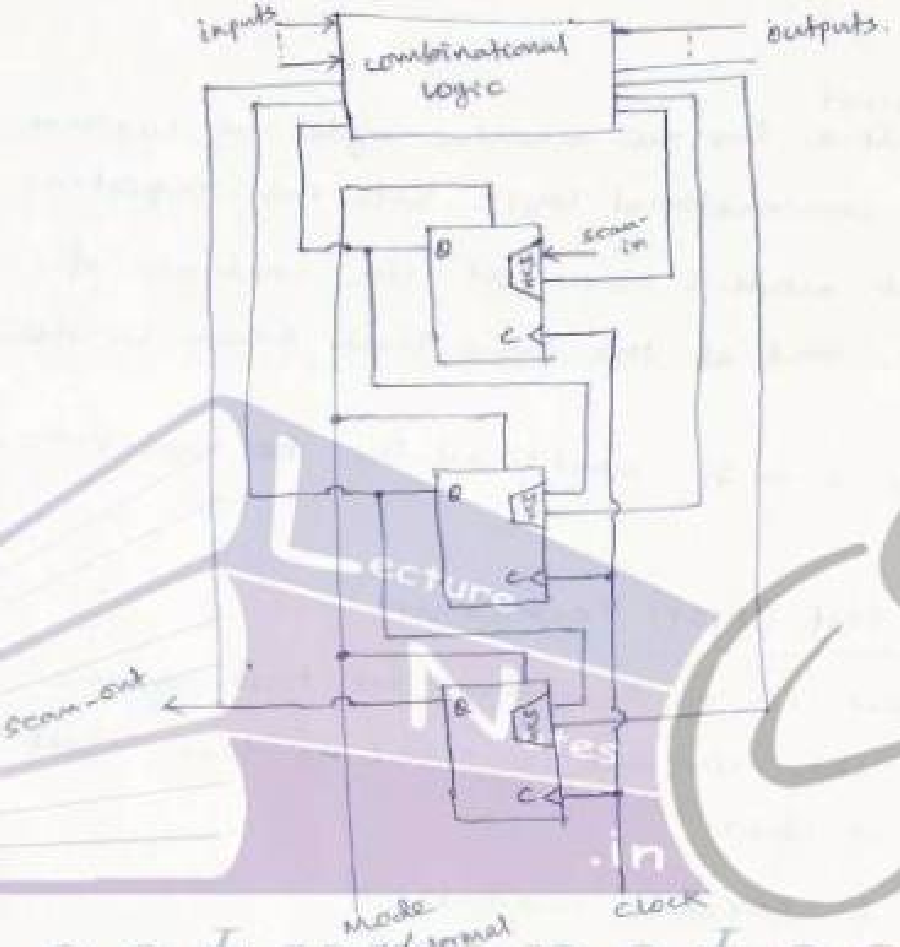
5> Avoid Delay-dependent logic:-

The use of delay-dependent logic should be avoided in design for testability since the delay dependent logic involves redundant combinational logic whose output is set to logic '0'.

Scan-Based Techniques:-

It is a structured approach for designing sequential logic circuits for testability of IC chips. It uses scan registers with both shift and parallel load capabilies. Using scan-based technique, the testing of sequential circuit is reduced to testing of combinational circuit.

In scan-based design, the storage elements of the sequential circuit are connected to form a long serial shift register called "scan-path", by using multiplexers and a mode control signal.



→ In test-mode, the scan-in signal is clocked into the scan-path, and the output of the last stage latch is scanned out.

In normal-mode, the scan-in path is disabled and the circuit functions as a sequential circuit.

Steps for Scan-Based Technique :-

1> set the mode to test and let latches accept data from scan-in input..

2> verify the scan path by shifting in and out the test data.

3> Scan-in (shift in) the desired state vector into the shift register.

4> Apply the test pattern to the Primary input pins.

5> Set the mode to normal and observe the primary outputs of the circuit after sufficient time for propagation.

6> Assert the circuit clock for one machine cycle to capture the outputs of the combinational logic into the registers.

7> Return to test mode; scan out the contents of shift registers, and at the same time scan in the next pattern.

8> Repeat steps 3 to 7 until all the ~~to~~ test patterns are applied.

Built-in Self Test (BIST) Technique :-

BIST technique is a technique for testing the IC chip where the circuit is used to test itself alongwith the IC chip.

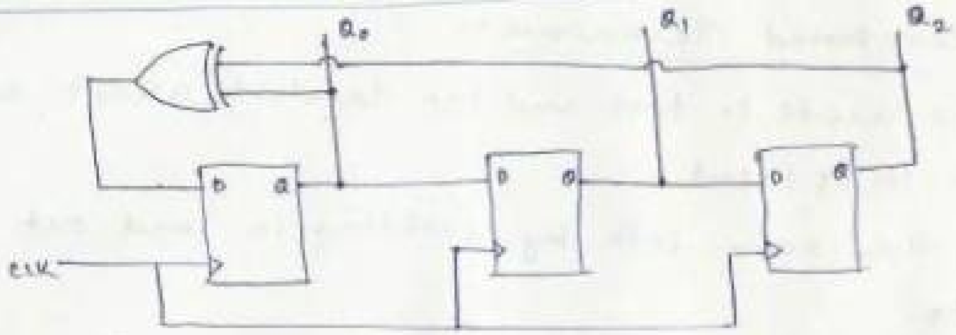
It involves 2 circuit modules :

1> Pseudorandom pattern generator (PRPG)

2> Output Response Analyser (ORA)

The implementation of both these modules are done by using Linear feedback shift Registers (LFSR).

PseudoRandom Pattern Generator :-



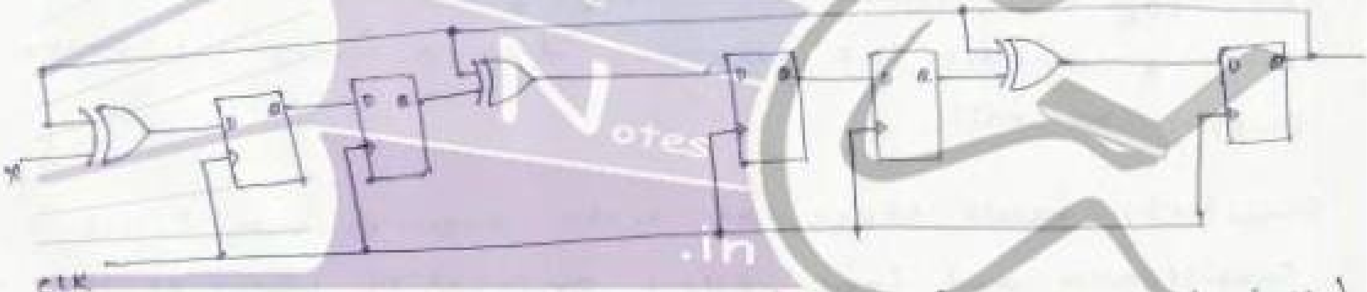
(Pseudorandom sequence generator using LFSR)

To test a circuit, test patterns have to be generated either by using a pseudorandom pattern generator, weighted test generator or an adaptive test generator.

Output Response Analyzer (ORA) :-

The output Response Analyzer can be designed by using a linear feedback shift Register. Here instead of the entire test data, a compact test data is compared with the predicted test data.

A typical example of compact test data formation technique is called 'Signature Analysis' which uses 'polynomial division' method, where the remainder becomes the signature. This signature is then compared with the expected signature to determine whether the device under test is faulty or not.



(polynomial division using LFSR for signature Analysis)

Built-in Logic Block Observer :-

The Built-in Logic Block observer (BILBO) register is a form of ORA which can be used in each cluster of partitioned registers.

The BILBO operation allows ~~multiple~~ monitors the circuit operation by ~~XOR~~ XOR into LFSR at multiple points.

(figure 15.14, page 635 : 3-bit built-in logic observer (BILBO) example)

Current monitoring I_{DDQ} test :-

This is the most commonly used technique for testing fabrication defects is called I_{DDQ} test.

In case of bridging fault, the static currents from the power supply is very high i.e much higher than the expected range of leakage currents.

This method can also detect other fabrication defects which are undetected by other test methods. Other types of fabrication defects are :-

1) Gate oxide short
2) channel punch through
3) p-N diode leakage
4) Transmission gate defect

The I_{DDQ} test consists of applying the test vector & ~~then~~ monitors the current drawn from the power supply rail.

Similarly ~~fault~~ stuck-at tests require ~~both~~ fault sensitization and fault effect propagation whereas the I_{DDQ} test has only fault sensitization.

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